On the geometry of linked projective spaces

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May 7, 2026

Contents

Abstract

Linked projective spaces are quiver Grassmannians of constant dimension vector equal to one for a certain class of quiver representations arising from degenerations of linear series. The goal of this thesis is to study the geometry of linked projective spaces. More precisely, we address the problem of computing their Chow class and of characterizing the type of their singularities.

The first main result is the computation of the Chow class of linked projective spaces. We show that it coincides with the Chow class of the small diagonal in a certain product of equidimensional projective spaces. The second main result shows that linked projective spaces are normal crossings schemes.

This thesis is divided into five chapters. The first chapter is the introduction. We explain in detail the motivation for the problems addressed and some further developments that we decided not to include. The second chapter is devoted to developing a combinatorial framework based on polymatroidal tilings. The third chapter deals with linked nets, the class of quiver representations that we are interested in. In chapter four, we explain how to associate to a linked net a family of polymatroids. We study their base polytopes: we characterize their faces in terms of combinatorial data coming from the linked net, and we then show that they form a regular polyhedral tiling of a certain subset of an affine space. Finally, the fifth chapter deals with linked projective spaces: we use the theory developed in the previous chapters to compute their Chow class. We show that linked projective spaces are strictly normal crossings.

Resumo

Os espaços projetivos ligados são Grassmannianos de representações de aljavas, com vetor de dimensão constante igual a um, associados a uma certa classe de representações de aljavas que surgem no estudo de degenerações de séries lineares. O objetivo desta tese é estudar a geometria dos espaços projetivos ligados. Mais precisamente, abordamos o problema de calcular suas classes de Chow e de caracterizar o tipo de suas singularidades.

O primeiro resultado principal é o cálculo da classe de Chow dos espaços projetivos ligados. Mostramos que ela coincide com a classe de Chow da diagonal pequena em um certo produto de espaços projetivos equidimensionais. O segundo resultado principal mostra que os espaços projetivos ligados são esquemas com cruzamentos normais.

Esta tese é dividida em cinco capítulos. O primeiro capítulo é a introdução. Explicamos em detalhe a motivação dos problemas abordados e alguns desenvolvimentos posteriores que decidimos não incluir. O segundo capítulo é dedicado ao desenvolvimento de um conjunto de ferramentas combinatórias baseado em tesselações polimatroidais. O terceiro capítulo trata de redes ligadas, a classe de representações de aljavas que nos interessa. No quarto capítulo, explicamos como associar, a uma rede ligada, uma família de polimatroides. Estudamos seus politopos base: caracterizamos suas faces em termos de dados combinatórios provenientes da rede ligada e mostramos, em seguida, que eles formam uma tesselação poliédrica regular de um certo subconjunto de um espaço afim. Por fim, o quinto capítulo trata dos espaços projetivos ligados: utilizamos a teoria desenvolvida nos capítulos anteriores para calcular suas classes de Chow. Mostramos que os espaços projetivos ligados são esquemas com cruzamentos normais estritos.

Acknowledgment

Nao terminado! nao proprio para ser lido.

O autor gostaria de agradecer à CAPES por ter financiado este trabalho.

Chapter 1

Introduction.

1.1 Motivation.

The main goal of this thesis is to study the geometry of the linked projective space. More precisely, we address the problem of computing its Chow class and characterizing the type of its singularities. Before explaining the results in this thesis, we will briefly discuss the motivation behind the study of the linked projective space.

Consider a family of linear series on smooth varieties parameterized by a punctured smooth curve. As we complete the family of varieties by adding a possibly singular fiber X, we may often complete the family of linear series by adding a linear series on X in infinitely many ways. In many situations, several of these linear series are necessary in order to describe the full geometry of the degeneration.

The case that has been most studied is when X is a nodal curve, most notably: for linear series of rank 1 by Harris and Mumford [15], with their theory of admissible covers; for X of compact type, by Eisenbud and Harris [8], with their theory of limit linear series; and for X with two components by Osserman [25], who later considered general nodal curves [26, 27].

In [11] and [10], Esteves, Santos and Vital interpreted Osserman's general approach in [26, 27] as that of organizing the data of all limits of linear series on X in terms of quiver representations, and extended it to higher dimensional varieties.

More precisely, consider a flat proper map $\mathcal{X} \to B$ over the spectrum B of a discrete valuation ring R. Let X be its special fiber. Assume X is connected and reduced, and that $\mathcal{X}$ is regular. (A semistable family satisfies these assumptions; see [20].)

Let (L_η, V_η) be a linear series on the generic fiber X_η of $\mathcal{X}/B$. Since $\mathcal{X}$ is regular, there is a line bundle extension $\mathcal{L}$ of L_η to $\mathcal{X}$. Let $X_0, \dots, X_n$ be the irreducible components of X; they are Cartier divisors of $\mathcal{X}$ whose sum is equivalent to zero. Each line bundle extension of L_η is isomorphic to $\mathcal{L}(\sum \ell_i X_i)$ for a tuple $(\ell_0, \dots, \ell_n) \in \mathbb{Z}^{n+1}$. Its isomorphism class depends only on the class u of the tuple in $\mathbb{Z}^{n+1}/\mathbb{Z}(1, \dots, 1)$. Fix a line bundle $\mathcal{L}_u$ in the isomorphism class.

Define a quiver Q with vertex set $\mathbb{Z}^{n+1}/\mathbb{Z}$ by putting an arrow, and only one, from vertex u to v if and only if $v - u = \bar{e}_i$ for some $i \in [n]$, where $e_0, \dots, e_n$ is the canonical basis of $\mathbb{Z}^{n+1}$. We get a representation $\mathfrak{L}$ of Q associating to each vertex u of Q the line bundle $L_u := \mathcal{L}_u|_X$, and to the arrow in Q from u to v with $v - u = \bar{e}_i$ the restriction to X of the map $\mathcal{L}_u \to \mathcal{L}_v$ given by summing X_i.

Finally, we get a subrepresentation $\mathfrak{V}$ of the representation $H^0(X, \mathfrak{L})$ obtained by taking global sections in $\mathfrak{L}$, which associates to a vertex u of Q the image in $H^0(X, L_u)$ of $V_\eta \cap H^0(\mathcal{X}, \mathcal{L}_u)$.

In the case the residue field $\mathbf{k}$ of the valuation R is algebraically closed, it was shown in [10][Prop. 10.9 and Thm. 10.12] that the map $\mathbb{P}(V_\eta) \to \mathrm{Hilb}_{X_\eta}$, associating to a nonzero section its zero scheme, degenerates to a naturally defined map $\mathbb{LP}(\mathfrak{V}) \to \mathrm{Hilb}_X$, where Hilb denotes the Hilbert scheme, and $\mathbb{LP}(\mathfrak{V})$ the *linked projective space* of $\mathfrak{V}$, the Grassmannian of subrepresentations of $\mathfrak{V}$ with constant dimension vector 1.

It is therefore natural to study the geometry of $\mathbb{LP}(\mathfrak{V})$. It follows from [10][Thm. 9.2] that $\mathbb{LP}(\mathfrak{V})$ is the special fiber of the Mustafin variety, or Deligne scheme, of a certain configuration of lattices; see [4]. This is a well-studied class of degenerations of projective spaces. Also, it was shown in [4][Thm. 2.3, p. 763] that $\mathbb{LP}(\mathfrak{V})$ is reduced and Cohen-Macaulay.

It is also natural to understand which representations $\mathfrak{V}$ of Q arise in this construction. Esteves, Santos and Vital [10][Prop. 4.1, p. 532, and Prop. 4.2, p. 533], proved that $\mathfrak{V}$ is what they call an *exact finitely generated linked net of vector spaces* over Q. Namely, a representation $\mathfrak{V}$ of Q, assigning to each vertex v a vector space V_v, and to each arrow a from vertex v to u a linear map $\varphi_a \colon V_v \to V_u$, is a *linked net* if, for each vertex v of Q and tuples $\ell, \ell' \in \mathbb{Z}^n_{\geq 0}$ of nonnegative integers, with $u := v + \bar{\ell}$ and $w := v + \bar{\ell}'$, the following properties are satisfied:

1. If $\ell_i > 0$ for all i, then for each path γ in Q connecting v to u with $\sum \ell_i$ arrows, the composition φ_γ of the φ_a for arrows $a \in \gamma$ is zero.
2. If $\ell_i = 0$ for some i, then the φ_γ for paths γ in Q with $\sum \ell_i$ arrows connecting v to u differ by multiplication by nonzero scalars. (Denote $\varphi^v_u := \varphi_\gamma$ for any such path γ.)
3. If $\ell + \ell' = (1, \dots, 1)$ then $\mathrm{Ker}(\varphi^v_u) \cap \mathrm{Ker}(\varphi^v_w) = 0$.

In addition, $\mathfrak{V}$ is called

(4) *exact* if $\mathrm{Ker}(\varphi^v_u) = \mathrm{Im}(\varphi^u_v)$ for each two distinct vertices u and v of Q with $u - v = \bar{\ell}$ for $\ell \in \mathbb{Z}^n$ with $0 \leq \ell_i \leq 1$ for all i, and

(5) *finitely generated* by a finite set H of vertices of Q if $\sum_{z \in H} \mathrm{Im}(\varphi^z_v) = V_v$ for each vertex v of Q.

1.2 First main result.

The main problem considered in this thesis is whether the linked projective space $\mathbb{LP}(\mathfrak{V})$ of an exact finitely generated linked net $\mathfrak{V}$ of vector spaces over Q is the special fiber of a Mustafin variety. Although we do not provide a complete answer to this question, the results presented here are indications of a possible positive answer. In particular, we show that $\mathbb{LP}(\mathfrak{V})$ has the same Chow class as the diagonal, and thus the same class as the special fiber of a Mustafin variety.

We will now explain our main results in more detail. As we noted above, the quiver Q is infinite, but $\mathfrak{V}$ is generated by a finite set of vertices H of Q, which we may pick minimum. This allows us to realize the linked projective space as a closed subscheme of a product of projective spaces:

$$\mathbb{LP}(\mathfrak{V}) \subseteq \prod_{v \in H} \mathbb{P}(V_v).$$

Since $\mathfrak{V}$ is exact, the vector spaces V_v have the same dimension. One of the main results of this thesis is Theorem 5.1.3, which states that $\mathbb{LP}(\mathfrak{V})$ has the same class as the small diagonal in the above product of projective spaces.

We prove Theorem 5.1.3 by computing the Chow classes of the irreducible components of $\mathbb{LP}(\mathfrak{V})$. As shown in [11], the irreducible components of $\mathbb{LP}(\mathfrak{V})$ are the schematic images $\mathbb{LP}(\mathfrak{V})_u$ of the natural rational maps

$$\mathbb{P}(V_u) \dashrightarrow \prod_{v \in H} \mathbb{P}(V_v)$$

induced by the linear maps φ_v^u. The Chow class of $\mathbb{LP}(\mathfrak{V})_u$, by a result of Binglin Li's [21], is the sum of products

$$\prod_{v \in H} h_v^{r-q(v)} \in A^* \left(\prod_{v \in H} \mathbb{P}(V_v) \right),$$

where each h_v is the pullback to the product of the hyperplane class in the factor $\mathbb{P}(V_v)$, and where q ranges over a certain subset M_u of the integer tuples in

$$\Omega_r := \left\{ q \in \mathbb{R}^H \mid q(v) \geq 0 \text{ for all } u \in H \text{ and } \sum_{v \in H} q(u) = r \right\},$$

for r equal to the dimension of the $\mathbb{P}(V_v)$.

We therefore need to show that the sets M_v are pairwise disjoint and that their union covers the full set of integer points in Ω_r. To this end, we introduce a general framework that produces tilings of Ω_{r+1} by polymatroids. We apply this results to the collection of polytopes $\mathbf{P}_u \subseteq \mathbb{R}^H$ for $u \in H$ associated to the linear maps $V_u \to \bigoplus_{v \in H} V_v$. Indeed, Theorem 4.2.2 and Theorem 4.3.1 state that the $\mathbf{P}_u$ and their faces form a polyhedral tiling of Ω_{r+1}. In the course of the proof, we also characterize the faces of $\mathbf{P}_u$ in terms of the combinatorics of H.

To conclude, we show that the scaled polytopes $r/(r+1)\mathbf{P}_v$ do not intersect, their union contains al the integer tuples in Ω_r, and the set of integer tuples in $r/(r+1)\mathbf{P}_v$ is M_v, for each $v \in H$.

However, our results are not sufficient to conclude that $\mathbb{LP}(\mathfrak{V})$ is a degeneration of the small diagonal. Even having the same Hilbert polynomial of the diagonal does not suffice; see [5].

As a complementary result, we show that the tilling given by the $\mathbf{P}_u$ and their faces is regular. This can be seen as another indication that linked projective spaces may be special fibers of Mustafin varieties; see [1]§ 9.2, where a connection to works by Kapranov and Lafforgue among others is explained.

1.3 Second main result.

Our second main result is Theorem 5.4.4, which states that $\mathbb{LP}(\mathfrak{V})$ is a normal crossings scheme. The proof is based on a local description of $\mathbb{LP}(\mathfrak{V})$ and on the theory of Mustafin varieties. To make this description precise, we briefly recall the definition of Mustafin varieties and then explain their role in our arguments.

A Mustafin variety is the total space of a degeneration of projective spaces induced by a finite collection of lattices in a vector space over a valued field. More precisely, let V be a vector space of dimension d over a field of fractions $\mathbf{K}$ of a discrete valuation ring R, and let $\mathbb{P}(V) := \mathrm{ProjSym}(V^*)$ denote the projective space parameterizing lines on V through 0. We call free R-modules $L \subseteq V$ of rank d *lattices*, and we define $\mathbb{P}(L) := \mathrm{ProjSym}(L^*)$, where $L^* = \mathrm{Hom}_R(L, R)$. We will mostly consider lattices up to *homothety*, i.e., $L \sim L'$ if $L = c \cdot L'$ for some $c \in \mathbf{K}^\times$. We denote the homothety class of L by $[L]$. A collection Γ of lattice classes in V, also called a *lattice configuration*, is *convex* if $[T^a L \cap T^b L'] \in \Gamma$ for every $[L], [L'] \in \Gamma$ and $a, b \in \mathbb{Z}$, where $T \in R$ is a parameter for the valuation.

Definition 1.3.1. ([4, Def. 1.1] or [14, Def. 1.1]) Let $\Gamma = \{[L_1], \ldots, [L_n]\}$ be a set of lattice classes in V. Then $\mathbb{P}(L_1), \ldots, \mathbb{P}(L_n)$ are projective spaces over R whose generic fibers are canonically isomorphic to $\mathbb{P}(V)$. The open immersions $\mathbb{P}(V) \hookrightarrow \mathbb{P}(L_i)$ give rise to a map

$$\mathbb{P}(V) \longrightarrow \mathbb{P}(L_1) \times_R \cdots \times_R \mathbb{P}(L_n).$$

We denote the closure of the image, endowed with the reduced scheme structure, by $\mathcal{M}(\Gamma)$. We call $\mathcal{M}(\Gamma)$ the *associated Mustafin variety* to Γ. Its special fiber $\mathcal{M}(\Gamma)_{\mathbf{k}}$ is a scheme over $\mathbf{k}$, where $\mathbf{k}$ is the residue field of R.

Mustafin varieties were introduced by Mustafin in [24] in order to generalize Mumford's work on uniformisation of curves to higher dimensions in [23]. Since then, in the case of convex lattice configurations, they have been studied by several authors under the name of Deligne schemes, see for instance [5], [12] and [19]. The term Mustafin varieties was later introduced by D. Cartwright, M. Häbich, B. Sturmfels and A. Werner in [4] to recognize Mustafin's contributions in [24]; their work treats arbitrary lattice configurations.

We prove that each point on $\mathbb{LP}(\mathfrak{V})$ admits a neighborhood isomorphic to an open subscheme of the linked projective space of a certain simpler exact finitely generated linked net $\mathfrak{V}_\Delta$. We then show that $\mathbb{LP}(\mathfrak{V}_\Delta)$ is the special fiber of the Mustafin variety $\mathcal{M}(\Gamma)$ associated to a suitable lattice configuration Γ. Finally, we prove that Γ is convex. The conclusion then follows from a result of Faltings's stating that Mustafin varieties associated to convex lattice configurations are strictly semi-stable schemes over the spectrum of R; see [12] or [13]. Thus, their special fibers are strict normal crossings schemes.

1.4 Further developments.

The framework and the intermediate results we obtain in this thesis have also been applied to others problems. Since these applications will not be included in this thesis, we at least summarize them below.

1.4.1 Mustafin varieties.

Motivated by the parallels between linked projective spaces and special fibers of Mustafin varieties associated with convex configurations, one is led to ask the following question: Given a convex configuration Γ, is the special fiber of the associated Mustafin variety $\mathcal{M}(\Gamma)$ the linked projective space of an exact finitely generated linked net $\mathfrak{V}$ over a suitable $\mathbb{Z}^n$-quiver Q?

The answer is affirmative, and this result is part of a joint work with E. Esteves and C. Urtecho. We briefly outline the argument below.

The proof proceeds in three steps. First, in [14], the authors show, using the work by Faltings [12], that Mustafin varieties are prelinked Grassmannians; we refer to 5.3.1 for details and definitions. As a consequence, for a convex configuration of lattices Γ, the special fiber of the Mustafin variety $\mathcal{M}(\Gamma)$ can be realized as the quiver Grassmanian of subrepresentations of constant dimension 1 of a suitable quiver representation. The set of vertices of the underlying quiver is Γ.

Second, we the apply work by Sturmfels and Yu [18] to realize this quiver as a subquiver of a suitable $\mathbb{Z}^n$-quiver. Third, we show that the quiver representation satisfies the criteria of Section 3.6, which implies that it extends to an exact finitely generated linked net whose linked projective space is the same as the quiver Grassmannian mentioned above.

1.4.2 The moduli scheme $G_d^0(X)$ for a curve X of compact type.

Let d be a nonnegative integer. The degree-d Abel–Jacobi of a smooth curve X is a morphism $\alpha_d\colon S^d(X) \to \mathrm{Pic}_X^d$ from the dth symmetric product of X to the degree-d component of the Picard scheme of X, taking a divisor D to the associated line bundle $\mathcal{O}_X(D)$. Its fibers can be identified with the projectivizations of the spaces of global sections of line bundles L on X, each nonzero global section of L giving a Cartier divisor, its zero scheme.

This framework breaks down if X is a singular curve. Points on $S^d(X)$ are cycles, not Cartier divisors, and even if we replace $S^d(X)$ by the Hilbert scheme Hilb_X^d, it parameterizes Cartier divisors only on an open subset. Even we replace Pic_X^d by a compactification (or more precisely, an appropriate compactified Jacobian), and obtain a well-defined global map α_d, as done by Coelho and Pacini in [6], its fibers are not necessarily projective spaces. In fact, not all subschemes of X of length d can be given as the zero scheme of a single section of a line bundle over X. But all limits of Cartier divisors along families of smooth curves degenerating to X can be given as intersections of zero schemes of sections of line bundles over X which are compatible under a quiver representation, as shown in [11].

So we may use the framework developed in this thesis to construct Abel–Jacobi maps for singular curves. We explain how, for a curve X of compact type. Before doing so, we need to introduce some notation.

Definition 1.4.1. [17, Def. 4.2] Let X be a nodal curve over an algebraically closed field $\mathbf{k}$. Let G be its dual graph and V its vertex set. An *enriched structure* on X consists of the following data. For each vertex $\mathfrak{a} \in V$ a line bundle $\mathcal{O}_\mathfrak{a}$ on X and a section $s_\mathfrak{a} \in H^0(X, \mathcal{O}_\mathfrak{a})$, satisfying the following conditions:

(E1) For each vertex $\mathfrak{a} \in V$, we have $\mathcal{O}_\mathfrak{a}|_{X_\mathfrak{a}} \simeq \mathcal{O}_{X_\mathfrak{a}}(-(X_\mathfrak{a}^c \cap X_\mathfrak{a}))$ and $\mathcal{O}_\mathfrak{a}|_{X_\mathfrak{a}^c} \simeq \mathcal{O}_{X_\mathfrak{a}^c}(X_\mathfrak{a}^c \cap X_\mathfrak{a})$;

(E2) $\bigotimes_{\mathfrak{a}\in V} \mathcal{O}_a \simeq \mathcal{O}_X$;

(E3) For each vertex $\mathfrak{a} \in V$, the section $s_\mathfrak{a}$ vanishes precisely along $X_\mathfrak{a}$.

Above, $X_\mathfrak{a}$ is the component associated to $\mathfrak{a}$, and $X_\mathfrak{a}^c := \overline{X - X_\mathfrak{a}}$. The bundles $\mathcal{O}_\mathfrak{a}$ have been shown to arise simultaneously as the degenerations of the trivial bundle on a family of smooth curves degenerating to X by Mainò [22].

For the remanding of this section, we fix an enriched structure $(\mathcal{O}_\mathfrak{a}, s_\mathfrak{a} \,|\, \mathfrak{a} \in V)$ on a nodal curve X. Let d be an integer and L be a degree-d line bundle on X. The *multidegree* of L is the tuple

$$\underline{\deg}(L) = (\deg(L|_{X_\mathfrak{a}}))_{\mathfrak{a}\in V}.$$

Order V, so that we may write $V = \{0, \ldots, n\}$. For each $i, j \in V$, let $x_i := \deg(L|_{X_i})$ and $x_{i,j} := \deg(\mathcal{O}_i|_{X_j})$. Put $v = (x_0, \ldots, x_n)$, the multidegree of L, and $v_i := (x_{i0}, \ldots, x_{in})$, the multidegree of $\mathcal{O}_i$, for $i = 0, \ldots, n$. Notice that $v_0 + \cdots + v_n = 0$ by (E2). Put

$$Q_0 := v + \mathbb{Z}v_0 + \cdots + \mathbb{Z}v_n \subseteq \mathbb{Z}^{n+1}.$$

Each $u \in Q_0$ can be written in a unique way as $u = v + \sum \ell_i v_i \in Q_v$ for integers ℓ_i with $\min(\ell_0, \dots, \ell_n) = 0$; it is the multidegree of

$$L_u := L \otimes \left(\bigotimes_{i=0}^{n} \mathcal{O}_i^{\otimes \ell_i} \right) \tag{1.4.1}$$

We call the sheaves L_u the *twists* of L. (In particular, $L_v = L$.) It follows from (E3) that the sheaf

$$\bigotimes_{i=0}^{n} \mathcal{O}_i^{\otimes \ell_i}$$

has a section, given as the tensor product of the $s_i^{\otimes n_i}$, which vanishes exactly along

$$\bigcup_{i:\ell_i \neq 0} X_i.$$

For each $u_1, u_2 \in Q_0$ there are unique integers ℓ_i with $\min(\ell_0, \dots, \ell_n) = 0$ such that $u_2 = u_1 + \sum \ell_i v_i$. Hence, there is a natural nonzero morphism

$$\varphi_{u_2}^{u_1} : \mathcal{L}_{u_1} \to \mathcal{L}_{u_2} \tag{1.4.2}$$

induced by the tensor product of the $s_i^{\otimes \ell_i}$.

Let $A_i \subseteq Q_0 \times Q_0$ be the subset of pairs (u_1, u_2) such that $u_2 = u_1 + v_i$ for each $i = 0, \dots, n$, and set $Q_1 := \bigcup A_i$. Then $Q_v := (Q_0, Q_1)$ is a quiver, called a $\mathbb{Z}^n$*-quiver*; see Section 3.1 for definition.

The collection of the L_u for $u \in Q_0$ and the maps $\varphi_{u_2}^{u_1}$ for $(u_1, u_2) \in Q_1$ gives us a representation of Q_v, denoted by L_Q. We call L_Q the *linked net generated by L and its twists.*

If X is of compact type, then Q_0 is the set of all multidegrees $(u_0, \dots, u_n) \in \mathbb{Z}^{n+1}$ with total degree $\sum u_i = d$. Assume X is of compact type. In this case, the isomorphism classes of the $\mathcal{O}_\mathfrak{a}$ are fixed by ((E1)). (But the sections $s_\mathfrak{a}$ are not.)

For each positive integer d, let $v = (v_0, \dots, v_n) \in \mathbb{Z}^{n+1}$ be a multi-degree with total degree $\sum v_i = d$, and let H be the set of effective multidegrees $(u_0, \dots, u_n) \in \mathbb{Z}_{\geq 0}^{n+1}$ with total degree d, together with all the bridges between them; see Section 3.4 for the definition of bridges. Define the functor $\mathcal{G}_d^0(X)$ as that parameterizing the data of a line bundle L on X of multi-degree v and 1-dimensional subspaces $W_u \subseteq H^0(X, L_u)$ for each $u \in H$ such that $\varphi_{u_2}^{u_1}(W_{u_1}) \subseteq W_{u_2}$ for all $u_1, u_2 \in H$, where the L_u are given in (1.4.1) and the $\varphi_{u_2}^{u_1}$ are those in (1.4.2). We show:

Theorem 1.4.2. *$\mathcal{G}_d^0(X)$ is representable over Pic_X^v, the connected component of multidegree-v line bundles of the Picard scheme of X, by a projective morphism $\alpha_d \colon G_d^0(X) \to \mathrm{Pic}_X^v$. The fiber of α_d over each $L \in \mathrm{Pic}_X^v$ is the linked projective space $\mathbb{LP}(H^0(L_Q))$, where L_Q is the linked net obtained from L and all its twists L_u. Also, there is a scheme-theoretic surjection $\Psi_d \colon G_d^0(X) \to \mathrm{Hilb}_X^d$. The morphism Ψ_d has a quasi-section defined over the locus of Cartier divisors in Hilb_X^d.*

The scheme $G_d^0(X)$ is related to the Abel–Jacobi map constructed by Coelho and Pacini in [6]. In fact, they constructed a natural map $\beta_d \colon S^d(X) \to \mathrm{Pic}_X^e$ for a certain multidegree e. Composing on the left by the Hilbert-to-Chow map, and on the right by an isomorphism obtained by twisting the line bundles, we get a morphism $\tilde{\alpha}_d \colon \mathrm{Hilb}^d_X \to \mathrm{Pic}^v_X$. We show that Ψ_d is a morphism of Pic^v_X-schemes, that is, that the following diagram commutes:

$$\begin{tikzcd} G^0_d(X) \arrow[r, "\Psi_d"] \arrow[d, "\alpha_d"] & \mathrm{Hilb}^d_X \arrow[d, "\tilde{\alpha}_d"] \\ \mathrm{Pic}^v_X & \mathrm{Pic}^v_X \arrow[l, "\beta^d_X"'] \end{tikzcd}.$$

1.4.3 Riemman theorem for curves of compact type with three irreducible components.

We extend the work by Esteves, Rodriguez and Vital on the Riemman problem for curves of compact type with two irreducible components. We explain:

In [9] the authors raise the following Riemann-Roch type question: For a linked net of linear series $(Q, \mathfrak{L}, \mathfrak{V})$ how large can $\mathfrak{V}$ be, given Q and $\mathfrak{L}$? The authors address it for a curve X of compact type with two irreducible components.

More precisely, let X be a curve of compact type with two irreducible components Y and Z, which are smooth and meet transversally at a single point N. Assume that X has arithmetic genus g. Given a line bundle L over X, the corresponding linked net $\mathfrak{L}_Q$ is a representation of the $\mathbb{Z}^1$-quiver Q in the category of line bundles over X whose associated line bundles L_u are gluings of the line bundles $L_{|Y}(-uN)$ and $L_{|Z}(uN)$ over Y and Z at N, and whose maps are the natural ones; see Section 1.4.2. In [9], the authors define $\mathfrak{h}^0(L)$ as the maximum dimension of a pure linked net $\mathfrak{V}$ over Q that is a subrepresentation of $H^0(X, L_Q)$. They show the following:

Theorem 1.4.3. [9][Thm. 5.9] *Let L be a line bundle over X with* $\deg(L) \geq 2g-1$. *Then*

$$\mathfrak{h}^0(L) = \deg(L) - g + 1.$$

We extend this result to the case where X is a nodal curve of compact type curve with 3 irreducible components. We consider a line bundle L over X, and L_Q the representation over the $\mathbb{Z}^2$-quiver Q defined in Section 1.4.2, and let as before $\mathfrak{h}^0(L)$ be the maximum dimension of a pure linked net $\mathfrak{V}$ over Q that is a subrepresentation of $H^0(X, L_Q)$. We show:

Theorem 1.4.4. *Let L be a line bundle over X with degree* $\deg(L) \geq 2g-1$. *Then*

$$\mathfrak{h}^0(L) = \deg(L) - g + 1.$$

Chapter 2

Polyhedral tilings.

In this section, we review and prove new facts about polyhedral tilings. The material in this section is based on [1].

2.1 Polytopes.

We review the theory of modular pairs and their base polytopes.

Let H be a finite nonempty set, and $\mathbb{R}^H$ the vector space whose elements are maps $q : H \to \mathbb{R}$. For each $I \subseteq H$, set $q(I) := \sum_{v \in I} q(v)$. For each integer d, we denote by $\mathbb{R}^H_d$ the affine subspace of $\mathbb{R}^H$ consisting of all q with $q(H) := d$.

We denote by 2^H the collection of subsets of H. For $I \subseteq H$, denote by I^c the complement of I in H, that is, $I^c := H - I$.

We say a function $\mu : 2^H \to \mathbb{R}$ is *supermodular* if $\mu(\emptyset) = 0$ and we have inequalities

$$\mu(I_1) + \mu(I_2) \leq \mu(I_1 \cup I_2) + \mu(I_1 \cap I_2) \text{ for each } I_1, I_2 \subseteq H.$$

Similarly, we say that a function $\mu : 2^H \to \mathbb{R}$ is *submodular* if $\mu(\emptyset) = 0$ and the above inequalities are all reversed. In any case, the quantity $\mu(H)$ is called the *range* of μ. Functions that are both supermodular and submodular are called *modular.* If $I = \{v\}$ is a singleton, by abuse of notation, we simply write $\mu(v)$ instead of $\mu(\{v\})$.

Example 2.1.1. For each $q \in \mathbb{R}^H_d$, the function $I \mapsto q(I)$ is modular.

For each $\mu : 2^H \to \mathbb{R}$, let $\mu^* : 2^H \to \mathbb{R}$ be the *adjoint function to* μ, defined by

$$\mu^*(I) := \mu(H) - \mu(H - I) \text{ for each } I \subseteq H.$$

Observe that μ is supermodular (resp. submodular) if and only if μ^* is submodular (resp. supermodular). When μ is supermodular, we refer to μ^* as the *submodular function adjoint to* μ and call the ordered pair (μ, μ^*) an *adjoint modular pair*, or simply a *modular pair.*

For each modular pair (μ, μ^*), we define the subset in $\mathbb{R}^H$:

$$\mathbf{P}_\mu := \mathbf{P}_{(\mu,\mu^*)} := \left\{ q \in \mathbb{R}^H \mid \mu(I) \leq q(I) \leq \mu^*(I) \text{ for each } I \subseteq H \right\}.$$

We say that $\mathbf{P}_\mu$ is the *base polytope associated to* μ, or simply the *polytope of* μ.

2.1.1 Partitions.

An *ordered partition* of H is an ordered sequence $\pi = (\pi_1, ..., \pi_s)$ of pairwise disjoint subsets of H whose union is equal to H. It is called *nontrivial* if $\pi_i \neq \emptyset$ for every i. If π is nontrivial, we call it a *bipartition* if $s = 2$, and a *s-partition* in general. If π is a bipartition, we put $\pi^c = (\pi_2, \pi_1)$.

The data of an ordered partition π as above is equivalent to the data of a filtration

$$F_\bullet := (F_0, \dots, F_s) \text{ where } \emptyset = F_0 \subseteq F_1 \subseteq F_2 \subseteq \cdots \subseteq F_s = H$$

via the correspondence $F_j := \pi_1 \cup \cdots \cup \pi_j$ for $j = 1, ..., s$ in one direction, and $\pi_j := F_j - F_{j-1}$ for $j = 1, ..., s$ in the other direction.

For each supermodular function μ and each two subsets $J_1, J_2 \subseteq H$ with $J_1 \subseteq J_2$, we define

$$\mu_{J_2/J_1}(I) := \mu((I \cap J_2) \cup J_1) - \mu(J_1) \text{ for each } I \subseteq H$$

Then, μ_{J_2/J_1} is a supermodular function.

For each ordered partition $\pi = (\pi_1, \dots, \pi_s)$ of H with $F_\bullet$ the corresponding filtration, we define the function μ_π on 2^H as the sum

$$\mu_\pi = \sum_{i=1}^{s} \mu_{F_i/F_{i-1}}.$$

Note that μ_π is supermodular. We say that μ_π is the *splitting of μ with respect to the ordered partition π*. We say that a supermodular function ν is a *splitting of μ* if it coincides with μ_π for some ordered partition π of H, and call the splitting *nontrivial* if $\nu \neq \mu$. The base polytopes of the splittings μ_π of a supermodular function μ can be computed as follows:

Proposition 2.1.2. *[1]*[Prop. 2.5.] Let μ be a supermodular function and $\pi = (\pi_1, \dots, \pi_s)$ an ordered partition of V. Let $\mathbf{F}_\bullet$ the corresponding filtration. Then, we have

$$\mathbf{P}_{\mu_\pi} = \mathbf{P}_\mu \cap \{q \in \mathbb{R}^H \mid q(F_j) = \mu(F_j) \text{ for } j = 1, \dots, s-1\}.$$

Let μ be a supermodular function 2^H. Denote by cd_μ the largest integer s for which there is a s-partition π of H such that $\mu = \mu_\pi$. We call cd_μ the *codimension* of μ.

The following proposition justifies the given name to cd_μ and make precise the relation between partitions, splittings, and faces.

Proposition 2.1.3. *[1]*[Prop. 2.7.] Let μ be a supermodular function. Then, the base polytope $\mathbf{P}_\mu$ is nonempty and we can recover μ (and μ^*) from $\mathbf{P}_\mu$ as follows:

$$\mu(I) = \min\{q(I) \mid q \in \mathbf{P}_\mu\} \quad \text{ and } \quad \mu^*(I) = \max\{q(I) \mid q \in \mathbf{P}_\mu\} \text{ for each } I \subseteq H.$$

In particular, we have $\mathbf{P}_{\mu_1} = \mathbf{P}_{\mu_2}$ if and only if $\mu_1 = \mu_2$. In addition, the base polytope $\mathbf{P}_{\mu_\pi}$ for ordered partition π of H are all the faces of $\mathbf{P}_\mu$. Finally, we have $\dim \mathbf{P}_\mu + \mathrm{cd}_\mu = |H|$.

2.1.2 Simpleness

A supermodular function μ is called *simple* if there is no s-partition π of H with $s > 1$ such that $\mu = \mu_\pi$, or equivalently, if $\mathbf{P}_\mu$ has dimension $|H| - 1$, or equivalently, $\mathrm{cd}_\mu = 1$.

Let μ, ν be two supermodular functions on 2^H of range n, and let $\pi = (I, J)$ be a bipartition of H. We say that π is a *separation* for the ordered pair (μ, ν) if $\mu(I) + \nu(J) \geq n$.

We say that the separation is strict if the inequality is strict. We say that the separation is *nontrivial* provided that it is either strict, or we have $\mu_\pi \neq \mu$ or $\nu_\pi \neq \nu$. Alternatively, π is a separation for (μ, ν) if and only if $\mu(I) \geq \nu^*(I)$, which is equivalent to $n \geq \nu^*(I) + \mu^*(J)$.

The following lemma will be used later.

Lemma 2.1.4. *Let $\mu : 2^H \to \mathbb{R}$ be a nonnegative supermodular function. Then, both μ and μ^* are nondecreasing. In addition, if*

(1) $\mu^*(v) \neq 0$ for all $v \in H$, and

(2) there is $v_0 \in H$ such that $\mu^*(v_0) = \mu^*(H)$,

then μ is simple.

Proof. Let $J \subseteq I$ be subsets of H. Since μ is nonnegative and supermodular,

$$\mu(J) \leq \mu(J) + \mu(I - J) \leq \mu(I),$$

so μ is nondecreasing. Applying this to $H - I$ and $H - J$, we obtain

$$\mu^*(J) = \mu(H) - \mu(H - J) \leq \mu(H) - \mu(H - I) = \mu^*(I).$$

Hence, also μ^* is nondecreasing.

Now, assume (1) and (2) hold. Suppose by contradiction that there is a s-partition $\pi = (\pi_1, \ldots, \pi_s)$ such that $\mu = \mu_\pi$. We may assume that $s = 2$. Indeed, let $\pi' = (\pi_1, \pi_1^c)$. It follows from [1][Prop. 2.2] that

$$\mu(\pi_2) + \cdots + \mu(\pi_s) = \mu_\pi(\pi_2) + \cdots + \mu_\pi(\pi_s) = \mu_\pi(\pi_1^c) = \mu(\pi_1^c).$$

Thus, again by [1][Prop. 2.2], we have that

$$\mu(H) = \mu(\pi_1) + \mu(\pi_1^c).$$

Hence, we may assume $s = 2$, and

$$\mu(H) = \mu(\pi_1) + \mu(\pi_2),$$

or equivalently

$$\mu^*(H) = \mu^*(\pi_1) + \mu^*(\pi_2).$$

Since π is a bipartition, either $v_0 \in \pi_1$ or $v_0 \in \pi_2$. Assume $v_0 \in \pi_1$. As μ^* is nondecreasing, we have

$$\mu^*(v_0) \leq \mu^*(\pi_1) \leq \mu^*(H).$$

Hence, $\mu^*(\pi_1) = \mu^*(H)$ by (2), and thus $\mu^*(\pi_2) = 0$. This implies that $\mu^*(u) = 0$ for all $u \in \pi_2$, contradicting (1). □

2.1.3 Families

Let $\mathcal{C}$ be a family of supermodular functions on 2^H of range d. We denote by $\mathbf{P}_\mathcal{C}$ the union of the polytopes $\mathbf{P}_\mu$ for $\mu \in \mathcal{C}$, and call it the *support of the collection* $\mathcal{C}$. Let $\Omega \subseteq \mathbb{R}^H_d$ be a subset. We say that $\mathcal{C}$ is *complete for* Ω if for each element $\mu \in \mathcal{C}$ and each bipartition π of H such that $\mathbf{P}_{\mu_\pi}$ intersects Ω, there is $\mu' \in \mathcal{C}$ distinct from μ_π such that $\mu'_{\pi^c} = \mu_\pi$. We say that $\mathcal{C}$ is *separated* if each pair (μ, ν) of distinct elements of $\mathcal{C}$ admits a nontrivial separation.

2.2 Polytopes associated to vector spaces.

Let $\mathbf{k}$ be a field and let H be a finite set. For each $v \in H$, let U_v be a finite-dimensional vector space over $\mathbf{k}$. Define $U := \bigoplus_{v\in H} U_v$. For each subset $I \subseteq H$, let $U_I := \bigoplus_{v\in I} U_v$, and denote by $\iota_I : U_I \to U$ and $\theta_I : U \to U_I$ the corresponding inclusion and projection maps, respectively. The composition $\theta_I \circ \iota_I$ is the identity. By convention, we set $U_\emptyset := (0)$. We have a natural exact sequence

$$0 \to U_I \xrightarrow{\iota_I} U \xrightarrow{\theta_{I^c}} U_{I^c} \to 0.$$

We view elements $\varphi \in \mathbf{k}^H$ as endomorphisms of U acting by scalar multiplication on each summand U_v; by abuse of notation, we denote such endomorphism φ. An endomorphism $\varphi \in \mathbf{k}^H$ is an automorphism if and only if $\varphi_v \neq 0$ for every $v \in H$.

Let $W \subseteq U$ be a vector subspace. For each $I \subseteq H$, we put $W_I := \iota_I \circ \theta_I(W)$ and denote by $\mu^*_W(I)$ its dimension, that is, $\mu^*_W(I) := \dim_{\mathbf{k}}(W_I)$. Similarly, we denote by W^I the intersection of W with $\iota_I(U_I)$, and by $\mu_W(I)$ its dimension. We have a short exact sequence

$$0 \to W^{I^c} \to W \to W_I \to 0,$$

where the second map is the inclusion and the third map is the composition $\iota_I \circ \theta_I$. Furthermore, if W has dimension n, then (μ_W, μ^*_W) is a modular pair of range n. Put $\mathbf{P}_W := \mathbf{P}_{\mu_W}$, the polytope of μ_W. For each $n \in \mathbb{Z}_{\geq 0}$, denote by Ω_n the standard simplex in $\mathbb{R}^H_n$, given by

$$\Omega_n := \left\{ q \in \mathbb{R}^H_n \,|\, q(v) \geq 0 \text{ for all } v \in H \text{ and } q(H) = n \right\}.$$

The polytope $\mathbf{P}_W$ is a subset of the standard simplex Ω_n in $\mathbb{R}^H_n$. Note as well that $\mathbf{P}_W = \mathbf{P}_{\varphi W}$ for every automorphism $\varphi \in \mathbf{k}^H$.

2.2.1 Splittings.

Let $\pi = (\pi_1, \ldots, \pi_s)$ be a nontrivial ordered partition of H. Denote by $F_\bullet$ the corresponding filtration of H. Let $W \subseteq U$ be a vector subspace. Consider the filtration

$$0 \subseteq W^{F_1} \subseteq W^{F_2} \subseteq \cdots \subseteq W^{F_{s-1}} \subseteq W,$$

and let

$$W_\pi := \bigoplus_{j=1}^{s} W^{F_j}/W^{F_{j-1}}$$

be the associated graded vector space. We regard W_π as a subspace of U as follows: For each $j \in [s]$, we have the following short exact sequence:

$$0 \to W^{F_{j-1}} \to W^{F_j} \to \theta_{\pi_j}(W^{F_j}) \to 0.$$

Hence, the map $\theta_{\pi_j} : U \to U_{\pi_j}$ induces an isomorphism $W^{F_j}/W^{F_{j-1}} \simeq \theta_{\pi_j}(W^{F_j})$. We identify W_π with $\bigoplus_{j=1}^{s} \theta_{\pi_j}(W^{F_j})$, which is a subspace of U under the natural identification $U = \bigoplus_{j=1}^{s} U_{\pi_j}$. It follows from [2][Prop. 5.2] that $\mu_{W,\pi} = \mu_{W_\pi}$, i.e., the splitting of μ_W with respect to the ordered partition π is equal to μ_{W_π}. The vector spaces W_π are called *splittings of W associated to π*.

2.2.2 Polyhedral tilings.

For a polytope $\mathbf{P}$ in $\mathbb{R}^H_n$, the *relative interior* of $\mathbf{P}$, denoted by $\mathring{\mathbf{P}}$, is the set of all points $q \in \mathbf{P}$ that do not belong to any proper face of $\mathbf{P}$.

Given a vector space $W \subseteq U$ of dimension n, the associated supermodular function μ_W is simple if and only if $\mathring{\mathbf{P}}_W$ is an open set in $\mathbb{R}^H_n$. In this case, we say that W is *simple*.

Let W_1 and W_2 be two vector subspaces of U of the same dimension. We say that W_1 and W_2 are *equivalent* if there exists $\varphi \in (\mathbf{k}^*)^H$ such that $\varphi W_1 \subseteq W_2$. In this case, one has $\mathbf{P}_{W_1} = \mathbf{P}_{W_2}$. We also say that $\mathbf{P}_{W_1}$ and $\mathbf{P}_{W_2}$ *intersect in codimension at most* 1 if there are faces $\mathbf{F}_1$ of $\mathbf{P}_{W_1}$ and $\mathbf{F}_2$ of $\mathbf{P}_{W_2}$ of codimension at most 1 such that $\mathring{\mathbf{F}}_1 \cap \mathring{\mathbf{F}}_2 \neq \emptyset$.

The following propositions are simplified versions of results by Amini and Esteves in a forthcoming paper. For the sake of completeness, their proofs will be sketched.

Proposition 2.2.1. *Let W_1 and W_2 be two vector subspaces of U of the same dimension. Let $\varphi \in \mathbf{k}^H$ be a nonzero endomorphism such that $\varphi W_1 \subseteq W_2$. Assume that W_1 and W_2 are simple. Then either φ is an isomorphism or the relative interiors $\mathring{\mathbf{P}}_{W_1}$ and $\mathring{\mathbf{P}}_{W_2}$ are disjoint.*

Proof. Set $I := \{v \in H \,|\, \varphi_v \neq 0\}$. Since $\varphi \neq 0$, we have $I \neq \emptyset$. If $I = H$, then φ is an isomorphism. Assume $I \neq H$. Then $\pi = (I, I^c)$ is a bipartition. The morphism φ factors through an injective morphism $c_\varphi \colon W_{1,I} \to W_2^I$, whence the inequality $\dim W_{1,I} \leq \dim W_2^I$, which implies that π is a separation of μ_{W_2} and μ_{W_1}. Since W_1 and W_2 are simple, the separation is nontrivial. Thus, by [1][Prop. 3.4], the relative interiors $\mathring{\mathbf{P}}_{W_1}$ and $\mathring{\mathbf{P}}_{W_2}$ are disjoint. □

Let $\mathcal{W}$ be a finite collection of simple vector subspaces of U of dimension n. We say that $\mathcal{W}$ is *complete* for a subset $\Omega \subseteq \mathbb{R}^H$ if for each $W \in \mathcal{W}$ and each bipartition π of H such that $\mathbf{P}_{W_\pi}$ intersects Ω, there exists $W' \in \mathcal{W}$ such that $\mu_{W'_{\pi^c}} = \mu_{W_\pi}$. Since W is simple, this implies $\mu_{W'} \neq \mu_W$. Thus, $\mathcal{W}$ is complete for Ω if the collection of associated supermodular functions is complete for Ω. Denote by $\mathbf{P}_\mathcal{W}$ the union of the polytopes $\mathbf{P}_W$ with $W \in \mathcal{W}$.

Proposition 2.2.2. *Let $\mathcal{W}$ be a finite nonempty collection of simple nonequivalent vector subspaces of U of dimension n. Assume:*

(1) Each facet of a polytope in $\mathcal{W}$ intersecting $\mathring{\Omega}_n$ is shared with another polytope of $\mathcal{W}$.

(2) For each $W_1, W_2 \in \mathcal{W}$, there is a nonzero $c \in \mathbf{k}^H$ such that $cW_1 \subseteq W_2$.

Then the collection of polytopes $\mathbf{P}_{W_\pi}$ for $W \in \mathcal{W}$ and π ordered partition of H is a tiling of Ω_n.

Proof. = Since $\mathcal{W}$ is finite, it follows from (1) and [1][Lemma. 3.7] that $\mathring{\Omega}_n \subseteq \mathbf{P}_\mathcal{W}$. Since $\mathbf{P}_W \subseteq \Omega_{r+1}$ for each $W \in \mathcal{W}$, we have $\Omega_n = \mathbf{P}_\mathcal{W}$. We need only show that for each two $W, W' \in \mathcal{W}$, the polytopes $\mathbf{P}_W$ and $\mathbf{P}_{W'}$ are disjoint or intersect in a common face. Assume $\mathbf{P}_W \cap \mathbf{P}_{W'} \neq \emptyset$. We will show that $\mathbf{P}_W \cap \mathbf{P}_{W'}$ is a face of $\mathbf{P}_W$; the argument for $\mathbf{P}_{W'}$ is analogous. Let $q \in \mathbf{P}_W \cap \mathbf{P}_{W'}$. Let $\mathbf{F}$ be the face of $\mathbf{P}_W$ that contains q in its relative interior. Since $\mathbf{P}_W \cap \mathbf{P}_{W'}$ is convex and q is arbitrary, it suffices to show that $\mathbf{F} \subseteq \mathbf{P}_{W'}$. Indeed, it would then follow that $\mathbf{P}_W \cap \mathbf{P}_{W'}$ is a union of faces, so, by convexity, a single face of $\mathbf{P}_W$. It follows from (2) and Proposition 2.2.1 that either W and W' are equivalent or the relative interiors $\mathring{\mathbf{P}}_W$ and $\mathring{\mathbf{P}}_{W'}$ are disjoint. Since the W in the collection $\mathcal{W}$ are pairwise nonequivalent, we have $\mathring{\mathbf{P}}_W \cap \mathring{\mathbf{P}}_{W'} = \emptyset$. We claim the existence of a sequence $W_0, \dots, W_m \in \mathcal{W}$ such that $\mu_{W_0} = \mu_W$ and $W_m = W'$, and, for each $j = 1, \dots, m$,

$$\mathbf{P}_{W_{j-1}} \cap \mathbf{P}_{W_j} \text{ is a common facet of } \mathbf{P}_{W_{j-1}} \text{ and } \mathbf{P}_{W_j} \text{ containing } \mathbf{F}. \tag{2.2.1}$$

If such sequence exists, then the proposition follows, as $\mathbf{F} \subseteq \mathbf{P}_{W'}$.

Put $W_0 := W$. Assume that we are given a sequence $W_1, \ldots, W_i \in \mathcal{W}$ for some $i \geq 0$ such that 2.2.1 holds for each $j = 1, \ldots, i$. Notice that $\mathbf{F}$ is a face of $\mathbf{P}_{W_j}$ for each j. If $W' = W_i$, the claim is proved. Suppose not. Then, since W_i and W' are nonequivalent, it follows from (2) and Proposition 2.2.1 that $\mathring{\mathbf{P}}_{W_i} \cap \mathring{\mathbf{P}}_{W'} = \emptyset$. Hence, there exists a facet $\mathbf{F}'$ of $\mathbf{P}_{W_i}$ containing $\mathbf{F}$ such that $\mathring{\mathbf{P}}_{W_i}$ and $\mathring{\mathbf{P}}_{W'}$ lie on opposite sides of the hyperplane H spanned by $\mathbf{F}'$. Since $\mathbf{F}'$ is a facet, $\mathbf{F}' = \mathbf{P}_{W_{i,\pi}}$ for a certain bipartition π of H.

We claim that $\mathbf{F}' \cap \mathring{\Omega}_n \neq \emptyset$. Assume by contradiction that $\mathbf{F}' \cap \mathring{\Omega}_n = \emptyset$. Then the hyperplane H is cut out by an equation of the form $q(v) = 0$ for a certain $v \in H$. Since $\mathring{\mathbf{P}}_{W_i}$ and $\mathring{\mathbf{P}}_{W'}$ are contained in $\mathring{\Omega}_n$, neither of them lies on the side $q(v) < 0$ of H. Hence, $\mathbf{F}' \cap \mathring{\Omega}_n \neq \emptyset$.

It follows from (1) that, there is $W_{i+1} \in \mathcal{W}$ distinct from W_i such that $\mathbf{P}_{W_{i+1,\pi^c}} = \mathbf{F}'$. As W_i and W_{i+1} are simple and nonequivalent, $\mathring{\mathbf{P}}_{W_{i+1}}$ and $\mathring{\mathbf{P}}_{W_i}$ lie on opposite sides of the hyperplane H, thus $\mathbf{P}_{W_{i+1}}$ lies on the same side as $\mathbf{P}_{W'}$. Since the collection $\mathcal{W}$ is finite, and we "approach" $\mathbf{P}_{W'}$ at each step, there is m such that $\mu_{W_m} = \mu_{W'}$. $\square$

Chapter 3

$\mathbb{Z}^n$-quivers and linked nets.

Here, we review and prove new results on $\mathbb{Z}^n$-quivers and linked nets. We refer to [10] and [11], which contain the necessary background.

3.1 $\mathbb{Z}^n$-quivers.

Let Q be a quiver. Let T be a nontrivial partition of the arrow set of Q. Each part $\mathfrak{a}$ is called an *arrow type*, and we say $a \in \mathfrak{a}$ has or is of type $\mathfrak{a}$.

Given a path γ in Q and an arrow type $\mathfrak{a}$, we denote by $\mathbf{t}_\gamma(\mathfrak{a})$ the number of arrows of that type that the path contains. We call $\mathbf{t}_\gamma$ the *type* of γ, and the collection of arrow types $\{\mathfrak{a}|\mathbf{t}_\gamma(\mathfrak{a}) > 0\}$ its *essential type*, denoted by $\mathrm{Ess}(\gamma)$. The path γ is called *admissible* if $\mathbf{t}_\gamma(\mathfrak{a}) = 0$ for some $\mathfrak{a}$ and *simple* if $\mathbf{t}_\gamma(\mathfrak{a}) \leq 1$ for every arrow type $\mathfrak{a}$.

A quiver Q with a nontrivial partition T of its set of arrows in $n+1$ parts is a $\mathbb{Z}^n$-*quiver* if the following three conditions are satisfied:

(Q1) There is exactly one arrow of each type leaving each vertex.

(Q2) Each vertex is connected to each other by an admissible path.

(Q3) Each maximal simple path is a circuit.

(Q4) Two admissible paths leaving the same vertex arrive at the same vertex if and only if they are of the same type.

Remark 3.1.1. These quivers are named after their realizations as lattices. For instance, letting $e_1, \dots, e_n$ be the canonical basis of $\mathbb{Z}^n$, and $e_0 := -\sum e_j$, the quiver Q^n whose vertex set is $\mathbb{Z}^n$, and arrow set is the disjoint union of

$$A_j := \{(u, u - e_j) \,|\, u \in \mathbb{Z}^n\} \subseteq \mathbb{Z}^n \times \mathbb{Z}^n,$$

for $j = 0, \dots, n$, is a $\mathbb{Z}^n$-quiver, with partition T given by the A_j.

Although $\mathbb{Z}^n$-quivers may be presented in various ways, any two $\mathbb{Z}^n$-quivers are isomorphic; see [10][Prop. 2.5]. We draw in Figure 3.1 and Figure 3.2 the $\mathbb{Z}^1$-quiver and the $\mathbb{Z}^2$-quiver, up to isomorphism.

A simple nonadmissible path is thus a circuit, called a *minimal circuit*. Two distinct vertices connected by a simple path are called *neighbors*. If v_1 and v_2 are neighbors and I is the essential type of a simple path connecting v_1 to v_2 we write $v_2 = I \cdot v_1$.

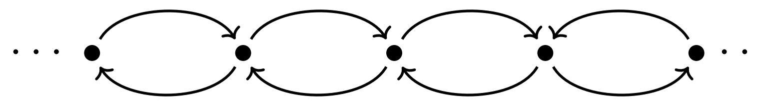

Figure 3.1: $\mathbb{Z}^1$-quiver.

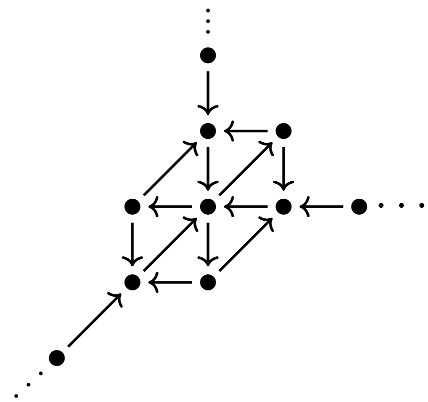

Figure 3.2: $\mathbb{Z}^2$-quiver.

Recall that the *hull* of a collection of vertices H of a $\mathbb{Z}^n$-quiver Q is the set $P(H)$ of all vertices v of Q such that for each arrow type there are $z \in H$ and a path γ connecting z to v not containing any arrow of that type.

For each vertex v in Q there is $w_v \in P(H)$ such that for each $z \in H$ there is an admissible path connecting z to v through w_v. Furthermore, w_v is unique if $P(H) = H$. We call w_v the *shadow* of v in H. See [10][Prop. 5.7] for further details.

Remark 3.1.2. If $Q = Q^n$, the hull of H is the set of integer tuples in the tropical convex hull of H by [10][Prop. 5.8, p. 537].

Lemma 3.1.3. *Let H and H' be sets of vertices of a $\mathbb{Z}^n$-quiver Q. Assume that $P(H) = H$ and $P(H') = H'$. Then $P(H \cap H') = H \cap H'$.*

Proof. Since $H \cap H' \subseteq P(H \cap H')$, we need only to show that $P(H \cap H') \subseteq H \cap H'$. Let $u \in P(H \cap H')$. For each arrow type there are $z \in H \cap H'$ and a path γ connecting z to v containing no arrow of that type. Hence, $u \in P(H)$ and $u \in P(H')$. As $P(H) = H$ and $P(H') = H'$, it follows that $u \in H \cap H'$, as required. $\square$

Definition 3.1.4. [10][Def. 5.3] Let Q be a $\mathbb{Z}^n$-quiver. For each vertex v of Q and each set I of arrow types, the *I-cone of v*, denoted $C_I(v)$, is the set of end vertices of all paths leaving v having essential type contained in I.

Lemma 3.1.5. *Let Q be a $\mathbb{Z}^n$-quiver. For each vertex v of Q and each set I of arrow types, $P(C_I(v)) = C_I(v)$.*

Proof. For each $\mathfrak{a} \in T$, set $I_{\mathfrak{a}} := T - \{\mathfrak{a}\}$. We claim that

$$C_I(v) = \bigcap_{\mathfrak{a} \in T - I} C_{I_{\mathfrak{a}}}(v).$$

Indeed, $u \in C_I(v)$ if and only if there exists an admissible path connecting v to u avoiding any arrow type in $T - I$, which implies that $u \in C_{I_{\mathfrak{a}}}(v)$ for each $\mathfrak{a} \in T - I$. As for the converse, if $u \in \bigcap_{\mathfrak{a} \in T-I} C_{I_\mathfrak{a}}(v)$, then for each $\mathfrak{a} \in T - I$ there is an admissible path $\gamma_\mathfrak{a}$ connecting v to u avoiding arrows of type $\mathfrak{a}$. By (Q4), it follows that there is an admissible path connecting v to u avoiding any arrow type in $T - I$.

Now, by Lemma 3.1.3, we need only show that $P(C_{I_\mathfrak{a}}(v)) = C_{I_\mathfrak{a}}(v)$. Let $u \in P(C_{I_\mathfrak{a}}(v))$. There are $w \in C_{I_\mathfrak{a}}(v)$ and a path μ connecting w to u avoiding $\mathfrak{a}$. Let γ be an admissible path connecting v to w. Then $\mu\gamma$ is a path connecting v to u avoiding $\mathfrak{a}$. Thus, $u \in C_{I_\mathfrak{a}}(v)$, as required. □

3.2 Extreme vertices.

Lemma 3.2.1. *Given distinct vertices u, v of Q, they are neighbors if and only if $P(\{u, v\}) = \{u, v\}$.*

Proof. There is a neighbor w of v with $v = I \cdot w$, where I is the essential type of an admissible path connecting u to v. We claim that $w \in P(\{u, v\}) - \{v\}$. Indeed, the essential type J of an admissible path connecting u to w is contained in I. Then, for each arrow type $\mathfrak{a}$, we have $\mathfrak{a} \notin I^c$ or $\mathfrak{a} \notin J$ and, thus, by definition $w \in P(\{u, v\}) - \{v\}$.

If u and v are not neighbors, also $w \neq u$, whence $P(\{u, v\}) \neq \{u, v\}$. As for the converse, by definition, for each $z \in P(\{u, v\})$ there are admissible paths γ_1 and γ_2 connecting u and v to z, respectively, with $\mathrm{Ess}(\gamma_1) \cap \mathrm{Ess}(\gamma_2) = \emptyset$. If $z \neq v$, an admissible path γ_2' from w to v has essential type containing $T - \mathrm{Ess}(\gamma_2)$. Since $\mathrm{Ess}(\gamma_2') \supseteq \mathrm{Ess}(\gamma_1)$, the concatenation $\gamma_2'\gamma_1$ is an admissible path connecting u to v. If u and v are neighbors, we must have $\mathrm{Ess}(\gamma_1) = \emptyset$, whence $z = u$. Thus $P(\{u, v\}) = \{u, v\}$.

□

Lemma 3.2.2. *Let $\mathfrak{a}$ be an arrow type of a $\mathbb{Z}^n$-quiver Q, and H a finite nonempty collection of vertices. Put $I_\mathfrak{a} := T - \{\mathfrak{a}\}$. The following statements hold:*

(1) There exists a vertex $u_\mathfrak{a} \in H$ such that $C_{I_\mathfrak{a}}(u_\mathfrak{a}) \cap H = \{u_\mathfrak{a}\}$.

(2) If $P(H) = H$, then $u_\mathfrak{a}$ is unique. Furthermore, every admissible path connecting each $u \in H$ to $u_\mathfrak{a}$ avoids $\mathfrak{a}$.

Proof. Set $I := I_\mathfrak{a}$. The set H is partially ordered by the following relation: $v \leq w$ if and only if $v \in C_I(w)$. This relation is clearly reflexive and transitive. Also, a vertex $u \in H$ satisfies $C_I(u) \cap H = \{u\}$ if and only if u is a minimal element for the order $\leq$. Since H is finite, minimal elements exist, and thus (1) holds, if $\leq$ is antisymmetric.

We prove now that $\leq$ is antisymmetric. Let $v, w \in H$ with $v \leq w$ and $w \leq v$. Then there are admissible paths γ and γ' connecting w to v and v to w, respectively, both avoiding $\mathfrak{a}$. As the concatenation $\gamma'\gamma$ connects w to itself, its type is constant. But $\gamma'\gamma$ avoids $\mathfrak{a}$. Thus γ and γ' are empty, that is, $v = w$.

As for (2), let $u, v \in H$ be minimal for $\leq$. By definition of hull, each vertex in $P(\{u, v\})$ is the end of admissible paths γ and γ' leaving u and v, respectively, with $\mathrm{Ess}(\gamma) \cap \mathrm{Ess}(\gamma') = \emptyset$. Thus $P(\{u, v\}) \subseteq C_I(u) \cup C_I(v)$. Assume $P(H) = H$. Then

$$P(\{u, v\}) = P(\{u, v\}) \cap H \subseteq (C_I(u) \cup C_I(v)) \cap H = \{u, v\}.$$

Also, since $C_I(u) \cap H = \{u\}$ and $C_I(v) \cap H = \{v\}$, we have that u and v are not neighbors. Then Lemma 3.2.1 yields $u = v$.

Let thus $u_\mathfrak{a}$ be the unique minimal element of H for $\leq$. Given any $u \in H$, since H is finite, $u_\mathfrak{a} \leq u$, or equivalently, there is an admissible path connecting u to $u_\mathfrak{a}$ avoiding $\mathfrak{a}$. □

Definition 3.2.3. Let Q be a $\mathbb{Z}^n$-quiver and H a finite nonempty collection of vertices of Q such that $P(H) = H$. For each arrow type $\mathfrak{a} \in T$, the vertex $u_{\mathfrak{a}} \in H$ is called the *extreme vertex* of H with respect to $\mathfrak{a}$. Denote by $\mathfrak{e}_H : T \to H$ the function taking an arrow type $\mathfrak{a}$ to the corresponding vertex $u_{\mathfrak{a}} \in H$.

3.3 Divisors on graphs

In this section, we will describe the relation between extreme points, as defined in Section 2.2, and v-reduced divisors.

Let G be a finite graph without loops. We denote by V and E, respectively, the sets of vertices and of edges of G. We write $e = wv$ for an edge that connects vertices v and w. Let $\mathrm{Div}(G)$ be the free Abelian group on the set of vertices of G. We write an element of $\mathrm{Div}(G)$ as

$$D = \sum_{v \in V(G)} D(v)v,$$

and we call it a *divisor*. There is a natural order on the group $\mathrm{Div}(G)$: $D \geq D'$ if and only if $D(v) \geq D'(v)$ for all $v \in V$. A divisor $D \in \mathrm{Div}(G)$ is called *effective* if $D \geq 0$.

The degree function $\deg : \mathrm{Div}(G) \to \mathbb{Z}$ is defined by $\deg(D) = \sum D(v)$. We define the subgroup $\mathrm{Div}^0(G) := \mathrm{Ker}(\deg)$ of $\mathrm{Div}(G)$ consisting of degree-zero divisors. More generally, for each $k \in \mathbb{Z}$, we define $\mathrm{Div}^k(G)$ as the set of degree-k divisors and $\mathrm{Div}^k_+(G)$ as the set of effective degree-k divisors.

The Laplacian operator Δ on $\mathrm{Div}^0(G)$ is given by the formula

$$\Delta(f) := \sum_{v \in V(G)} \Delta_v(f)\, v,$$

where

$$\Delta_v(f) := \deg(v) f(v) - \sum_{e=wv} f(w),$$

with $\deg(v)$ denoting the number of edges incident to the given vertex v.

Divisors in the image of Δ are called *principal divisors*. They form a subgroup $\mathrm{Prin}(G)$ of $\mathrm{Div}(G)$.

We define an equivalence relation on $\mathrm{Div}(G)$ by declaring that $D \sim D'$ if and only if $D - D' \in \mathrm{Prin}(G)$. Given $D \in \mathrm{Div}(G)$, we define the *linear system* associated to D to be the set

$$|D| = \{E \in \mathrm{Div}(G) \mid E \geq 0,\ E \sim D\}.$$

Two divisors D and D' such that $D \sim D'$ are called *linearly equivalent*.

We view D as a tally of how many chips $D(v)$ each vertex v of G has. A vertex v of G such that $D(v) < 0$ is said to be in *debt* with respect to D. A *lending move*, or simply a *move*, consists of a vertex v giving as many chips to each $w \in V - \{v\}$ as the number $e(v, w)$ of edges between v and w. More precisely, we say that a vertex v moves D to $D^v \in \mathcal{D}iv(G)$, where $D^v(w) := D(w) + e(v, w)$ for $w \in V$ distinct from v, and $D^v(v) := D(v) - \sum e(v, w)$. More generally, a sequence of vertices $v_1, \ldots, v_m$ of V *moves* D to D' if $D' = (\ldots((D^{v_1})^{v_2})^{v_3} \ldots)^{v_m}$. The divisor D' does not depend on the order of the vertices $v_1, \ldots, v_m$. Thus for each subset $A = \{v_1, \ldots, v_m\}$ of A, we say that A *moves* D

to D' if $v_1, \dots, v_m$ moves D to D'. By [3][Lemma 4.3], two divisors D and D' are linearly equivalent if and only if there is a sequence of moves taking D to D'.

For $A \subset V$ and $v \in A$, let $\mathrm{outdeg}_A(v)$ denote the number of edges of G connecting v to a vertex in $V \setminus A$. Select a vertex $v_0 \in V$. We say that a function $f : V \setminus \{v_0\} \to \mathbb{Z}$ is a *G-parking function* relative to the base vertex v_0 (or simply *parking function*, if G and v_0 are clear from the context) if the following two conditions are satisfied:

1. $f(v) \geq 0$ for all $v \in V \setminus \{v_0\}$.

2. For every non-empty set $A \subseteq V \setminus \{v_0\}$, there exists a vertex $v \in A$ such that $f(v) < \mathrm{outdeg}_A(v)$.

We say that $D \in \mathrm{Div}(G)$ is v_0-*reduced* if the map

$$v \in V \setminus \{v_0\} \longmapsto D(v) \in \mathbb{Z}$$

is a G-parking function relative to v_0. In other words, we say that a divisor $D \in \mathrm{Div}(G)$ is v_0-reduced if and only if

1. no vertex $v \neq v_0$ is in debt;

2. for every nonempty subset A of $V - \{v_0\}$, A moves D to a divisor with respect to which some vertex A is in debt.

Let D be a divisor on G. The set Q of all divisors linearly equivalent to D is the vertex set of a $\mathbb{Z}^n$-quiver Q, where we put an arrow, which we say is of type v, from vertex D_1 to D_2 if and only if $D_2 = (D_1)^v$. By [3], for each vertex v_0 of G, there exists a unique v_0-reduced divisor $D' \in \mathrm{Div}(G)$ such that $D' \sim D$.

Proposition 3.3.1. *Let D be a divisor on G. Then $P(|D|) = |D|$. In addition, the divisor corresponding to the extreme vertex of $|D|$ with respect to v_0 is v_0-reduced.*

Proof. Let $D' \in P(|D|)$. For each vertex u of G, there exists $D'' \in |D|$ and a sequence $v_1, \dots, v_m$ moving D'' to D' such that $v_i \neq u$ for all $i = 1, \dots, m$. Then

$$D'(u) = D''(u) + \sum_{i=1}^{m} e(v_i, u) \geq 0.$$

Thus $D' \in |D|$.

As for the second statement, let D' be the divisor corresponding to the extreme vertex of $|D|$ with respect to v_0. Since $D' \in |D|$, and thus $D'(v) \geq 0$ for each vertex v of G, we need only show that every nonempty subset A of $V - \{v_0\}$ moves D' to a divisor with respect to which some vertex of A is in debt.

Let D_A be the divisor obtained from D' by the move of A. We claim that $D_A \notin |D|$. Indeed, assume by contradiction that $D_A \in |D|$. Since D' is v_0-extreme, there is an admissible path γ connecting D_A to D' avoiding v_0. Also, since $v_0 \notin A$, there is a simple admissible μ path avoiding v_0 and connecting D' to D_A. Thus, the path $\mu\gamma$ is admissible, in contradiction to (Q4). Hence, $D_A \notin |D|$ and, thus, D_A is not effective. Since D_A is obtained from D by the move by A, the values $D(u)$ are nonnegative for $u \notin A$. Thus, $D(u) < 0$ for some $u \in A$. $\square$

3.4 Bridges.

Definition 3.4.1. [11][Def. 7.6] Two distinct vertices v_1 and v_2 of a $\mathbb{Z}^n$-quiver Q are said to be *weakly neighbors* if there are a vertex v of Q and simple admissible paths γ_1 and γ_2 connecting v to v_1 and v_2, respectively, with no arrow type in common. We call v a *bridge* of v_1 and v_2.

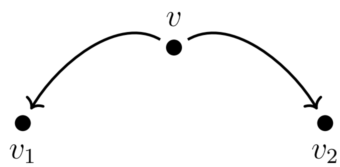

Figure 3.3: A bridge.

The following lemma follows directly from the definition.

Lemma 3.4.2. *Let Q be a $\mathbb{Z}^n$-quiver with set of arrow types T. Let H and H' be sets of vertices of Q. Assume that the set of bridges of all pairs of vertices in H (resp. H') is in H (resp. H'). Then the set of bridges of all pairs of vertices in $H \cap H'$ is in $H \cap H'$.*

Lemma 3.4.3. *Let Q be a $\mathbb{Z}^n$-quiver with set of arrows types T. Let $\mathfrak{a} \in T$ be an arrow type and v a vertex of Q. Put $I_\mathfrak{a} := T - \{\mathfrak{a}\}$. Then the set of bridges of all pairs of vertices in $C_{I_\mathfrak{a}}(v)$ is in $C_{I_\mathfrak{a}}(v)$.*

Proof. Let $v_1, v_2 \in C_{I_\mathfrak{a}}(v)$. If v_1 and v_2 are neighbors, then by [11, Prop. 7.7] their bridges are themselves. Assume that v_1 and v_2 are not neighbors. Let u be a bridge between v_1 and v_2. Let γ be an admissible path connecting v to u. If $\mathfrak{a} \notin \mathrm{Ess}(\gamma)$, then $u \in C_{I_\mathfrak{a}}(v)$. Assume, by contradiction, that $\mathfrak{a} \in \mathrm{Ess}(\gamma)$. Let γ_1 and γ_2 be admissible paths connecting v to v_1 and v_2, respectively. Since $v_i \in \mathcal{C}_{I_\mathfrak{a}}(v)$, there is an admissible path connecting v to v_i avoiding $\mathfrak{a}$, and thus, by (Q4), γ_i avoids $\mathfrak{a}$ for $i = 1, 2$. Since u is a bridge, there are admissible simple paths μ_1 and μ_2 connecting u to v_1 and v_2, respectively, such that $\mathrm{Ess}(\mu_1) \cap \mathrm{Ess}(\mu_2) = \emptyset$. Then, for $i = 1, 2$, the path $\mu_i\gamma$ is not admissible. Indeed, assume by contradiction that $\mu_i\gamma$ is admissible. Since, since $\mu_i\gamma$ and γ_i are admissible, it follows from (Q4), that $\mu_i\gamma$ and γ_i have the same type, which contradicts the assumption $\mathfrak{a} \in \mathrm{Ess}(\gamma)$. Hence $\mu_i\gamma$ is not admissible, and thus $\mathrm{Ess}(\gamma) \cup \mathrm{Ess}(\mu_i) = T$ for $i = 1, 2$. As γ is admissible, we have $\mathrm{Ess}(\mu_1) \cap \mathrm{Ess}(\mu_2) \neq \emptyset$, contradicting the assumption that u is bridge. $\square$

Lemma 3.4.4. *Let Q be a $\mathbb{Z}^n$-quiver. Let H be a nonempty finite set of vertices of Q. Let $\mathfrak{a}$ be an arrow type of Q and $I_\mathfrak{a} := T - \{\mathfrak{a}\}$. Then there exists a vertex v of Q such that $H \subset C_{I_\mathfrak{a}}(v)$.*

Proof. Let u be a vertex of Q. Let k be the maximum of the $\mathfrak{t}_\gamma(\mathfrak{a})$ for admissible paths connecting u to a $w \in H$. Now, let $v := \mathfrak{a}^k \cdot u$. We claim that $H \subseteq C_{I_\mathfrak{a}}(w)$. Assume, by contradiction, that there is $w \in H - C_{I_\mathfrak{a}}(v)$. Then there is an admissible path connecting v to w containing $\mathfrak{a}$. It follows from the definition of v that there is an admissible path connecting u to w passing through v having at least $k+1$ arrows of type $\mathfrak{a}$. This contradicts the maximality k.

$\square$

Proposition 3.4.5. *Let H be a set of vertices of a $\mathbb{Z}^n$-quiver Q such that $P(H) = H$. Then there is a finite set H' such that the following holds:*

1. *the set of bridges of all pairs of vertices in H' is in H',*

2. *$H \subseteq H'$,*

3. *$P(H') = H'$.*

Proof. Assume that H is nonempty. By Lemma 3.4.4, for each arrow type $\mathfrak{a}$ of Q there is a vertex $v_{\mathfrak{a}}$ of Q such that $H \subseteq C_{I_{\mathfrak{a}}}(v_{\mathfrak{a}})$. Put

$$H' := \bigcap_{\mathfrak{a} \in T} C_{I_{\mathfrak{a}}}(v_{\mathfrak{a}}).$$

The first statement follows from Lemma 3.4.2 and Lemma 3.4.3. The second statement is clear. The third statement follows from Lemma 3.1.3 and Lemma 3.1.5. □

3.5 Linked nets

Let $\mathbf{k}$ be a field and call its elements *scalars.* For each representation $\mathfrak{V}$ of a quiver Q in the category of finite dimensional vector spaces over $\mathbf{k}$, denote by $V_v^{\mathfrak{V}}$ the vector space associated to each vertex v of Q, and denote by $\varphi_\gamma^{\mathfrak{V}}$ the composition of morphisms associated to each path γ in Q. (If γ is the trivial path, $\varphi_\gamma^{\mathfrak{V}} = \mathrm{id}$ by convention.) We write V_v and φ_γ if $\mathfrak{V}$ is clear from the context. The representation $\mathfrak{V}$ is called *pure* if the V_v have the same dimension.

Let H be a set of vertices in Q and $m \in \mathbb{N} \cup \{\infty\}$. We say $\mathfrak{V}$ is *m-generated* by H if for each vertex v of Q there exist $w_1, \dots, w_l \in H$ for $l \in \mathbb{N}$ with $l \leq m$ and paths γ_i connecting w_i to v for each i, such that

$$(\varphi_{\gamma_1}, \dots, \varphi_{\gamma_m}) : \bigoplus_{i=1}^{m} V_{w_i} \longrightarrow V_v$$

is surjective.

If Q is a $\mathbb{Z}^n$-quiver, we say $\mathfrak{V}$ is a *weakly linked net* over Q if $\mathfrak{V}$ satisfies the following two conditions:

(N1) If γ_1 and γ_2 are two paths connecting the same two vertices and γ_2 is admissible then φ_{γ_1} is a scalar multiple of φ_{γ_2};

(N2) $\varphi_\gamma = 0$ for each minimal circuit γ.

We say that $\mathfrak{V}$ is a *linked net* if in addition a third condition is verified:

(N3) If γ_1 and γ_2 are two admissible paths leaving the same vertex without arrow type in common, then $\ker(\varphi_{\gamma_1}) \cap \ker(\varphi_{\gamma_2}) = 0$.

For each vector w in a vector space, we will denote by $[w]$ the set of its nonzero scalar multiples. And for each linear map φ, we denote $\ker[\varphi] := \ker(\varphi)$ and $\mathrm{Im}[\varphi] := \mathrm{Im}(\varphi)$.

Given two vertices u_1 and u_2 of Q, let $\varphi_{u_2}^{u_1} := [\varphi_\gamma] = \mathbf{k}^* \varphi_\gamma$ for any admissible path γ connecting u_1 to u_2. We say $\mathfrak{V}$ is *exact* if $\mathrm{Im}(\varphi_{u_2}^{u_1}) = \ker(\varphi_{u_1}^{u_2})$ for each two neighbors u_1 and u_2.

Remark 3.5.1. Let $\mathfrak{V}$ be an exact linked net of vector spaces over a $\mathbb{Z}^n$-quiver Q. Then, exactness, together with the rank-nullity theorem, implies that $\mathfrak{V}$ is pure.

The following lemma extends [11][Prop. 10.7].

Lemma 3.5.2. *Let $\mathfrak{V}$ be a nontrivial finitely generated exact linked net of vector spaces over* $\mathbf{k}$ *over a $\mathbb{Z}^n$-quiver Q. Let H be the intersection of all collections of vertices 1-generating $\mathfrak{V}$. Then H is finite, and $\mathfrak{V}$ is 1-generated by H. Furthermore, for each $v, u \in H$, we have $\varphi_u^v \neq 0$. Also, $P(H) = H$.*

Proof. That H is finite follows from [10][Prop. 6.4]. That $\mathfrak{V}$ is 1-generated by H follows from [10][Prop. 6.3].

As for the second statement, let $u_{\mathfrak{a}} := \mathfrak{e}_{P(H)}(\mathfrak{a})$ for each $\mathfrak{a} \in T$, where T is the set of arrow types of Q. We claim first that $\varphi_{u_{\mathfrak{a}}}^v \neq 0$ for each $v \in H$ and $\mathfrak{a} \in T$. Indeed, let $v \in H$ and $\mathfrak{a} \in T$. Let γ be an admissible path connecting v to $u_{\mathfrak{a}}$ and J be its essential type. By [10][Lem. 6.6], it is enough to show that $\varphi_{J \cdot v}^v \neq 0$.

Let w be the shadow of $J \cdot v$ in $P(H)$. Then w is v or a neighbor of v, and $\varphi_{J \cdot v}^v = \varphi_{J \cdot v}^w \varphi_w^v$. Since $\mathfrak{V}$ is 1-generated by H, it follows from [10][Prop. 6.4] that $\mathfrak{V}$ is 1-generated by $P(H)$. Hence $\varphi_{J \cdot v}^w$ is surjective. Moreover, since $\mathfrak{V}$ is exact, $\mathfrak{V}$ is pure, and thus $\varphi_{J \cdot v}^w$ is an isomorphism. It is thus enough to show that $\varphi_w^v \neq 0$.

Clearly, $\varphi_v^v \neq 0$ because $\mathfrak{V}$ is nontrivial. We may thus assume $w \neq v$. Then v and w are neighbors. By exactness of $\mathfrak{V}$, it is now enough to show that φ_v^w is not surjective.

Let $H' := (H - \{v\}) \cup \{w\}$. Since $H \not\subseteq H'$, we have that $\mathfrak{V}$ is not 1-generated by H'. Thus there is a vertex u of Q such that φ_u^z is not surjective for any $z \in H'$. In particular, φ_u^w is not surjective. Since H 1-generates $\mathfrak{V}$, we must have that φ_u^v is surjective. Were φ_v^w surjective, also the composition $\varphi_u^v \varphi_v^w$ would be surjective, whence nonzero, and thus equal to φ_u^w by [10][Lem. 6.2], contradicting the fact that φ_u^w is not surjective. Our first claim is proved.

We claim second that $\varphi_{u_{\mathfrak{a}}}^v \neq 0$ for every $v \in P(H)$ and $\mathfrak{a} \in T$. Indeed, let $v \in P(H)$ and $\mathfrak{a} \in T$. Let γ' be an admissible path connecting v to $u_{\mathfrak{a}}$. It avoids $\mathfrak{a}$. Since $v \in P(H)$, there are $z \in H$ and an admissible path γ connecting z to v avoiding $\mathfrak{a}$. Then $\gamma'\gamma$ is admissible, and thus $\varphi_{u_{\mathfrak{a}}}^z = \varphi_{u_{\mathfrak{a}}}^v \varphi_v^z$. Since $\varphi_{u_{\mathfrak{a}}}^z \neq 0$ by our first claim, also $\varphi_{u_{\mathfrak{a}}}^v \neq 0$, proving our second claim.

We claim third that $\varphi_u^v \neq 0$ for each $u, v \in P(H)$. Indeed, let $u, v \in P(H)$. Clearly, $\varphi_v^v \neq 0$ because $\mathfrak{V}$ is nontrivial. We may thus assume $u \neq v$. Let γ be an admissible path connecting v to u, and let $\mathfrak{a} \in T - \mathrm{Ess}(\gamma)$. Let γ' be an admissible path connecting u to $u_{\mathfrak{a}}$. Then also $\mathfrak{a} \notin \mathrm{Ess}(\gamma')$ by Lemma 3.2.2, and thus $\gamma'\gamma$ is admissible. Then $\varphi_{u_{\mathfrak{a}}}^v = \varphi_{u_{\mathfrak{a}}}^u \varphi_u^v$. Since $\varphi_{u_{\mathfrak{a}}}^v \neq 0$ by our second claim, also $\varphi_u^v \neq 0$, proving our third claim, and thus the second statement of the lemma.

Finally, we prove that $P(H) = H$. Let $z \in P(H)$. Since H 1-generates $\mathfrak{V}$, there is $w \in H$ such that φ_z^w is an isomorphism. We will show that $w = z$. Let γ be an admissible path connecting w to z. We need to show that $\mathrm{Ess}(\gamma) = \emptyset$. Let $\mathfrak{a} \in T$. Let μ be an admissible path connecting z to $u_{\mathfrak{a}}$. Then $\varphi_\mu \neq 0$ by our second claim. Since φ_γ is an isomorphism, $\varphi_\mu \varphi_\gamma \neq 0$. Thus $\mu\gamma$ is admissible by [10][Lem. 6.1]. But then $\mathfrak{a} \notin \mathrm{Ess}(\mu\gamma)$ by Lemma 3.2.2, whence $\mathfrak{a} \notin \mathrm{Ess}(\gamma)$. As $\mathfrak{a}$ was arbitrary, the proof of the lemma is complete.

□

Definition 3.5.3. Let $\mathfrak{V}$ be an exact finitely generated linked net of vector spaces. Let H be the intersection of all collections of vertices 1-generating $\mathfrak{V}$. We call H the *minimum set of generators* of $\mathfrak{V}$.

3.6 Extending representations defined on subquivers.

Let Q be a $\mathbb{Z}^n$-quiver, and H be a set of vertices of Q. Assume that $P(H) = H$. Let $Q(H)$ be the quiver with set of vertices H, and arrows only between neighboring vertices, one in each direction. The *type* and the *essential type* of an arrow in $Q(H)$ are the type and essential type of any admissible path in Q connecting the same vertices. The *type*(resp. *essential type*) of a path in $Q(H)$ is the sum (resp. union) of the types (resp. essential types) of its arrows. A path in $Q(H)$ is said to be *admissible* if its essential type is a proper set of arrow types.

Example 3.6.1. In Figure 3.4, at the left-hand side, we draw in green a finite set of vertices H such that $P(H) = H$, and at the right-hand side the corresponding quiver $Q(H)$.

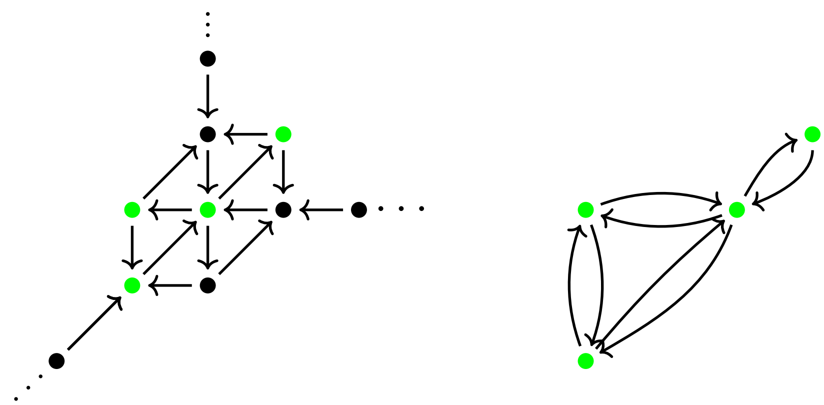

Figure 3.4: Quiver $Q(H)$ obtained from the green finite set H of vertices on a Z^2-quiver.

In what follows, we determine conditions under which a representation defined on $Q(H)$ extends to a (exact, pure) linked net over Q generated by H.

Definition 3.6.2. Let $\mathfrak{V}$ be a representation of $Q(H)$. We say $\mathfrak{V}$ is a *weakly linked* representation if:

(P1) If γ_1 and γ_2 are two paths in $Q(H)$ connecting the same two vertices and γ_2 is admissible, then φ_{γ_1} is a scalar multiple of φ_{γ_2}.

(P2) $\varphi_\gamma = 0$ for each path γ in $Q(H)$ whose corresponding path in Q is a minimal circuit for Q.

We call $\mathfrak{V}$ a *linked* representation if furthermore:

(P3) $\ker(\varphi_{\gamma_1}) \cap \ker(\varphi_{\gamma_2}) = 0$ for each two arrows γ_1 and γ_2 in $Q(H)$ leaving the same vertex $v \in H$ such that the shadow of $I \cdot v$ in H is v where $I = \mathrm{Ess}(\gamma_1) \cap \mathrm{Ess}(\gamma_2)$.

Remark 3.6.3. If H contains all bridges between vertices of H, then (P3) is equivalent to $\ker(\varphi_{\gamma_1}) \cap \ker(\varphi_{\gamma_2}) = 0$ for each two arrows γ_1 and γ_2 in $Q(H)$ leaving the same vertex v with disjoint essential types.

Definition 3.6.4. A weakly linked representation $\mathfrak{V}$ of $Q(H)$ is *exact* if for each arrow a of $Q(H)$,

$$\ker(\varphi_a) = \mathrm{Im}(\varphi_{\overline{a}}),$$

where $\overline{a}$ denotes the arrow in the opposite direction. Also, $\mathfrak{V}$ is pure if the V_v have the same dimension.

Let $\mathfrak{V}$ be a weakly linked net over a $\mathbb{Z}^n$-quiver Q, and H be a nonempty finite set of vertices such that $P(H) = H$. We define a representation $\mathfrak{V}_{|H}$ of $Q(H)$ as follows: First, for each vertex v of H, set $V_v := V_v^{\mathfrak{V}}$. Second, given an arrow ρ of $Q(H)$ connecting v to u, let φ_ρ be any map with class φ_u^v.

Different choices of maps φ_ρ produce different $\mathfrak{V}|_H$. However, they are projectively equivalent, see Definition 3.6.5. Clearly, a representation of $Q(H)$ is a weakly linked representation, a linked representation, exact or pure if and only if it is projectively equivalent to a representation with the corresponding properties. Thus, when we write $\mathfrak{V}|_H$, we will not mention explicitly which φ_ρ are chosen.

Definition 3.6.5. Let $\mathfrak{V}_1$ and $\mathfrak{V}_2$ be two representations of a quiver Q. We say that they are *projectively equivalent* if there is an isomorphism $f_v : V_v^{\mathfrak{V}_1} \to V_v^{\mathfrak{V}_2}$ for each vertex v of Q such that $[\varphi_a^{\mathfrak{V}_2}][f_v] = [f_u][\varphi_a^{\mathfrak{V}_2}]$ for each arrow a, where v and u are its tail and head, respectively.

The following lemma follows directly from the definitions.

Lemma 3.6.6. *Let $\mathfrak{V}_1$ and $\mathfrak{V}_2$ be weakly linked nets over a $\mathbb{Z}^n$-quiver Q. If $\mathfrak{V}_1$ and $\mathfrak{V}_2$ are projectively equivalent, then so are $\mathfrak{V}_1|_H$ and $\mathfrak{V}_2|_H$.*

Proposition 3.6.7. *Let $\mathfrak{V}$ be a weakly linked net over a $\mathbb{Z}^n$-quiver Q. Let H be a finite set of vertices of Q such that $P(H) = H$. Then:*

(1) $\mathfrak{V}|_H$ is weakly linked representation of $Q(H)$

(2) If $\mathfrak{V}$ is exact (resp. pure), so is $\mathfrak{V}|_H$

(3) If $\mathfrak{V}$ is a linked net, and H contains all bridges between vertices of H, then $\mathfrak{V}_{|H}$ is a linked representation

(4) If $\mathfrak{V}$ is 1-generated by H and is a linked net, then $\mathfrak{V}_{|H}$ is a linked representation.

Proof. First, let γ_1 and γ_2 be two paths in $Q(H)$ connecting the same vertices, and assume that γ_2 is admissible. These paths are given by concatenations of arrows, which corresponds to concatenations of admissible simple paths in Q. More precisely, γ_1 corresponds to $\rho = \rho_m \cdots \rho_1$ and γ_2 to $\theta = \theta_{m'} \cdots \theta_1$ in Q, both connecting the same two vertices in H. Since $\mathfrak{V}$ is a weakly linked net and θ is admissible, φ_ρ is a scalar multiple of φ_θ, and thus φ_{γ_1} is an scalar multiple of φ_{γ_2}.

Second, if γ is a path in $Q(H)$ whose corresponding path μ in Q is a minimal circuit for Q, then, since $\mathfrak{V}$ is a weakly linked net, we have that $\varphi_\mu = 0$, and thus $\varphi_\gamma = 0$, as required. Statement (1) is proved.

Statement (2) follows directly from the definitions.

We prove (3). Assume that $\mathfrak{V}$ is a linked net and H contains all bridges between vertices of H. Let γ_1 and γ_2 be admissible simple paths in Q leaving a vertex v of H and arriving at vertices v_1 and v_2 of H, respectively. Let $I := \mathrm{Ess}(\gamma_1) \cap \mathrm{Ess}(\gamma_2)$ and $u := I \cdot v$. Assume that the shadow of u in H is v. Let τ be an admissible path connecting v to u and γ_1' and γ_2' be admissible paths connecting u to v_1 and v_2, respectively. As γ_1' and γ_2' are simple, and have no arrow type in common, u is a bridge between v_1 and v_2, and thus $u \in H$. It follows that $u = v$, as the shadow is unique. We must have $I = \emptyset$. Since $\mathfrak{V}$ is a linked net, by (N3) we have

$$\ker(\varphi_{\gamma_1}) \cap \ker(\varphi_{\gamma_2}) = 0$$

as required.

We prove now (4). Assume that $\mathfrak{V}$ is 1-generated by H. Let γ_1 and γ_2 be simple admissible paths of Q leaving a vertex v of H and arriving at vertices v_1 and v_2 of H, respectively. Let $I := \mathrm{Ess}(\gamma_1) \cap \mathrm{Ess}(\gamma_2)$ and $u := I \cdot v$. Assume that the shadow of u in H is v. Let τ be an admissible path connecting v to u and γ_1' and γ_2' be admissible paths connecting u to v_1 and v_2, respectively. Since $\mathfrak{V}$ is 1-generated by H, the map φ_τ is an isomorphism. By [10][Lem. 6.6], we have

$$\ker(\varphi_{\gamma_1}) \cap \ker(\varphi_{\gamma_2}) = \ker(\varphi_{\gamma_1'}\varphi_\tau) \cap \ker(\varphi_{\gamma_2'}\varphi_\tau) = \varphi_\tau^{-1}(\ker(\varphi_{\gamma_1'}) \cap \ker(\varphi_{\gamma_2'})).$$

Since $\mathrm{Ess}(\gamma_1') \cap \mathrm{Ess}(\gamma_2') = \emptyset$ and $\mathfrak{V}$ is a linked net, it follows from (N3) that $\ker(\varphi_{\gamma_1'}) \cap \ker(\varphi_{\gamma_2'}) = 0$, as required. □

Let H be set of vertices of Q such that $P(H) = H$. Let $\mathfrak{V}$ be a weakly linked representation of $Q(H)$. We define a representation $\mathfrak{V}^E$ on Q as follows. First, for each vertex v of Q, set $V_v^{\mathfrak{V}^E} := V_{w_v}$, where w_v is the shadow of v in H. Second, given an arrow a of Q, let v_1 and v_2 be its tail and head, and w_1 and w_2 their shadows in H, respectively. If there is no admissible path from w_1 to v_2 through v_1, put $\varphi_a^{\mathfrak{V}^E} = 0$; otherwise, let $\varphi_a^{\mathfrak{V}^E}$ be any map with class $[\varphi_\rho^{\mathfrak{V}}]$, where ρ is an admissible path in $Q(H)$ from w_1 to w_2. Different choices of ρ produce different $\mathfrak{V}^E$. However, all of them are projectively equivalent. Also, observe that $\mathfrak{V} = (\mathfrak{V}^E)|_H$, and that $\mathfrak{V}^E$ is 1-generated by H.

Lemma 3.6.8. *Let $\mathfrak{V}$ be a pure finitely generated weakly linked net, and let H be a set of generators such that $P(H) = H$. Then $(\mathfrak{V}_{|_H})^E$ and $\mathfrak{V}$ are projectively equivalent.*

Proof. Put $\mathfrak{W} := (\mathfrak{V}_{|_H})^E$. We define a morphism of representations $f : \mathfrak{W} \to \mathfrak{V}$ as follows. For each vertex v of Q, we have $V_v^{\mathfrak{W}} = V_{w_v}^{\mathfrak{V}}$ where w_v is the shadow of v in H. Let γ be an admissible path connecting w_v to v. Define $f_v := \varphi_\gamma : V_v^{\mathfrak{W}} \to V_v^{\mathfrak{V}}$. By [10][Prop. 6.4], the map f_v is an isomorphism. We need only show that the f_v satisfy the relation $[\varphi_a^{\mathfrak{V}}][f_v] = [f_u][\varphi_a^{\mathfrak{W}}]$ for each arrow a of Q, with tail and head v and u, respectively. Let w_v and w_u be the shadows of v and u in H, respectively. If there is no admissible path connecting w_v to u passing through v, then $\varphi_a^{\mathfrak{W}} = 0$; on the other hand, by [10][Lem 6.1], we have $[\varphi_a^{\mathfrak{V}}][f_v] = \varphi_u^v\varphi_v^{w_v} = 0$. If there is an admissible path connecting w_v to u passing through v, then $[\varphi_a^{\mathfrak{W}}] = \varphi_{w_u}^{w_v}$. By [10][Lem 6.2],

$$[f_u][\varphi_a^{\mathfrak{W}}] = \varphi_u^{w_u}\varphi_{w_u}^{w_v} = \varphi_u^{w_v} \qquad \text{and} \qquad [\varphi_a^{\mathfrak{V}}][f_v] = \varphi_u^v\varphi_v^{w_v} = \varphi_u^{w_v},$$

as required □

The following lemma appears in [11], and will be used in the proof of Lemma 3.6.10.

Lemma 3.6.9. [11][Lem. 7.4] *Let H be a nonempty set of vertices of a $\mathbb{Z}^n$-quiver Q such that $P(H) = H$. Let $\mu := \alpha_m \cdots \alpha_1$ for paths α_i of Q. For $i = 1, \dots, m$ let v_i be the initial vertex of α_i and v_{m+1} be the final vertex of α_m. For $i = 1, \dots, m+1$, let w_i be the shadow of v_i in H and γ_i an admissible path connecting w_i to v_i. For each $i = 1, \dots, m$ let ρ_i be an admissible path connecting w_i to w_{i+1}. Put $\rho := \rho_m \cdots \rho_1$. Then the length of $\mu\gamma_1$ is at least that of $\gamma_{m+1}\rho$, with equality if and only if $\alpha_i\gamma_i$ is admissible for every i. In particular, if $\mu\gamma_1$ is admissible then equality holds and ρ is admissible. Conversely, if ρ is admissible, and equality holds, then $\mu\gamma_1$ is admissible.*

We draw the configuration of Lemma 3.6.9 in Figure 3.5.

$$\begin{tikzcd}
v_1 \arrow[r, "\alpha_1"] & v_2 \arrow[r, "\alpha_2"] & v_3 & \cdots & v_m \arrow[r, "\alpha_m"] & v_{m+1} \\
w_1 \arrow[u, "\gamma_1"] \arrow[r, "\rho_1"'] & w_2 \arrow[u, "\gamma_2"] \arrow[r, "\rho_2"'] & w_3 \arrow[u, "\gamma_3"] & \cdots & w_m \arrow[u, "\gamma_m"] \arrow[r, "\rho_m"'] & w_{m+1} \arrow[u, "\gamma_{m+1}"]
\end{tikzcd}$$

Figure 3.5

Lemma 3.6.10. *Let $\mathfrak{V}$ be a weakly linked representation of $Q(H)$. Then:*

1. *$\mathfrak{V}^E$ is a weakly linked net over Q generated by H.*
2. *If $\mathfrak{V}$ is exact (pure), then $\mathfrak{V}^E$ is exact (pure).*
3. *If $\mathfrak{V}$ is a linked representation, then $\mathfrak{V}^E$ is a linked net.*

Proof. Set $\mathfrak{W} := \mathfrak{V}^E$.

We first show that $\mathfrak{W}$ is a weakly linked net. Let μ_1 and μ_2 be paths in Q connecting the same two vertices u_1 and u_2. Assume that μ_2 is admissible. We will show that $\varphi^{\mathfrak{W}}_{\mu_1}$ is a scalar multiple of $\varphi^{\mathfrak{W}}_{\mu_2}$. If $\varphi^{\mathfrak{W}}_{\mu_1} = 0$, we are done. Assume $\varphi^{\mathfrak{W}}_{\mu_1} \neq 0$.

Let z_1 and z_2 be the shadows of u_1 and u_2 in H, respectively. Write $\mu_1 = \alpha_m \cdots \alpha_1$ for arrows α_i of Q. For $i = 1, \ldots, m$, let v_i be the tail of α_i, and w_i be its shadow in H. Let v_{m+1} be the head of α_m, and w_{m+1} be its shadow in H. Notice that $v_1 = u_1$ and $v_{m+1} = u_2$, and hence $w_1 = z_1$ and $w_{m+1} = z_2$. For $i = 1, \ldots, m+1$, let γ_i be an admissible path connecting w_i to v_i; see Figure 3.5. Since $\varphi^{\mathfrak{W}}_{\mu_1} \neq 0$ and $\varphi^{\mathfrak{W}}_{\mu_1} = \varphi^{\mathfrak{W}}_{\alpha_m} \cdots \varphi^{\mathfrak{W}}_{\alpha_1}$, we have that $\varphi^{\mathfrak{W}}_{\alpha_i} \neq 0$ for each $i = 1, \ldots, m$. Thus, by definition of $\mathfrak{W}$, the concatenation $\alpha_i \gamma_i$ is admissible for each $i = 1, \ldots, m$, and

$$\varphi^{\mathfrak{W}}_{\mu_1} = \varphi^{\mathfrak{W}}_{\alpha_m} \cdots \varphi^{\mathfrak{W}}_{\alpha_1} = \varphi^{\mathfrak{V}}_{\rho_m} \cdots \varphi^{\mathfrak{V}}_{\rho_1} = \varphi^{\mathfrak{V}}_{\rho},$$

where ρ_i is an admissible path connecting w_i to w_{i+1} for $i = 1, \ldots, m$, and $\rho := \rho_m \cdots \rho_1$. (Notice, for later use, that the ρ_i are also simple by [11, Lem. 7.3], and are thus arrows in $Q(H)$, if nontrivial.) Since $\varphi^{\mathfrak{V}}_{\rho} \neq 0$, the path ρ is admissible. Thus $[\varphi^{\mathfrak{V}}_{\rho}] = \varphi^{\mathfrak{V}, w_1}_{w_{m+1}} = \varphi^{\mathfrak{V}, z_1}_{z_2}$.

Now, since $\alpha_i \gamma_i$ is admissible for every i, the length of $\mu_1 \gamma_1$ is equal to that of $\gamma_{m+1} \rho$ by Lemma 3.6.9. Furthermore, as ρ is admissible, Lemma 3.6.9 implies that $\mu_1 \gamma_1$ is admissible, and thus so is μ_1. Since μ_2 is admissible, μ_1 and μ_2 have the same type, and thus also $\mu_2 \gamma_1$ is admissible. By Lemma 3.6.9 again, the analogous reasoning for μ_2 instead of μ_1 holds, without the need of having $\varphi^{\mathfrak{W}}_{\mu_2} \neq 0$. The upshot is that also $[\varphi^{\mathfrak{W}}_{\mu_2}] = \varphi^{\mathfrak{V}, z_1}_{z_2}$. Thus, $\varphi^{\mathfrak{W}}_{\mu_1}$ is a scalar multiple of $\varphi^{\mathfrak{W}}_{\mu_2}$.

Let now μ be a minimal circuit of Q. Let u be its inital and final points, z its shadow in H, and γ an admissible path in Q connecting z to u. Suppose by contradiction that $\varphi^{\mathfrak{W}}_{\mu} \neq 0$. We proceed as above, as if $\mu_1 = \mu$. More precisely, we get that

$$\varphi^{\mathfrak{W}}_{\mu} = \varphi^{\mathfrak{V}}_{\rho_n} \cdots \varphi^{\mathfrak{V}}_{\rho_0} = \varphi^{\mathfrak{V}}_{\rho}, \tag{3.6.1}$$

where the ρ_i are admissible simple paths connecting vertices of H, thus arrows in $Q(H)$ if nontrivial, and $\rho := \rho_n \ldots \rho_0$ is a circuit in Q with z as initial and final points. Furthermore, we may apply Lemma 3.6.9 to conclude that the length of $\mu\gamma$ is equal to the length of $\gamma\rho$, and thus that also ρ is a minimal circuit of Q. Since $\mathfrak{V}$ is weakly linked, we must have $\varphi^{\mathfrak{V}}_{\rho} = 0$, and thus $\varphi^{\mathfrak{W}}_{\mu} = 0$ by (3.6.1), a contradiction. Thus $\mathfrak{W}$ is a weakly linked net.

Finally, for each vertex v of Q, we have that $\varphi_{\gamma}^{\mathfrak{W}}$ is an isomorphism (the identity) for each admissible path connecting the shadow of v in H to v. Thus $\mathfrak{W}$ is 1-generated by H. The proof of Statement 1 is complete.

As for Statement 2, if $\mathfrak{V}$ is pure, then $\mathfrak{W}$ is pure, as the spaces associated to the vertices are the ones of $\mathfrak{V}$. Assume $\mathfrak{V}$ is exact. Let v_1 and v_2 be neighboring vertices of Q. Let μ_1 be an admissible simple path connecting v_1 to v_2 and μ_2 a reverse path. For each $i = 1, 2$, let w_i be the shadow of v_i and γ_i an admissible path connecting w_i to v_i; see Figure 3.6.

$$\begin{tikzcd} v_1 \arrow[r, bend left, "\mu_1"] & v_2 \arrow[l, bend left, "\mu_2"] \\ w_1 \arrow[u, "\gamma_1"] \arrow[r] & w_2 \arrow[u, "\gamma_2"'] \end{tikzcd}$$

Figure 3.6

Since $\mathfrak{W}$ is a weakly linked net,

$$\ker(\varphi_{\mu_2}^{\mathfrak{W}}) \supseteq \operatorname{Im}(\varphi_{\mu_1}^{\mathfrak{W}}). \tag{3.6.2}$$

We need to show equality. If both $\mu_1\gamma_1$ and $\mu_2\gamma_2$ are admissible, then $\varphi_{\mu_1}^{\mathfrak{W}}$ has class $\varphi_{w_2}^{\mathfrak{V},w_1}$, whereas $\varphi_{\mu_2}^{\mathfrak{W}}$ has class $\varphi_{w_1}^{\mathfrak{V},w_2}$. Since μ_1 is simple, by [11, Lem. 7.3], w_1 and w_2 are neighbors, and thus equality follows from the exactness of $\mathfrak{V}$.

If $\mu_2\gamma_2$ is not admissible, then the essential type of μ_1 is contained in that of γ_2. Since μ_1 is simple, there is an admissible path γ_2' connecting w_2 to v_1 such that $\mu_1\gamma_2'$ has the same type as γ_2; see Figure 3.7

$$\begin{tikzcd} v_1 \arrow[r, bend left, "\mu_1"] & v_2 \arrow[l, bend left, "\mu_2"] \\ w_1 \arrow[u, "\gamma_1"] \arrow[r] & w_2 \arrow[u, "\gamma_2"'] \arrow[ul, bend left, "\gamma_2'"] \end{tikzcd}$$

Figure 3.7

It follows that there is an admissible path from w_1 to v_1 passing through w_2, and then w_2 is the shadow of v_1 in H, that is, $w_1 = w_2$. Since the analogous argument holds if $\mu_1\gamma_1$ is not admissible, we may assume that $w_1 = w_2$.

Finally, equality in (3.6.2) follows because either $\varphi_{\mu_1}^{\mathfrak{W}}$ or $\varphi_{\mu_2}^{\mathfrak{W}}$ is the identity map. Indeed, by [10, Lem. 6.6], either $\mu_1\gamma_1$ or $\mu_2\gamma_2$ is admissible. The proof of Statement 2 is complete.

Finally, we prove Statement 3. Assume that $\mathfrak{V}$ is a linked representation. Since $\mathfrak{W}$ is a weakly linked net, by [10, Lem. 6.6], we need to show that

$$\ker(\varphi_{\mu_1}^{\mathfrak{W}}) \cap \ker(\varphi_{\mu_2}^{\mathfrak{W}}) = 0 \tag{3.6.3}$$

for each two simple admissible paths μ_1 and μ_2 in Q leaving the same vertex v, with no arrow type in common. Let v_1 and v_2 denote the respective final vertices of μ_1 and μ_2. Let

w, w_1, w_2 denote the respective shadows of v, v_1, v_2 in H. Let ρ_1 and ρ_2 be admissible paths in Q connecting w to w_1 and to w_2, respectively. Let γ be an admissible path connecting w to v.

Since μ_1 and μ_2 have no arrow type in common, $\mu_1\gamma$ or $\mu_2\gamma$ is admissible. Suppose first that one of them is not, say $\mu_1\gamma$ is not admissible. By definition of $\mathfrak{W}$, we have $\varphi^{\mathfrak{W}}_{\mu_1} = 0$. Also, the essential type of μ_2 is contained in that of γ, so the shadow of v_2 in H is w, i.e., $w_2 = w$. It follows that $\varphi^{\mathfrak{W}}_{\mu_2}$ is the identity.

Suppose now that both $\mu_1\gamma$ and $\mu_2\gamma$ are admissible. Then

$$\ker(\varphi^{\mathfrak{W}}_{\mu_1}) \cap \ker(\varphi^{\mathfrak{W}}_{\mu_2}) = \ker(\varphi^{\mathfrak{V}}_{\rho_1}) \cap \ker(\varphi^{\mathfrak{V}}_{\rho_2}). \tag{3.6.4}$$

If $w = w_1$ or $w = w_2$, the above intersection is zero, as one of the maps would be the identity.

Suppose $w_1 \neq w$ and $w_2 \neq w$. The paths ρ_i are simple by [11, Lem 7.3]. Since μ_1 and μ_2 have no arrow type in common, it follows that the intersection of the essential types of ρ_1 and ρ_2 is contained in the essential type of γ. Thus we may choose the ρ_i and γ', γ'' such that $\rho_i = \rho'_i\gamma'$ and $\gamma = \gamma''\gamma'$ for paths γ' such that ρ'_1 and ρ'_2 have no arrow type in common. The shadow in H of the common initial vertex z of the ρ'_i is w. Then, since $\mathfrak{V}$ is a linked representation,

$$\ker(\varphi^{\mathfrak{V}}_{\rho_1}) \cap \ker(\varphi^{\mathfrak{V}}_{\rho_2}) = 0.$$

But then, (3.6.3) follows from (3.6.4). □

Proposition 3.6.11. *Let $\mathfrak{V}$ be an exact linked quiver representation of $Q(H)$. Then there exists a finitely generated exact linked net $\mathfrak{W}$ of Q such that $\mathfrak{W}_{|_H}$ is projectively equivalent to $\mathfrak{V}$.*

Proof. By Lemma 3.6.10, $\mathfrak{V}^E$ is a finitely generated exact linked net of Q generated by H. And $\mathfrak{V} = (\mathfrak{V}^E)|_H$. □

Proposition 3.6.12. *Let $\mathfrak{V}$ and $\mathfrak{W}$ be two finite generated exact linked net over Q with minimum set of generators H and H', respectively. Then $\mathfrak{V}$ and $\mathfrak{W}$ are projectively equivalent if and only if $H = H'$ and, $\mathfrak{V}_{|_H}$ and $\mathfrak{W}_{|_{H'}}$, are projectively equivalent.*

Proof. If $\mathfrak{V}$ and $\mathfrak{W}$ are projectively equivalent, by Lemma 3.6.6, so are $\mathfrak{V}_{|_H}$ and $\mathfrak{W}_{|_H}$ are. Suppose that $H \neq H'$. Let $v \in H - H'$. Let w_v be the shadow of v in H', then $\varphi^{w_v,\mathfrak{W}}_v : V^{\mathfrak{W}}_{w_v} \to V^{\mathfrak{W}}_v$ is an isomorphism. Since $\mathfrak{V}$ and $\mathfrak{W}$ are projectively equivalent, then there is a collection of maps $f_v : V^{\mathfrak{V}}_v \to V^{\mathfrak{W}}_v$ be a collection of maps as in Definition 3.6.5. We have

$$[f_v][\varphi^{w_v,\mathfrak{W}}_v] = [\varphi^{w_v,\mathfrak{V}}_v][f_{w_u}].$$

It follows that $\varphi^{w_v,\mathfrak{V}}_v$ is an isomorphism contradiction that $v \in H$. This shows that $H \subseteq H'$. Showing that $H' \subseteq H$ is analogous. □

Chapter 4

Polyhedral tilings arising from linked nets.

4.1 Linked nets and Polytopes I.

4.1.1 Adjoint modular pairs associated to linked nets.

Let $\mathfrak{V}$ be a nontrivial finitely generated exact linked net of vector spaces over a $\mathbb{Z}^n$-quiver Q, and let H be its minimum set of generators.

In the following, for each $v, u \in H$, we fix an admissible path γ_u^v connecting v to u. When the initial (resp.final) vertex is clear we write γ_u (resp. γ^v).

For each $v \in H$, and for each nonempty subset $I \subseteq H$, define

$$\Psi_I^v : V_v \longrightarrow U_I := \bigoplus_{u \in I} V_u \tag{4.1}$$

as $\Psi_I^v(s) := (\varphi_{\gamma_u}(s)|u \in I)$. Set $U := U_H$. Let $W_v \subseteq U$ be the image of Ψ_H^v, that is,

$$W_v := \{\Psi_H^v(s) \,|\, s \in V_v\}\,. \subseteq U \tag{4.2}$$

The maps Ψ_I^v depend on the choice of admissible paths γ_u connecting v to u for each $u \in I$. Different choices produce different subspaces W_v. However, since $\mathfrak{V}$ is a linked net, all the W_v obtained via this process lie in the same orbit by the natural action of $\prod_{v \in H} \mathbf{k}^\times$ defined in Section 2.2.

For each $v \in H$, we denote by $(\mu_v^{\mathfrak{V}}, \mu_v^{*,\mathfrak{V}})$ the modular pair corresponding to W_v, defined in Section 2.2, and denote by $\mathbf{P}_{\mathfrak{V},v}$ the corresponding base polytope. If $\mathfrak{V}$ is clear from the context, we simply write (μ_v, μ_v^*) and $\mathbf{P}_v$. The pair (μ_v, μ_v^*) does not depend on the choice of admissible paths connecting v to $u \in H$. In fact, for each $I \subseteq H$,

$$\mu_v(I) = \dim\left(\bigcap_{u \in H-I} \ker(\varphi_u^v)\right) \text{ and } \mu_v^*(I) = \dim\left(V_v / \bigcap_{u \in I} \ker(\varphi_u^v)\right) \tag{4.3}$$

The following statement follows directly from Lemma 2.1.4 and Lemma 3.5.2.

Proposition 4.1.1. *Let $\mathfrak{V}$ be a nontrivial finitely generated exact linked net of vector spaces, and let H be its minimum set of generators. For each $v \in H$, the supermodular function μ_v is simple.*

4.1.2 Face structure.

Let $\mathfrak{V}$ be a nontrivial finitely generated exact linked net of vector spaces of dimension $r+1$ over a $\mathbb{Z}^n$-quiver Q, and let H be its minimum set of generators. Recall that $P(H) = H$. For each $v \in H$, let $\mathbf{P}_v$ be the polytope of the modular pair (μ_v, μ_v^*). In this section, we give a characterization of the faces of $\mathbf{P}_v$ that intersect $\mathring{\Omega}_{r+1}$, where Ω_{r+1} is the standard simplex in $\mathbb{R}^H_{r+1}$, as defined in Section 2.2.

Polygons.

Let H be a set of vertices of Q such that $P(H) = H$. Recall that, for each vertex u of Q, there is a unique vertex $w_u \in H$, called *shadow of* u *in* H, such that for every $v \in H$ there is an admissible path connecting v to u passing through w_u [10][Prop. 5.7]. For each $v \in H$, let R_v^H be the set of vertices of Q whose shadow in H is v; we write R_v when there is no ambiguity. We call R_v^H the *shadow region of* v *in* H. The collection of all shadow regions form a partition of the set of vertices of Q which we call the *shadow partition of* H.

Recall that a *polygon* is a collection of pairwise neighboring vertices of Q. A polygon with m vertices is called a *m-gon*. Given a vertex v of Q, a way of constructing a m-gon containing v is to pick an ordered m-partition $T = I_1 \cup \cdots \cup I_m$ of the set of arrow types T of Q, and put $v_1 := v$ and $v_{i+1} := I_i \cdot v_i$ for $i = 1, \dots, m-1$. Then $\{v_1, \dots, v_m\}$ is a m-gon. We say that $v_1, \dots, v_m$ form an *oriented m-gon*.

Let Δ be a m-gon in Q and $v \in \Delta$. There is a unique ordering $v_1, \dots, v_m$ of the vertices of Δ such that $v = v_1$ and $v_1, \dots, v_m$ form an oriented m-gon [10][Prop. 5.9]. Recall that for each polygon Δ in Q, we have $P(\Delta) = \Delta$ [10][Prop. 5.10]. Therefore, the shadow partition of Δ induces a partition π of H whose parts are the $R_v^\Delta \cap H$ for $v \in \Delta$. Order the partition π according ordering of Δ, i.e., put $\pi_i := R_{v_i}^\Delta \cap H$ for $i = 1, \dots, m$. We call π the *ordered partition induced by* Δ *and* v. If $\Delta \subseteq H$, then $v_i \in \pi_i$ for each i. In this case, the ordered partition π is a m-partition.

The following lemmas are straightforward consequences of the definitions. Since they will be used repeatedly in the sequel, we state and prove them here for reference.

Lemma 4.1.2. *Let $v_1, \dots, v_m$ be vertices of a $\mathbb{Z}^n$-quiver Q forming an oriented m-gon Δ. Put $v_{m+1} := v_1$, and let $i \in \{1, \dots, m\}$ and $u \notin R_{v_i}^\Delta$. Then there exists an admissible path connecting v_i to u passing through v_{i+1}.*

Proof. Let v_j be the shadow of u in Δ. It is sufficient to show that there is an admissible path from v_i to v_j passing through v_{i+1}. Since $v_1, \dots, v_m$ form an oriented m-gon, there exists a m-partition $T = I_1 \cup \cdots \cup I_m$ of the set of arrows T of Q such that $v_{i+1} = I_i \cdot v_i$ for $i = 1, \dots, m$. We need only show that $\mathrm{Ess}(\gamma_{v_j}^{v_i})$ contains I_i. By assumption, either $j > i$ or $j < i$. If $j > i$, then $\mathrm{Ess}(\gamma_{v_j}^{v_i}) = I_i \cup \cdots \cup I_{j-1}$. If $j < i$, then $\mathrm{Ess}(\gamma_{v_j}^{v_i}) = I_i \cup \cdots \cup I_m \cup I_1 \cup \cdots \cup I_{j-1}$. In any case, $\mathrm{Ess}(\gamma_{v_j}^{v_i})$ contains I_i, as required.

□

Lemma 4.1.3. *Let $\mathfrak{V}$ be an exact linked net over a $\mathbb{Z}^n$-quiver Q. Let $v_1, \dots, v_m$ form an oriented m-gon. Put $v_{m+1} := v_1$. Then the sequence*

$$0 \to \ker(\varphi_{v_i}^{v_1}) \to \ker(\varphi_{v_{i+1}}^{v_1}) \to \ker(\varphi_{v_{i+1}}^{v_i}) \to 0 \tag{4.4}$$

is exact for each $i = 1, \dots, m-1$, where the second map is the inclusion and the third map is $\varphi_{v_i}^{v_1}$.

Proof. We need only show that the third map is surjective. Since $\mathfrak{V}$ is exact, we have that

$$\varphi_{v_i}^{v_1}(\ker(\varphi_{v_{i+1}}^{v_1})) = \varphi_{v_i}^{v_1}(\mathrm{Im}(\varphi_{v_1}^{v_{i+1}})) = \mathrm{Im}(\varphi_{v_i}^{v_{i+1}}) = \ker(\varphi_{v_{i+1}}^{v_i}).$$

□

Faces induced by polygons.

Let $\mathfrak{V}$ be an exact finitely generated nontrivial linked net of vector spaces of dimension $r+1$ over a $\mathbb{Z}^n$-quiver Q, and let H be its minimum set of generators. Let $\Delta \subseteq H$ be a polygon. We associate to Δ a modular pair $(\mu_\Delta, \mu_\Delta^*)$ as follows. First, let V_Δ be the following direct sum of vector spaces:

$$V_\Delta := \bigoplus_{v\in\Delta}\left(\bigcap_{w\in H-R_v^\Delta} \ker(\varphi_w^v)\right). \tag{4.5}$$

Next, define the linear map $\Psi_H^\Delta : V_\Delta \to U$ sending $(s_v \mid v \in \Delta)$ to $(\varphi_u^{w_u}(s_{w_u}) \mid u \in H)$, where w_u is the shadow of u in Δ for each $v \in H$. Let $W_\Delta := \Psi_H^\Delta(V_\Delta)$, the image of Ψ_H^Δ, and $(\mu_\Delta, \mu_\Delta^*)$ the associated modular pair, as defined in Section 2.2. Let $\mathbf{P}_\Delta$ be the base polytope of μ_Δ.

Lemma 4.1.4. *Let H be the minimum set of generators of $\mathfrak{V}$. Let $\Delta \subseteq H$ be a m-gon and $v \in \Delta$. Let π be the m-partition of H induced by Δ and v, and let $W_{v,\pi}$ be the splitting of W_v associated to π, and $\mathbf{P}_{v,\pi}$ the corresponding face of $\mathbf{P}_v$. The following statements hold:*

(1) $\mathrm{cd}_{\mu_\Delta} = m$;

(2) $W_{v,\pi}$ *and* W_Δ *are equivalent; in particular,* $\mathbf{P}_{v,\pi} = \mathbf{P}_\Delta$;

(3) $\mathring{\mathbf{P}}_\Delta \cap \mathring{\Omega}_{r+1} \neq \emptyset$.

Proof. Let $v_1, \dots, v_m$ be the vertices of Δ, ordered in such a way that $v_1 = v$ and $v_1, \dots, v_m$ form an oriented m-gon. Put $v_{m+1} := v_1$. Then,

$$V_\Delta = \bigoplus_{i=1}^{m}\left(\bigcap_{w\in H-\pi_i} \ker(\varphi_w^{v_i})\right),$$

where $\pi_i = R_{v_i}^\Delta \cap H$ for each $i = 1, \dots, m$. Since $v_{i+1} \notin \pi_i$, it follows from Lemma 4.1.3 and [10, Lem. 8.1] that

$$\bigcap_{w\in H-\pi_i} \ker(\varphi_w^{v_i}) = \ker(\varphi_{v_{i+1}}^{v_i}).$$

By exactness, we have that

$$V_\Delta = \bigoplus_{i=1}^{m} \ker(\varphi_{v_{i+1}}^{v_i}) = \bigoplus_{i=1}^{m} \mathrm{Im}(\varphi_{v_i}^{v_{i+1}}).$$

It follows that

$$W_\Delta = \bigoplus_{i=1}^{m} \Psi_{\pi_i}^{v_i}(\mathrm{Im}(\varphi_{v_i}^{v_{i+1}}))$$

where $\Psi^{v_i}_{\pi_i}$ is the map defined in (4.1). Since $v_i \in \pi_i$, it follows from Lemma 3.2.2 and Lemma 2.1.4 that, for each $i = 1, \ldots, m$, the subspace $\Psi^{v_i}_{\pi_i}(\mathrm{Im}(\varphi^{v_{i+1}}_{v_i})) \subseteq U_{\pi_i}$ is simple, whence $\mathrm{cd}_{\mu_\Delta} = m$, finishing the proof of (1).

As for (2), let $F_\bullet$ be the filtration of H corresponding to π, and let

$$0 \subseteq W_v^{F_1} \subseteq \cdots \subseteq W_v^{F_{m-1}} \subseteq W_v$$

be the corresponding filtration of W_v. It is enough to show that $\Psi^{v_i}_{\pi_i}(\ker(\varphi^{v_i}_{v_{i+1}}))$ and $\theta_{\pi_i}(W_v^{F_i})$ are equivalent in U_{π_i} for every i. Recall that each $W_v^{F_i}$ is the image of $\bigcap_{z\in H-F_i}\ker(\varphi^v_z)$ in U via Ψ^v_H. Since $v_{i+1} \notin F_i$, it follows from Lemma 4.1.2 and [10][Lem. 8.1] that

$$\bigcap_{z\in H-F_i}\ker(\varphi^v_z) = \ker(\varphi^v_{v_{i+1}}).$$

Thus,

$$\theta_{\pi_i}(W_v^{F_i}) = \Psi^v_{\pi_i}(\ker(\varphi^v_{v_{i+1}})).$$

For each $w \in \pi_i$, the concatenation $\gamma^{v_i}_w\gamma^v_{v_i}$ is admissible. Since γ^v_w is admissible, there is $c_w \in \mathbf{k}^*$ such that

$$\varphi_{\gamma^v_w} = c_w\varphi_{\gamma^{v_i}_w}\varphi_{\gamma^v_{v_i}}.$$

Set $c := (c_w) \in (\mathbf{k}^*)^{\pi_i}$. Then $c\Psi^{v_i}_{\pi_i}\circ\varphi^v_{v_i} = \Psi^v_{\pi_i}$, whence $\Psi^v_{\pi_i}(\ker(\varphi^v_{v_{i+1}}))$ and $\Psi^{v_i}_{\pi_i}\circ\varphi^v_{v_i}(\ker(\varphi^v_{v_{i+1}}))$ are equivalent. It follows from Lemma 4.1.3 that

$$\Psi^{v_i}_{\pi_i}\circ\varphi^v_{v_i}(\ker(\varphi^v_{v_{i+1}})) = \Psi^{v_i}_{\pi_i}(\ker(\varphi^{v_i}_{v_{i+1}})),$$

as required.

As for (3), suppose by contradiction that $\mathring{\mathbf{P}}_\Delta \cap \mathring{\Omega}_{r+1} = \emptyset$. Since $\mathbf{P}_{v,\pi} = \mathbf{P}_\Delta$, it follows from [1, Prop. 2.7] that $\mu^*_{v,\pi}(I) = 0$ for some nonempty $I \subsetneq H$. As $\mu^*_{v,\pi}$ is nondecreasing, for each $u \in I$, we have $\mu^*_{v,\pi}(u) = 0$. Fix $u \in I$ and let v_i be its the shadow in Δ. Then $u \in \pi_i$. Now, we have

$$\mu^*_{v,\pi}(u) = \dim\varphi^v_u(\ker(\varphi^v_{v_{i+1}})) = \dim\varphi^{v_i}_u\varphi^v_{v_i}(\ker(\varphi^v_{v_{i+1}})).$$

By Lemma 4.1.3 and exactness,

$$\mu^*_{v,\pi}(u) = \dim(\varphi^{v_i}_u(\ker(\varphi^{v_i}_{v_{i+1}})) = \dim(\varphi^{v_i}_u(\mathrm{Im}(\varphi^{v_{i+1}}_{v_i})) = \dim(\mathrm{Im}(\varphi^{v_{i+1}}_u)).$$

Hence, $\varphi^{v_{i+1}}_u = 0$, contradicting the minimality of H. □

Proposition 4.1.5. *Let $\mathfrak{V}$ be a nontrivial exact finitely generated linked net of vector spaces over a $\mathbb{Z}^n$-quiver Q, and H its minimum set of generators. Then, for each $v \in H$ and each m-partition $\pi = (\pi_1, \ldots, \pi_m)$ of H such that $\mathbf{P}_{v,\pi} \cap \mathring{\Omega}_{r+1} \neq \emptyset$, there is a m-gon $\Delta \subseteq H$ such that $v \in \Delta$ and the ordered partition of H induced by Δ and v coincides with π. In particular, $\mathbf{P}_{v,\pi} = \mathbf{P}_\Delta$.*

Proof. Let $\mathbf{F} := \mathbf{P}_{v,\pi}$ be the face of $\mathbf{P}_v$ corresponding to the m-partition π. We divide the proof in two steps. First, we will construct a certain oriented m-gon Δ containing v, and then we will show that the ordered partition induced by Δ and v coincides with π.

Let $F_\bullet$ be the filtration corresponding to π. By [1, Prop. 2.5], since $\mathring{\mathbf{F}} \cap \mathring{\Omega}_{r+1} \neq \emptyset$, we have $\mu_v(F_j) \neq 0$ for all $j = 1, \ldots, m-1$. Then,

$$\bigcap_{u\in H-F_j}\ker(\varphi^v_u) \neq 0 \text{ for all } j = 1, \ldots, m-1.$$

It follows from [10, Lem. 8.1] that

$$\bigcap_{u\in H-F_j}\mathrm{Ess}(\gamma_u^v)\neq\emptyset \text{ for all } j=1,\dots,m-1.$$

Now, for each $j=1,\dots,m-1$, define

$$I_j:=\bigcap_{u\in H-F_j}\mathrm{Ess}(\gamma_u^v).$$

Let w_{j+1} denote the shadow of $I_j\cdot v$ in H. Then, $\varphi_{I_j\cdot v}^{w_{j+1}}$ is an isomorphism. So, by [10, Lem. 8.1], we have

$$\bigcap_{u\in H-F_j}\ker(\varphi_u^v)=\ker(\varphi_{I_j\cdot v}^v)=\ker(\varphi_{w_{j+1}}^v). \tag{3.2}$$

Thus,

$$W_v^{F_j}=\Psi_H^v\left(\ker(\varphi_{w_{j+1}}^v)\right).$$

Since $F_j\subseteq F_{j+1}$, we have $I_j\subseteq I_{j+1}$. Note that this inclusion is strict. Indeed, assume by contradiction that $I_j=I_{j+1}$. Then, $w_{j+1}=w_{j+2}$, whence

$$\mu_v(F_j)=\dim\ker(\varphi_{w_{j+1}}^v)=\dim\ker(\varphi_{w_{j+2}}^v)=\mu_v(F_{j+1}).$$

Now, for each $q\in\mathbf{F}$, by [1][Prop.2.5], we have

$$\mu_v(F_j)=q(F_j)\leq q(F_{j+1})=\mu_v(F_{j+1}),$$

which implies $q(F_{j+1}-F_j)=0$ for all $q\in\mathbf{F}$. This contradicts the assumption $\mathbf{F}\cap\mathring{\Omega}_{r+1}\neq\emptyset$.

Set $w_1:=v$. Since $v,I_1\cdot v,\dots,I_{m-1}\cdot v$ form an oriented m-gon, by [11][Lem 7.3], their shadows $w_1,w_2,\dots,w_m$ form an oriented m-gon, which we denote by Δ. Let π' be the partition of H induced by Δ and v. We will show that π' coincides with π.

We claim that for each $j=1,\dots,m$ and $u\in\pi_j$, the shadow of u in Δ is w_j. If the claim holds, then by definition of π', we have $\pi_j\subseteq\pi'_j$ for every j. It follows that $\pi=\pi'$, finishing the proof of the statement.

Let $u\in\pi_j$ and w_i be its shadow in Δ. Assume by contradiction that $i\neq j$. By Lemma 4.1.2 there is an admissible path connecting w_j to u passing through w_{j+1}. Then $\ker(\varphi_{w_{j+1}}^{w_j})\subseteq\ker(\varphi_u^{w_j})$. Since $w_1,\dots,w_m$ is an oriented m-gon, we have

$$\mu_{v,\pi}^*(u)=\dim\varphi_u^v(\ker(\varphi_{w_{j+1}}^v))=\dim\varphi_u^{w_j}\varphi_{w_j}^v(\ker(\varphi_{w_{j+1}}^v))=\dim\varphi_u^{w_j}(\ker(\varphi_{w_{j+1}}^{w_j}))=0,$$

where the first equality follows from (3.2), the second from $I_{j-1}\subseteq\mathrm{Ess}(\gamma_u^v)$ and the fact that w_j is the shadow of $I_{j-1}\cdot v$, and the third from Lemma 4.1.3. This contradicts the assumption that $\mathbf{P}_{v,\pi}\cap\mathring{\Omega}_{r+1}\neq\emptyset$.

□

4.2 Tilings induced by linked nets.

Let $\mathfrak{V}$ be a nontrivial finitely generated exact linked net of vector spaces over a $\mathbb{Z}^n$-quiver Q, and let H be its minimum set of generators. As in Section 4.1, for each $v,u\in H$, we fix a choice of an admissible path γ_u^v connecting v to u. For each $v\in H$, we let (μ_v,μ_v^*) be the corresponding modular pair, as defined in Section 4.1.1. Let $\mathcal{C}_H(\mathfrak{V})$ be the collection of all supermodular functions μ_v for $v\in H$. Define

$$\mathbf{P}_{\mathfrak{V}}:=\mathbf{P}_{\mathcal{C}_H(\mathfrak{V})}=\bigcup_{v\in H}\mathbf{P}_v.$$

Lemma 4.2.1. *For each $v, u \in H, v \neq u$, there exists a nonzero $c \in \mathbf{k}^H$ such that $cW_v \subseteq W_u$. Also, $\mu_v \neq \mu_u$. Finally, the collection $\mathcal{C}_H(\mathfrak{V})$ is separated.*

Proof. Let γ be an admissible path connecting v to u. Let I be the set of $w \in H$ such that the concatenation $\gamma_w^u \gamma$ is admissible. Since γ_w^v is admissible, there exists $c_w \in \mathbf{k}$ such that

$$c_w \varphi_{\gamma_w^v} = \varphi_{\gamma_w^u} \varphi_\gamma.$$

Since H is minimum and $\gamma_w^u \gamma$ is admissible, for each $w \in I$, we have $\varphi_{\gamma_w^u} \varphi_\gamma \neq 0$, and thus $c_w \neq 0$. Put $c_w := 0$ for $w \in H - I$, and let $c := (c_w)_{w \in H} \in \mathbf{k}^H$. We claim that $cW_v \subseteq W_u$. Indeed, an element of W_v is of the form $(\varphi_{\gamma_w^v}(s))_{w \in H}$ for $s \in V_v$. Note that $(\varphi_{\gamma_w^u} \varphi_\gamma(s))_{w \in H} \in W_u$. If $w \notin I$, then $\varphi_{\gamma_w^u} \varphi_\gamma(s) = 0$. If $w \in I$, then

$$\varphi_{\gamma_w^u} \varphi_\gamma(s) = c_w \varphi_{\gamma_w^v}(s).$$

Hence, $c(\varphi_{\gamma_w^v}(s))_{w \in H} = (\varphi_{\gamma_w^u} \varphi_\gamma(s))_{w \in H} \in W_u$, as required. Note that $c_v = 0$ and $c_u \neq 0$. Hence, it follows from Proposition 2.2.1 that $\mu_v \neq \mu_u$.

Note that c induces an injective map $W_{v,I} \to W_u^I$, whence $\mu_v^*(I) \leq \mu_u(I)$. Then (I, I^c) is a nontrivial separation for (μ_u, μ_v). □

Theorem 4.2.2. *The collection of polytopes $\mathbf{P}_{\mu_v}$ and their faces, associated to an exact finitely generated nontrivial linked net of vector spaces $\mathfrak{V}$ of dimension $r+1$, gives a polyhedral tiling of the standard simplex Ω_{r+1}. In particular, $\mathbf{P}_{\mathfrak{V}} = \Omega_{r+1}$.*

Proof. We will check the conditions in Proposition 2.2.2. Condition (2) follows immediately from Lemma 4.2.1.

As for Condition (1), let $v \in H$ and let $\pi = (\pi_1, \pi_2)$ be a bipartition of H such that $\mathbf{P}_{v,\pi}$ intersects $\mathring{\Omega}_{r+1}$. By Proposition 4.1.5 there is a 2-gon $\Delta = \{v, u\} \subseteq H$ such that the ordered partition induced by Δ and v coincides with π. Now, the bipartition π^c is equal to the bipartition induced by Δ and u. It follows from Lemma 4.1.4 that

$$\mathbf{P}_{v,\pi} = \mathbf{P}_\Delta = \mathbf{P}_{u,\pi^c}.$$

Hence, by [1, Prop. 2.7], $\mu_{v,\pi} = \mu_\Delta = \mu_{u,\pi^c}$. Thus, $\mathcal{W}$ is complete for $\mathring{\Omega}_{r+1}$.

□

Recall that an *abstract simplicial complex* is a nonempty family $\mathcal{S}$ of subsets of a given set H such that, for every $X \in \mathcal{S}$, and every nonempty $Y \subseteq X$, we have $Y \in \mathcal{S}$. The elements in $\mathcal{S}$ are called *faces*. The *vertex set* of $\mathcal{S}$ is the union of all faces of $\mathcal{S}$. A map from a simplicial complex $\mathcal{S}$ to a simplicial complex $\mathcal{S}'$ is a function f from the sets of vertices of $\mathcal{S}$ to that of $\mathcal{S}'$ such that for each face X of $\mathcal{S}$, the image $f(X)$ is a face of $\mathcal{S}'$.

The following proposition follows immediately from the definitions.

Proposition 4.2.3. *Let $\mathfrak{V}$ be a nontrivial finitely generated exact linked net of vector spaces over a $\mathbb{Z}^n$-quiver Q, and H be its minimum set of generators. The following abstract simplicial complexes of H are equal:*

1. *The family $\mathcal{S}_1$ of subsets $\Delta \subseteq H$ such that Δ is a polygon.*

2. *The family $\mathcal{S}_2$ of subsets $X \subseteq H$ such that $\bigcap_{v \in X} \mathbf{P}_v \cap \mathring{\Omega}_{r+1} \neq \emptyset$.*

Furthermore, $\bigcap_{v \in \Delta} \mathbf{P}_v = \mathbf{P}_\Delta$ for each polygon $\Delta \subseteq H$.

4.3 Regularity of the tiling.

Let $\mathfrak{V}$ be a nontrivial exact finitely generated linked net of vector spaces of dimension $r+1$ over a $\mathbb{Z}^n$-quiver Q, and let H be its minimum set of generators. Denote by $\mathcal{C}_H(\mathfrak{V})$ the collection of modular pairs defined in Section 4.2.

Recall that a polyhedral tiling of a convex domain Ω in a real vector space $\mathbb{R}^H$ is called *regular* if there is a convex function on Ω which has domains of linearity given by the tiles in the tiling.

The *dual graph* of a polyhedral tiling has as set of vertices the full dimensional polytopes, with an edge connecting a pair of vertices if the corresponding tiles share a facet.

Theorem 4.3.1. *The tiling of Ω_{r+1} induced by $\mathcal{C}_H(\mathfrak{V})$ is regular.*

Proof. Put $\mathcal{C} := \mathcal{C}_H(\mathfrak{V})$. Each function $f\colon H \to \mathbb{R}$ gives rise to a $\mathbb{R}$-linear function on $\mathbb{R}^H$, denoted $(f,\cdot)$ that sends q to

$$(f,q) := \sum_{v\in H} f(v)q(v).$$

Fix a vertex $v_0 \in H$, as well as $f_0 \in \mathbb{R}^H$ and $\lambda_0 \in \mathbb{R}$. We claim that there are $f_v \in \mathbb{R}^H$ and $\lambda_v \in \mathbb{R}$ such that the following properties hold:

1. The restrictions of the affine functions $\lambda_v - (f_v,\cdot)$ to $\mathbf{P}_v$ for $v \in H$ patch to a global function Φ defined on $\mathbf{P}_{\mathcal{C}} = \Omega_{r+1}$.

2. The function Φ is convex and piecewise linear, and its domains of linearity are exactly the polytopes $\mathbf{P}_v$ for $v \in H$.

To prove the claim, put $f_{v_0} := f_0$ and $\lambda_{v_0} := \lambda_0$. For each $v \in H$, there is a sequence of vertices $v_0,\dots,v_p$ in H such that $v_p = v$ and, for each $i = 1,\dots,p$, v_{i+1} is a neighbor of v_i. Let $\pi_i = (F_i, F_i^c)$ be the ordered partition induced by the oriented 2-gon $\Delta_i =: \{v_i, v_{i+1}\}$ and v_i, as in Section 4.1.2. Then, by Lemma 4.1.4, the π_i-splitting of μ_{v_i} is the π_i^c-splitting of $\mu_{v_{i+1}}$, and is equal to μ_{Δ_i}. Equivalently,

$$\mathbf{P}_{v_i,\pi_i} = \mathbf{P}_{\Delta_i} = \mathbf{P}_{v_{i+1},\pi_i^c}.$$

We define

$$f_v := f_{v_0} + \sum_{i=0}^{p-1} 1_{F_i} \qquad \text{and} \qquad \lambda_v := \lambda_{v_0} + \sum_{i=0}^{p-1} \mu_{v_i}(F_i)$$

We will prove that the restriction of $\lambda_v - (f_v,\cdot)$ to the hyperplane $\mathbb{R}^H_{r+1}$ does not depend on the choice of the vertices v_i. Assuming this for the moment, we will show Property 1.

First, we show that the $\lambda_v - (f_v,\cdot)$ patch to a function Φ. Let $v, v' \in H$ neighbors. Then $\mathbf{P}_{v'}$ and $\mathbf{P}_v$ share a facet. We assume that the sequence $v_0,\dots,v_{p-1},v_p$ chosen for v satisfies $v' = v_{p-1}$. Then, for each point $q \in \mathbf{P}_{v'} \cap \mathbf{P}_v$, we have

$$(f_v - f_{v'}, q) = q(F_{p-1}) = \mu_{v'}(F_{p-1}) = \lambda_v - \lambda_{v'},$$

showing that the functions $\lambda_v - (f_v,\cdot)$ patch to a function Φ. Since Ω_{r+1} is convex, it is enough to show that the function Φ is strictly convex around $\mathbf{P}_v \cap \mathbf{P}_{v'}$ for each $v, v' \in H$ whose associate polytopes share a common facet. Here, strict convexity means that there exists an affine function ϕ which coincide with Φ on the intersection $\mathbf{P}_v \cap \mathbf{P}_{v'}$, but $(\Phi - \phi)|_{\mathbf{P}_{v'} - \mathbf{P}_v} > 0$

and $(\Phi-\phi)|_{\mathbf{P}_v-\mathbf{P}_{v'}}>0$. As before, we assume that the sequence $v_0,\dots,v_{p-1},v_p$ chosen for v satisfies $v'=v_{p-1}$. Then $f_v-f_{v'}=1_{F_{p-1}}$ and

$$\lambda_v-\lambda_{v'}=\mu_{v_{p-1}}(F_{p-1})=\mu_{v'}(F_{p-1})=\mu^*_v(F_{p-1}).$$

We claim that

$$\phi:=-\frac{1}{2}\left((f_v,\cdot)-\lambda_v+(f_{v'},\cdot)-\lambda_{v'}\right)$$

works as desired. Indeed, since $(f_v,\cdot)-\lambda_v$ and $(f_{v'},\cdot)-\lambda_{v'}$ coincide on the facet, we have that

$$\phi|_{\mathbf{P}_v\cap\mathbf{P}_{v'}}=\Phi|_{\mathbf{P}_v\cap\mathbf{P}_{v'}}.$$

In addition,

$$\begin{aligned}(\Phi-\phi)|_{\mathbf{P}_v-\mathbf{P}_{v'}}&=\lambda_v-(f_v,\cdot)+\frac{1}{2}\left((f_v,\cdot)-\lambda_v+(f_{v'},\cdot)-\lambda_{v'}\right)\\&=\frac{1}{2}\left(-(f_v,\cdot)+\lambda_v+(f_{v'},\cdot)-\lambda_{v'}\right)\\&=-\frac{1}{2}\big((1_{F_{p-1}},\cdot)-\mu_{v'}(F_{p-1})\big)>0\end{aligned}$$

where the inequality follows since for $q\in\mathbf{P}_v-\mathbf{P}_{v'}$ we have $q(F_{p-1})<\mu^*_v(F_{p-1})$. That $(\Phi-\phi)|_{\mathbf{P}_{v'}-\mathbf{P}_v}>0$ is analogous.

It remains only to prove that the restriction of $(f_v,\cdot)-\lambda_v$ to the hyperplane $\mathbb{R}^H_{r+1}$ does not depend on the choice of the sequence v_i.

Assume that we have two sequences $v_0,\dots,v_p$ and $v'_0,\dots,v'_{p'}$ with $v_0=v'_0$ and $v_p=v'_{p'}$. We can put them together to get a sequence $v_0,\dots,v_p,v_{p+1}:=v'_{p'-1},v'_{p'-2},\dots,v_{p+p'}:=v'_0$. If $v_0,\dots,v_p$ and $v'_0,\dots,v'_{p'}$ are equal If $v_0,\dots,v_p$ and $v_0,\dots,v'_{p'}$ are equal then it is easy to check that So, it is sufficient to prove that $(f_{v_{p+p'}},\cdot)-\lambda_{v_{p+p'}}$ coincides with $(f_{v_0},\cdot)-\lambda_{v_0}$ in $\mathbb{R}^H_{r+1}$. It is thus enough to prove the coincide in general. Hence, it is enough to show the claim along any cycle $v_0,\dots,v_p=v_0$. This cycle represents an actual cycle in the dual graph of the tiling. We can decompose this cycle into smaller cycles $v_0,\dots,v_p=v_0$ such that $\mathbf{F}:=\bigcap\mathbf{P}_{v_i}\neq\emptyset$. It is sufficient to show the claim for this smaller cycles. For $q\in\mathbf{F}$,

$$(f_{v_p}-f_{v_0},q)=\sum_{i=0}^{p-1}q(F_i)=\sum_{i=0}^{p-1}\mu_{v_i}(F_i)=\lambda_{v_p}-\lambda_{v_0},$$

as desired.

□

4.4 Linked nets and Polytopes II

Let $\mathfrak{V}$ be a nontrivial exact finitely generated linked net of dimension $r+1$ over a $\mathbb{Z}^n$-quiver Q, and H be its minimum set of generators. Let T be the set of arrow types of Q. For each arrow type $\mathfrak{a}$, denote by $u_{\mathfrak{a}}\in H$ the extreme vertex of H with respect to $\mathfrak{a}$. Given a vertex v of Q and a collection of arrow types I, we put $\varphi^v_I:=\varphi^v_w$ where $w=I\cdot v$.

Fix, for each $v\in H$ and $\mathfrak{a}\in T$, an admissible path $\gamma^v_{u_{\mathfrak{a}}}$ connecting v to $u_{\mathfrak{a}}$. For each $v\in H$ and for each nonempty $I\subseteq T$, define

$$\Psi^v_I:V_v\to U_I:=\bigoplus_{\mathfrak{a}\in I}V_{u_{\mathfrak{a}}}\tag{3.3}$$

by $\Psi_I^v(s) := (\varphi_{\gamma^v_{u_\mathfrak{a}}}(s) | \mathfrak{a} \in I)$. Set $U := U_T$. Let $M_v \subseteq U$ be the image of Ψ_T^v, that is,

$$M_v := \{\Psi_T^v(s) \,|\, s \in V_v\}$$

The maps Ψ_I^v depend on the choices of the $\gamma^v_{u_\mathfrak{a}}$. Different choices produce different subspaces M_v. However, since $\mathfrak{V}$ is a linked net, all the M_v obtained via this process lie in the same orbit by the natural action of $\prod_{\mathfrak{a}\in T} \mathbf{k}^\times$ defined in Section 2.2.

For each $v \in H$, we denote by $(\nu_v^{\mathfrak{V}}, \nu_v^{*,\mathfrak{V}})$ the modular pair corresponding to M_v, defined in Section 2.2. We write (ν_v, ν_v^*) when $\mathfrak{V}$ is clear from the context.

The following lemma will be used several times in the continuation.

Lemma 4.4.1. *Let $\mathfrak{V}$ and H be as above. For each $v \in H$ and $I \subseteq T$,*

$$\ker(\varphi^v_{T-I}) = \bigcap_{\mathfrak{a}\in I} \ker(\varphi^v_{u_\mathfrak{a}})$$

. In particular, Ψ_T^v is injective.

Proof. It follows from [10][Lem. 8.1] that

$$\bigcap_{\mathfrak{a}\in I} \ker(\varphi^v_{T-\{\mathfrak{a}\}}) = \ker(\varphi^v_{T-I}).$$

Hence, we need only show that

$$\ker(\varphi^v_{u_\mathfrak{a}}) = \ker(\varphi^v_{T-\{\mathfrak{a}\}}).$$

Let $J := (T - \{\mathfrak{a}\}) - \mathrm{Ess}(\gamma^v_{u_\mathfrak{a}})$, and put $w := J \cdot u_\mathfrak{a}$. The shadow of w in H is $u_\mathfrak{a}$. Indeed, since $u_\mathfrak{a}$ is extreme with respect to $\mathfrak{a}$, for each $u \in H$ there is an admissible γ path from u to $u_\mathfrak{a}$ avoiding $\mathfrak{a}$. By (Q4), any admissible path μ from $u_\mathfrak{a}$ to w has essential type J, and thus avoids $\mathfrak{a}$. Then the concatenation $\mu\gamma$ is admissible. It follows that, for each $u \in H$, there is an admissible path connecting u to w passing through $u_\mathfrak{a}$, and thus $u_\mathfrak{a}$ is the shadow of w in H.

Now, since $\mathfrak{V}$ is pure and 1-generated by H, $\varphi^{u_\mathfrak{a}}_w$ is an isomorphism. Then, it follows from [10][Lem 6.6 (4)] that

$$\ker(\varphi^v_{u_\mathfrak{a}}) = \ker(\varphi^v_w) = \ker(\varphi^v_{T-\{\mathfrak{a}\}}).$$

$\square$

Proposition 4.4.2. *The pair (ν_v, ν_v^*) is a modular of rank $r+1$, and for each $I \subseteq T$, we have*

$$\nu_v(I) = \dim\ker(\varphi^v_I).$$

Proof. That (ν_v, ν_v^*) is a modular pair and has rank $r+1$ follows directly from the definition. We now show the second part. It follows from Lemma 4.4.1, that

$$\nu_v(I) = \dim M_v^I = \dim \Psi_T^v\left(\bigcap_{\mathfrak{a}\in T-I} \ker(\varphi^v_{u_a})\right) = \dim \Psi_T^v\left(\ker(\varphi^v_I)\right) = \dim\ker(\varphi^v_I),$$

as required. $\square$

4.4.1 Face structure.

Let $\mathfrak{V}$ be a nontrivial finitely generated exact linked net of vector spaces of dimension $r+1$ over a $\mathbb{Z}^n$-quiver Q, and let H be its minimum set of generators. Let $\Delta \subseteq H$ be a polygon. We will associate to Δ a modular pair $(\nu_\Delta, \nu_\Delta^*)$.

For each $v \in \Delta$, let I_v^Δ be the set of arrow types $\mathfrak{a} \in T$ such that the shadow $w_\mathfrak{a}$ of $u_\mathfrak{a}$ in Δ is v. By [10][Prop. 5.7], for each two different vertices u and v in Δ, we have $I_v^\Delta \cap I_u^\Delta = \emptyset$. In addition, for each $\mathfrak{a} \in T$, we have $\mathfrak{a} \in I_{w_{u_\mathfrak{a}}}^\Delta$. Hence, the collection of all I_v^Δ for $v \in \Delta$ is a $\#\Delta$-partition of T.

Lemma 4.4.3. *Let $\mathfrak{V}$ and $\Delta \subseteq H$ be as above. Let $v \in \Delta$, and put $I := I_v^\Delta$. Then, the restriction of the map Ψ_I^v,*

$$\Psi_I^v : s \in \bigcap_{w \in H - R_v^\Delta} \ker(\varphi_w^v) \longrightarrow (\varphi_{u_\mathfrak{a}}^v(s) \mid \mathfrak{a} \in I) \in \bigoplus_{\mathfrak{a} \in I} V_{u_\mathfrak{a}}$$

is injective.

Proof. Since

$$\ker(\Psi_I^v) = \bigcap_{\mathfrak{a} \in I} \ker(\varphi_{u_\mathfrak{a}}^v),$$

it is sufficient to show that

$$\bigcap_{\mathfrak{a} \in I} \ker(\varphi_{u_\mathfrak{a}}^v) \cap \bigcap_{w \in H - R_v^\Delta} \ker(\varphi_w^v) = 0.$$

By Lemma 4.1.2, there exist $u \in \Delta$, different from v, such that for each $w \in H - R_v^\Delta$ there is an admissible path connecting v to w passing through u. Then,

$$\bigcap_{w \in H - R_v^\Delta} \ker(\varphi_w^v) = \ker(\varphi_u^v).$$

Let γ be an admissible path connecting v to u, and put $J := \mathrm{Ess}(\gamma)$. We claim that $I = J$. If the claim holds, then the statement follows. Indeed, by Lemma 4.4.1 and [11][Lem 8.1], we have

$$\bigcap_{\mathfrak{a} \in I} \ker(\varphi_{u_\mathfrak{a}}^v) \cap \bigcap_{w \in H - R_v^\Delta} \ker(\varphi_w^v) = \ker(\varphi_{T-I}^v) \cap \ker(\varphi_I^v) = 0,$$

as required. We will now show the claim. Let $\mathfrak{a} \in I$. Assume by contradiction that $\mathfrak{a} \notin J$. Since $u_\mathfrak{a}$ is extreme with respect to $\mathfrak{a}$, there is an admissible path μ connecting u to $u_\mathfrak{a}$ avoiding $\mathfrak{a}$. As $\mathfrak{a} \notin J$, then $\mu\gamma$ is an admissible path connecting v to $u_\mathfrak{a}$ passing through u, contradicting the assumption that v is the shadow of $u_\mathfrak{a}$ in Δ. It follows that $I \subseteq J$. As for the converse, let $\mathfrak{a} \in J$. We must show that the shadow of $u_\mathfrak{a}$ in Δ is v. Since $\mathfrak{a} \in J$, for each $w \in \Delta$ there is a path θ connecting w to v avoiding $\mathfrak{a}$. As $u_\mathfrak{a}$ is extreme with respect to $\mathfrak{a}$ there is path μ connecting v to $u_\mathfrak{a}$ avoiding $\mathfrak{a}$. Hence, the concatenation $\mu\theta$ avoids $\mathfrak{a}$ and, thus, is an admissible path connecting w to $u_\mathfrak{a}$ passing through v, as required. □

Let V_Δ be the following direct sum of vector spaces:

$$V_\Delta := \bigoplus_{v \in \Delta} \bigcap_{w \in H - R_v^\Delta} \ker(\varphi_w^v).$$

Next, define the linear map $\Psi_T^{\Delta} : V_{\Delta} \to U$ sending $(s_v \mid v \in \Delta)$ to $(\Psi_{I_v}^v(s_v) \mid v \in \Delta)$. Put M_{Δ} be the image of Ψ_T^{Δ}. Let $(\nu_{\Delta}, \nu_{\Delta}^*)$ be the associated modular pair, as defined in Section 2.2. Denote by $\mathbf{P}_{\Delta}$ the corresponding polytope. By Lemma 4.4.3, the modular pair $(\nu_{\Delta}, \nu_{\Delta}^*)$ has rank $r+1$. Indeed, enough to show that V_{Δ} has dimension $r+1$. This follows from 4.1.3 and [11][Prop. 10.1].

Recall that, for each $v \in \Delta$, there is a unique ordering $v_1, \ldots, v_m$ of the vertices of Δ such that $v = v_1$ and $v_1, \ldots, v_m$ for an oriented m-gon. Let π be the ordered partition $(I_{v_1}^{\Delta}, \ldots, I_{v_m}^{\Delta})$ of T. We call π the *ordered partition of T induced by Δ and v.*

Lemma 4.4.4. *Let $\mathfrak{V}, \Delta \subseteq H, v \in \Delta$ and π be as above. The following statements hold:*

(1) $\mathrm{cd}_{\mu_{\Delta}} \geq \#\Delta$.

(2) M_{Δ} *is equivalent to* $M_{v,\pi}$*; in particular* $\mathbf{P}_{c,\pi} = \mathbf{P}_{\Delta}$.

(3) $\mathbf{P}_{\Delta} \cap \mathring{\Omega}_{r+1} \neq \emptyset$.

Proof. Statement (1) follows directly from the definition of V_{Δ} and Ψ_T^{Δ}. As for Statement (3), assume by contradiction that $\mathbf{P}_{\Delta} \cap \mathring{\Omega}_{r+1} = \emptyset$. Then, there is a nonempty $I \subseteq T$ such that $\mu_{\Delta}^*(I) = 0$. Thus, $\mu_{\Delta}^*(I \cap I_{v_i}^{\Delta}) = 0$ for all $v_i \in \Delta$. Hence, we may assume that $I \subseteq I_{v_i}^{\Delta}$ for some i. By Lemma 4.1.2, for each $u \in H - R_{v_i}$ there is an admissible path connecting v_i to u passing through v_{i+1}. Then, by [10, Lem. 8.1] and exactness, we have

$$\bigcap_{w \in H - R_{v_i}^{\Delta}} \ker(\varphi_w^{v_i}) = \ker(\varphi_{v_{i+1}}^{v_i}) = \mathrm{Im}(\varphi_{v_i}^{v_{i+1}}).$$

Thus

$$0 = \nu_{\Delta}^*(I) = \dim \Psi_I^v \left(\bigcap_{w \in H - R_{v_i}^{\Delta}} \ker(\varphi_w^{v_i}) \right) = \dim \Psi_I^v(\ker(\varphi_{v_{i+1}}^{v_i})) = \dim \Psi_I^v \varphi_{v_i}^{v_{i+1}}(V_{v_{i+1}}).$$

Note that, for each $\mathfrak{a} \in I$, as the shadow of $u_{\mathfrak{a}}$ is v_i, there is an admissible path connecting v_{i+1} to $u_{\mathfrak{a}}$ passing through v_i. Then, it follows that $\varphi_{u_{\mathfrak{a}}}^{v_{i+1}} = \varphi_{u_a}^{v_i} \varphi_{v_i}^{v_{i+1}} = 0$. This is a contradiction because H is the minimal set of generators of $\mathfrak{V}$ and $v_{i+1} \in H$. Hence, $\mathbf{P}_{\Delta} \cap \mathring{\Omega}_{r+1} \neq \emptyset$.

We will now show that M_{Δ} is equivalent to $M_{v,\pi}$. Let $F_{\bullet}$ the filtration associated to π. It is enough to show that $\theta_{\pi_j}(M_v^{F_j})$ is equivalent to $\Psi_{\pi_j}^{v_j}(\ker(\varphi_{v_{j+1}}^{v_j}))$ in U_{π_j}. Recall that $M_v^{F_j}$ is the image of $\bigcap_{\mathfrak{a} \in T - F_j} \ker(\varphi_{u_{\mathfrak{a}}}^v)$ in U via $\Psi_{F_j}^v$. Then, by Lemma 4.4.1,

$$\theta_{\pi_j}(M_v^{F_j}) = \Psi_{\pi_j}^v \left(\bigcap_{\mathfrak{a} \in T - F_j} \ker(\varphi_{u_{\mathfrak{a}}}^v) \right) = \Psi_{\pi_j}^v \left(\ker(\varphi_{F_j}^v) \right),$$

and, [10][Lem. 8.1], we have

$$\ker(\Psi_{\pi_j}^v) = \bigcap_{\mathfrak{a} \in \pi_j} \ker(\varphi_{u_{\mathfrak{a}}}^v) = \bigcap_{\mathfrak{a} \in \pi_j} \ker(\varphi_{T - \{\mathfrak{a}\}}^v) = \ker(\varphi_{T - \pi_j}^v).$$

Hence, by [10, Lem 8.1], we have

$$\ker(\Psi_{\pi_j}) \cap \ker(\varphi_{F_j}^v) = \ker(\varphi_{T - \pi_j}^v) \cap \ker(\varphi_{F_j}^v) = \ker(\varphi_{F_{j-1}}^v).$$

Thus,

$$\theta_{\pi_j}(M_v^{F_j}) = \Psi^v_{\pi_j}(\ker(\varphi^v_{F_j})/\ker(\varphi^v_{F_{j-1}})) = \Psi^v_{\pi_j}(\ker(\varphi^v_{v_{j+1}})/\ker(\varphi^v_{v_j}))$$

For each $\mathfrak{a} \in \pi_j$, the shadow of $u_\mathfrak{a}$ in Δ is v_j. Then $\varphi^{v_j}_{u_a}\varphi^v_{v_j} = \varphi^v_{u_\mathfrak{a}}$. Thus, there is $c \in (\mathbf{k}^*)^{\pi_j}$ such that

$$\Psi^v_{\pi_j} = c\Psi^{v_j}_{\pi_j}\varphi^v_{v_j}.$$

It follows from 4.1.3 that

$$\theta_{\pi_j}(M_v^{F_j}) = \Psi^v_{\pi_j}(\ker(\varphi^v_{v_{j+1}})/\ker(\varphi^v_{v_j})) = c\Psi^{v_j}_{\pi_j}\varphi^v_{v_j}(\ker(\varphi^v_{v_{j+1}})/\ker(\varphi^v_{v_j})) = c\Psi^{v_j}_{\pi_j}(\ker(\varphi^{v_j}_{v_{j+1}})),$$

as required. □

Proposition 4.4.5. *Let $\mathfrak{V}$ be an exact finitely generated nontrivial linked net over a $\mathbb{Z}^n$-quiver Q, and let H be its minimal set of generators. Let $v \in H$. For each nontrivial ordered partition π of T such that $\mathbf{P}_{v,\pi} \cap \mathring{\Omega}_{r+1} \neq \emptyset$, there exists a polygon $\Delta \subseteq H$ containing v such that π is the ordered partition of T induced by Δ and v. In addition, $\mathbf{P}_{v,\pi} = \mathbf{P}_\Delta$.*

Proof. Let $\pi = (\pi_1, \dots, \pi_m)$ be an m-partition such that $\mathbf{P}_{v,\pi} \cap \mathring{\Omega}_{r+1} \neq \emptyset$. Let $F_\bullet$ be the associated filtration of T. For each $i = 0, \dots, m-1$, put $v_i := F_i \cdot v$. Notice that $v_0 = v$. We claim that $v_i \in H$. Indeed, let w_i the shadow of v_i in H. Since v and v_i are neighbors, w_i is a neighbor of v and v_i. Thus, there is $I \subseteq T$ such that $I \cdot v = w_i$ and $I \subsetneq F_i$. Then, by Proposition 4.4.2,

$$\nu_v(I) = \dim\ker(\varphi^v_I) = \dim\ker(\varphi^v_{w_i}).$$

Since $\mathfrak{V}$ is 1-generated by H and w_i is the shadow of v_i in H, the map $\varphi^{w_i}_{v_i}$ is an isomorphism and $\varphi^v_{v_i} = \varphi^{w_i}_{v_i}\varphi^v_{w_i}$. Then, by Proposition 4.4.2, we have

$$\nu_v(F_i) = \dim\ker(\varphi^v_{F_i}) = \dim\ker(\varphi^v_{v_i}) = \dim\ker(\varphi^{w_i}_{v_i}\varphi^v_{w_i}) = \dim\ker(\varphi^v_{w_i}) = \nu_v(I).$$

It follows that $\mathbf{P}_{v,\pi} \subseteq \{q \in \mathbb{R}^T_{r+1} \,|\, q(F_i - I) = 0\}$. Since $\mathbf{P}_{v,\pi} \cap \mathring{\Omega}_{r+1} \neq \emptyset$, we must have $I = F_i$, and this $v_i \in H$, as required.

Now, let Δ be the oriented m-gon $\{v_0, \dots, v_{m-1}\}$. We claim that the partition $\pi' = (I^\Delta_{v_0}, \dots, I^\Delta_{v_m-1})$ induced by Δ coincides with π. It is enough to show that $\pi_i \subseteq \pi'_i$ for every $i = 1, \dots, m$. Let $\mathfrak{a} \in \pi_i$. We need to show that the shadow of $u_\mathfrak{a}$ in Δ is v_{i-1}. Since $u_\mathfrak{a}$ is extreme with respect to $\mathfrak{a}$, there is a path connecting v_{i-1} to $u_\mathfrak{a}$ avoiding $\mathfrak{a}$. As $\mathfrak{a} \in \pi_i$, for each $v_j \in \Delta$, there exists a path connecting v_j to v_{i-1} avoiding $\mathfrak{a}$. Hence, the shadow of $u_\mathfrak{a}$ in Δ is v_{i-1}, and thus $\mathfrak{a} \in \pi'_i$, as required.

□

Let $\mathcal{E}$ be the collection of ν_v for $v \in H$ and such that ν_v is simple.

Lemma 4.4.6. *For each $v, u \in H$, there exists a nonzero $c \in \mathbf{k}^T$ such that $cM_v \subseteq M_u$. Also, $\nu_v \neq \nu_u$ for all $u, v \in \mathcal{E}$. Finally, the collection $\mathcal{E}$ is separated.*

Proof. Let $I \subseteq T$ be the set of $\mathfrak{a} \in T$ such that there is an admissible path from v to $u_\mathfrak{a}$ passing through u. For each $\mathfrak{a} \in I$, there exists nonzero $c_\mathfrak{a} \in \mathbf{k}$ such that

$$\varphi^u_{u_\mathfrak{a}}\varphi^v_u = c_\mathfrak{a}\varphi^v_{u_\mathfrak{a}}.$$

If $\mathfrak{a} \notin I$, then $\varphi^u_{u_\mathfrak{a}}\varphi^v_u = 0$. Put $c_w := 0$ if $w \notin I$. We have $cM_v \subseteq M_u$.

The support of c is I. If M_v is equivalent to M_u, that is, $I = T$. There is no path from v to $u_\mathfrak{a}$ passing through u for any $\mathfrak{a}$ in the essentia; type of an admissible path connecting v to u. Then $u = v$.

Note that c induced an injective map $M_{v,I} \to M_u^I$, whence $\nu_v^*(I) \leq \nu_u * (I)$. Then (I, I^c) is a nontrivial separation for (ν_u, ν_v).

□

The following theorem is a consequence of Lemma 4.4.4 and Lemma 4.4.6, via Proposition 2.2.2.

Theorem 4.4.7. *Let $\mathfrak{V}$ be an exact finitely generated linked net nontrivial, and let H be its set of generators. Then the collection of polytopes $\mathbf{P}_{\mu_v}$ and their faces form polyhedral tiling of the standard simplex Ω_{r+1}.*

Proof. We will check the conditions of Proposition 2.2.2. Condition(2) follows immediately from Lemma 4.4.6.

As for Condition (1), we will show that for each $v \in H$ such that ν_v is simple, and for each facet $\mathbf{F}$ of $\mathbf{P}_{\nu_v}$ intersecting $\mathring{\Omega}_{r+1}$, there exists $u \in H$ such that ν_u is simple and $\mathbf{F}$ is a facet of $\mathbf{P}_{\nu_u}$. Let π be a bipartition such that $\mathbf{F} = \mathbf{P}_{\nu_v,\pi}$. By Proposition 4.4.5, there exists an ordered 2-gon $\Delta_1 = \{v, v_1\} \subseteq H$ such that π is the ordered partition of T induced by Δ_1 and v. Then $v_1 = I_v^\Delta \cdot v$. Also, $\mathbf{P}_{\nu_v,\pi} = \mathbf{P}_{\Delta_1}$. It follows from Lemma 4.4.4 (3) that

$$\mathbf{P}_{\nu_v,\pi} = \mathbf{P}_{\Delta_1} = \mathbf{P}_{\nu_{v_1},\pi^c}.$$

If ν_{v_1} is simple, we are done. Assume ν_{v_1} is not simple. Since $\mathbf{P}_{\nu_{v_1},\pi^c}$ is a facet of $\mathbf{P}_{\nu_v}$, we have that $\mathrm{cd}_{\nu_{v_1}} = 2$ and, $\nu_{v_1} = \nu_{v_1,\pi^c}$. Hence, by [1][Prop. 2.2 (4)], $\nu_{v_1} = \nu_{v_1,\pi}$. It follows that $\mathbf{P}_{\nu_{v_1},\pi} \cap \mathring{\Omega}_{r+1} \neq \emptyset$, and then by Proposition 4.4.5 there exists an ordered 2-gon $\Delta_2 = \{v_1, v_2\} \subseteq H$ such that π is the ordered partition of T induced by Δ_2 and v_1. Also, $\mathbf{P}_{\nu_{v_1},\pi} = \mathbf{P}_{\Delta_2}$ and, by Lemma 4.4.4 (3) we have that

$$\mathbf{P}_{\nu_{v_1},\pi} = \mathbf{P}_{\Delta_2} = \mathbf{P}_{\nu_{v_2},\pi^c}.$$

It follows that

$$\mathbf{P}_{\nu_v,\pi} = \mathbf{P}_{\nu_{v_1},\pi^c} = \mathbf{P}_{\nu_{v_1},\pi} = \mathbf{P}_{\nu_{v_2},\pi^c}.$$

Also $v_2 = I_{v_1}^{\Delta_2} \cdot v_1$. Since π is induced by Δ_1 and v_1, and by Δ_2 and v_2, we have that $I_v^{\Delta_1} = I_{v_1}^{\Delta_2} = \pi_1$. Then $v_2 = \pi^2 \cdot v$. Repeating this process we obtain a sequence of vertices $v_1, \ldots, v_p, \ldots$ such that $v_i = \pi_1^i \cdot v$. This process stop since H is finite and $\mathbf{F} \cap \mathring{\Omega}_{r+1} \neq 0$.

□

Chapter 5

The linked projective space.

5.1 The linked projective space.

Let $\mathfrak{V}$ be a a finitely generated exact linked net of dimension $r+1$ over a $\mathbb{Z}^n$-quiver Q in the category of finite-dimensional vector spaces over an algebraically closed field $\mathbf{k}$. We denote by $\mathbb{LP}(\mathfrak{V})$ the quiver Grassmannian of subrepresentations of constant dimension 1 of $\mathfrak{V}$. Let H be the minimum set of generators of $\mathfrak{V}$, and let $\mathbb{LP}_H(\mathfrak{V})$ be the subscheme of $\prod_{v\in H}\mathbb{P}(V_v)$ defined by

$$\mathbb{LP}_H(\mathfrak{V}) := \left\{ (s_v|v \in H) \in \prod_{v\in H} \mathbb{P}(V_v) \;\middle|\; \varphi^v_w(s_v) \wedge s_w = 0 \text{ for all } v, w \in H \right\}.$$

The natural map $\phi^{\mathfrak{V}}_H \colon \mathbb{LP}(\mathfrak{V}) \to \mathbb{LP}_H(\mathfrak{V})$ induced by restriction is a bijection. See for further details [11, Prop. 5.5 and Def. 5.6].

Definition 5.1.1. Let $\mathfrak{V}$ be a nontrivial finitely generated exact linked net of vector spaces over a $\mathbb{Z}^n$-quiver Q. Give $\mathbb{LP}(\mathfrak{V})$ the scheme structure induced from the bijection $\Psi^{\mathfrak{V}}_H$, where H is the minimum set of generators of $\mathfrak{V}$. We call $\mathbb{LP}(\mathfrak{V})$ the *linked projective space* associated to $\mathfrak{V}$.

It was shown in [11][thm. 6.4 and Thm. 8.2] that $\mathbb{LP}(\mathfrak{V})$ is generically smooth, a local complete intersection and reduced. The irreducible components of $\mathbb{LP}(\mathfrak{V})$ can be described as the scheme-theoretic images of the rational maps

$$\mathbb{P}(V_v) \dashrightarrow \prod_{u\in H} \mathbb{P}(V_u)$$

for v ranging over H. It follows from Lemma 3.5.2 that each map is indeed defined on a dense open subset, with image equal to

$$\mathbb{LP}(\mathfrak{V})^*_v := \{\mathfrak{M} \subseteq \mathfrak{V} \,|\, \mathfrak{M} \text{ is generated by } v\}.$$

Denote its closure by $\mathbb{LP}(\mathfrak{V})_v$.

Remark 5.1.2. In analogy with [4], we define the *reduction complex* of $\mathbb{LP}(\mathfrak{V})$ as the simplicial complex with one vertex for each component of $\mathbb{LP}(\mathfrak{V})$, where a set of vertices forms a simplex if and only if the intersection of the corresponding components is nonempty.

By [11][Prop. 6.3], the reduction complex of $\mathbb{LP}(\mathfrak{V})$ is isomorphic to the simplicial complex formed by the polygons in H. In addition, by Proposition 4.2.3, the latter is equal to the simplicial complex where a set of vertices forms a simplex if and only if the intersection of the corresponding polytopes intersects $\mathring{\Omega}_{r+1}$.

Chow ring.

Put $\mathbb{P} := \prod_{v \in H} \mathbb{P}(V_v)$. Let $A^*(\mathbb{P})$ be the Chow ring of $\mathbb{P}$. Recall that

$$A^*(\mathbb{P}) = \frac{\mathbb{Z}[h_v \,|,\, v \in H]}{(h_v^{r+1} | v \in H)},$$

where h_v denotes the pullback, via the projection map, of the hyperplane class on $\mathbb{P}(V_v)$ for each $v \in H$.

Given a reduced closed subscheme $X \subseteq \mathbb{P}$ with pure dimension r, its Chow class in $A^*(\mathbb{P})$ is given by

$$[X] = \sum_{q \in \Omega_r(\mathbb{Z})} \deg_{\mathbb{P}}^q(X) \prod_{v \in H} h_v^{r-q(v)} \in A^*(\mathbb{P}),$$

where $\deg_{\mathbb{P}}^q(X)$ is equal to the number of points on the intersection of X with a product $\prod L_v$ of general linear subspaces $L_v \subset \mathbb{P}(V_v)$ of dimension $r - q(v)$ for $v \in H$, and $\Omega_r(\mathbb{Z}) := \Omega_r \cap \mathbb{Z}^H$, with

$$\Omega_r := \{q \in \mathbb{R}^H \,|\, q(v) \geq 0 \text{ for all } v \in H \text{ and } q(H) = r\}.$$

In particular, the Chow class of the small diagonal in $\prod_{v \in H} \mathbb{P}(V_v)$ is

$$\sum_{q \in \Omega_r(\mathbb{Z})} \prod_{v \in H} h_v^{r-q(v)} \in A^*(\mathbb{P}).$$

Chow class of the linked projective space.

Here we show Theorem 5.1.3, which gives a formula for the Chow class of the linked projective space, showing that it has the same class as the small diagonal.

Theorem 5.1.3. *Let $\mathfrak{V}$ be a nontrivial exact finitely generated linked net of vector spaces of dimension $r+1$ over a $\mathbb{Z}^n$-quiver Q, and let H be its minimum set of generators. Then*

$$[\mathbb{LP}(\mathfrak{V})] = \sum_{q \in \Omega_r(\mathbb{Z})} \prod_{v \in H} h_v^{r-q(v)} \in A^*\left(\prod_{v \in H} \mathbb{P}(V_v)\right).$$

Proof. We will show that, for each $q \in \Omega_r(\mathbb{Z})$, the monomial

$$\prod_{v \in H} h_v^{r-q(v)}$$

appears in the class $[\mathbb{LP}(\mathfrak{V})]$ with multiplicity 1. Since $\mathbb{LP}(\mathfrak{V})$ is reduced, its Chow class can be written as

$$[\mathbb{LP}(\mathfrak{V})] = \sum_{v \in H} [\mathbb{LP}(\mathfrak{V})_v].$$

For each $v \in H$, let (μ_v, μ^*) the modular pair defined in Section 4.1.1, and let $\mathbf{P}_v$ be its corresponding polytope. Set

$$M_v := \{q \in \Omega_r(\mathbb{Z}) \ | \ q(I) \leq \mu_v^*(I) - 1 \text{ for each } I \subsetneq H, I \neq \emptyset\}.$$

It follows from [21] that

$$[\mathbb{LP}(\mathfrak{V})_v] = \sum_{q \in M_v} \prod h_v^{r-q(v)}.$$

It is thus enough to show that $\bigcup_{v\in H} M_v = \Omega_r(\mathbb{Z})$ and $M_v \cap M_u = \emptyset$ for each $u, v \in H$ with $u \neq v$.

Let $q \in \Omega_r(\mathbb{Z})$. Then $q' := (r+1)q/r \in \Omega_{r+1}$. It follows from Theorem 4.2.2 that there exists $v \in H$ such that $q' \in \mathbf{P}_v$. We claim that $q \in M_v$. We have to show that $q(I) \leq \mu_v^*(I) - 1$ for each $I \subsetneq H, I \neq \emptyset$. Since μ_v is simple, we have $\mu_v^*(I) \neq 0$. As $q' \in \mathbf{P}_v$, we have

$$q(I) \leq r\mu_v^*(I)/(r+1) < \mu_v^*(I).$$

Since $q(I) \in \mathbb{Z}$, we have $q(I) \leq \mu_v^*(I) - 1$, as required.

Assume now by contradiction that $q \in M_u \cap M_v$. By Theorem 4.2.1, there exists a bipartition $\pi = (I, J)$ such that $\mu_v^*(I) + \mu_u^*(J) \leq r+1$. Hence, we have

$$r = q(I) + q(J) \leq (\mu_v^*(I) - 1) + (\mu_u^*(J) - 1) \leq r - 1,$$

a contradiction. Thus $M_u \cap M_v = \emptyset$.

□

5.2 Normal crossings schemes.

Definition 5.2.1. Let X be a Noetherian scheme. Let D be an effective Cartier divisor, and let $D_i \subseteq D, i \in I$ be its irreducible components (considered as reduced closed subschemes of D or X). We say that D is a *strict normal crossings divisor in* X if:

1. for every $x \in D$, the local ring $\mathcal{O}_{X,x}$ is regular.
2. D is reduced,
3. for each finite subset $J \subset I$,
$$D_J := \bigcap_{j\in J} D_j$$
is regular and equidimensional of dimension $\dim X - \#J$

Definition 5.2.2. A reduced effective Cartier divisor D in a scheme X which is regular along D is a *normal crossings divisor* if for all $x \in D$, there exists an étale neighborhood $X' \to X$ such that $D' := X' \times_X D$ is a simple normal crossings divisor in X

Let R be a discrete valuation ring with field of fractions $\mathbf{K}$, uniformizer T and residue field $\mathbf{k}$. Denote by S the affine scheme $\mathrm{Spec}(R)$.

Let us recall the definition of a semi-stable S-variety from [7, 2.16], its local description and its relation with normal crossings divisors.

Definition 5.2.3. [7, 2.15] A *S-variety* is an irreducible, reduced scheme X, which is flat, separated and of finite type over S.

If X is a S-variety, we denote by $X_{\mathbf{K}} := X \times_S \mathrm{Spec}\,\mathbf{K}$ the generic fiber of X and by $X_{\mathbf{k}} := X \times_S \mathrm{Spec}\,\mathbf{k}$ the special fiber.

Definition 5.2.4. [7, 2.16] Let X be a S-variety. Let $X_i, i \in I$, be the irreducible components of $X_{\mathbf{k}}$. Put $X_J := \bigcap_{j\in J} X_j$ (scheme-theoretic intersection) for each nonempty subset J of I. The S-variety X is called *strictly semi-stable over* S if the following properties hold:

(a) $X_{\mathbf{K}}$ is smooth over $\mathbf{K}$,

(b) $X_{\mathbf{k}}$ is geometrically reduced,

(c) for each $i \in I$, X_i is a Cartier divisor on X, and

(d) for each nonempty $J \subseteq I$, the scheme X_J is smooth over $\mathbf{k}$ and equidimensional of dimension $\dim X - \#J$.

Also, X is called *semi-stable over* S if étale locally X is strictly semi-stable over S.

Following [7, 2.16], we describe locally a strictly semi-stable S-variety. Let $x \in X_{\mathbf{k}}$ be a point on the special fiber of X. Suppose that x lies on the components $X_1, \ldots, X_r$ and not on the other components of $X_{\mathbf{k}}$. Let $t_i \in \mathcal{O}_{X,x}$ be an element such that $V(t_i) = X_i \cap \mathrm{Spec}(\mathcal{O}_{X,x})$. Consider $\hat{\mathcal{O}}_{X,x}$, the completion of $\mathcal{O}_{X,x}$. Then, there exists a complete local R-algebra B, formally smooth over R such that

$$\hat{\mathcal{O}}_{X,x} \simeq \frac{B[[t_1, \ldots, t_r]]}{(t_1 \cdots t_r - T)}.$$

Definition 5.2.5. A scheme X is called *(strictly) normal crossings scheme* if every point has a Zariski open neighborhood which is isomorphic to a (strictly) normal crossings divisor.

Remark 5.2.6. Any (strict) semi-stable S-variety is in fact regular, and its special fiber is a divisor with (strict) normal crossings, see [7, 2.16] or [16, Rmk 1.1.1], and hence a strict normal crossings scheme.

5.3 Mustafin varieties

In this section we review the theory of Mustafin varieties. We recall the set-up given in the introduction. Let R be a discrete valuation ring with field of fractions $\mathbf{K}$, uniformizer T and residue field $\mathbf{k}$.

Let V be a vector space of dimension d over $\mathbf{K}$. We define $\mathbb{P}(V) := \mathrm{ProjSym}(V^*)$, thus parameterizing lines on V. We call free R-modules $L \subseteq V$ of rank d *lattices*, and define $\mathbb{P}(L) := \mathrm{ProjSym}(L^*)$, where $L^* = \mathrm{Hom}_R(L, R)$. Note that we will mostly consider lattices up to *homothety*, i.e., $L \sim L'$ if $L = c \cdot L'$ for some $c \in \mathbf{K}^\times$. We denote the homothety class of L by $[L]$. We will now define Mustafin varieties.

Definition 5.3.1. ([4, Def. 1.1.] or [14, Def. 1.1]) Let $\Gamma = \{[L_1], \ldots, [L_n]\}$ be a set of lattice classes in V. Then $\mathbb{P}(L_1), \ldots, \mathbb{P}(L_n)$ are projective spaces over R whose generic fibers are canonically isomorphic to $\mathbb{P}(V)$. The open immersions $\mathbb{P}(V) \hookrightarrow \mathbb{P}(L_i)$ give rise to a map

$$\mathbb{P}(V) \longrightarrow \mathbb{P}(L_1) \times_R \cdots \times_R \mathbb{P}(L_n).$$

We denote the closure of the image, endowed with the reduced scheme structure, by $\mathcal{M}(\Gamma)$. We call $\mathcal{M}(\Gamma)$ the *associated Mustafin variety to* Γ. Its special fiber $\mathcal{M}(\Gamma)_{\mathbf{k}}$ is a scheme over $\mathbf{k}$.

Consider the diagonal map

$$\Delta : \mathbb{P}(V) \to \mathbb{P}(V)^n = \mathbb{P}(V) \times_{\mathbf{K}} \cdots \times_{\mathbf{K}} \mathbb{P}(V).$$

The image of Δ is the subvariety of $\mathbb{P}(V)^n$ cut out by the ideal generated by the 2×2-minors of the matrix $X = (x_{ij})_{i=1,\ldots,d; j=1,\ldots,n}$ of unknowns, where the jth column corresponds to chosen coordinates on the jth factor.

Every $g \in \mathrm{GL}(V)$ determines a dual map $g^t : V^* \to V^*$ and a morphism $g : \mathbb{P}(V) \to \mathbb{P}(V)$. If $g_1, \ldots, g_n$ are elements of $\mathrm{GL}(V)$, the image of

$$\mathbb{P}(V) \xrightarrow{\Delta} \mathbb{P}(V)^n \xrightarrow{g_1 \times \cdots \times g_n} \mathbb{P}(V)^n$$

is the subvariety of the product $\mathbb{P}(V)^n$ given by the multi-homogeneous prime ideal

$$I_2((g_1^{-t}, \ldots, g_n^{-t})(X)) \subseteq \mathbf{K}[X]$$

generated by the 2×2 minors of the matrix $(g_1^{-1}, \ldots, g_n^{-1})(X)$, whose jth column equals

$$g_j^{-t} \begin{pmatrix} x_{1j} \\ \vdots \\ x_{dj} \end{pmatrix}.$$

Consider a reference lattice $L = Re_1 + \cdots + Re_d \subseteq V$. For each set of lattice classes $\Gamma = \{[L_1], \ldots, [L_n]\}$ we choose matrices $g_1, \ldots, g_n \in \mathrm{GL}(V)$ such that $L = g_i L_i$ for all i. The following diagram commutes:

$$\begin{tikzcd}
\mathbb{P}(V) \arrow[r, "(g_1 \times \cdots \times g_n) \circ \Delta"] \arrow[d] & \mathbb{P}(V)^n \arrow[d] \\
\prod_R \mathbb{P}(L_i) \arrow[r, "(g_1 \times \cdots \times g_n)"] & \mathbb{P}(L)^n
\end{tikzcd}$$

Hence, the Mustafin variety $\mathcal{M}(\Gamma)$ is isomorphic to the subscheme of $\mathbb{P}(L)^n \simeq (\mathbb{P}_R^{d-1})^n$ cut out by the multi-homogeneous ideal $I_2((g_1^t, \ldots, g_n^t)(X)) \cap R[X]$ in $R[X]$.

Remark 5.3.2. The construction of the Mustafin variety $\mathcal{M}(\Gamma)$ depends only on the homothety classes of the lattices Λ_i. Hence, Γ can be regarded as a configuration in the Bruhat-Tits building $\mathfrak{B}_d$ associated with the group $\mathrm{PGL}(V)$

We will now explain the relation between Mustafin varieties and tropical convexity.

Call two distinct lattices classes $[L]$ and $[M]$ in V *adjacent* if there exist representatives $L' \in [L]$ and $M' \in [M]$ such that $TM' \subseteq L' \subseteq M'$.

Definition 5.3.3. A finite set Γ of R-lattice classes in V is called *convex* if, for each two classes $[L], [L'] \in \Gamma$ and each representatives L and L', the class of the intersection $L \cap L'$ is again in Γ. For an arbitrary finite set of R-lattice classes, the intersection of all convex sets containing Γ is called its *convex closure*, and is denoted by $\mathrm{Conv}(\Gamma)$.

The following proposition was first shown by Faltigns in [12, 5][Section 5] using a moduli description for $\mathcal{M}(\Gamma)$. Also, F. Gora gave a proof by describing Mustafin varities as sequence of blowups, see [13, Prop. 2.25]

Proposition 5.3.4. *([12][*Section 5] or [13, 2.25][Prop. 2.25*) For a finite convex set Γ of R-lattices in V, the Mustafin variety $\mathcal{M}(\Gamma)$ is semi-stable.*

In [4], the notion of convexity for a finite set Γ of R-lattice classes is reformulated in terms of tropical geometry as follows. A subset P of $\mathbb{R}^d$ is called *tropically convex* if it is closed under linear combinations in the min-plus algebra, i.e., for each two vectors $x = (x_1, \dots, x_d)$ and $y = (y_1, \dots, y_d)$ in P, and scalars $\lambda, \mu \in \mathbb{R}$, we also have

$$(\min(x_1 + \lambda, y_1 + \mu), \dots, \min(x_d + \lambda, y_d + \mu)) \in P.$$

We write the tropical arithmetic operations as $x \oplus y := \min(x, y)$ and $x \otimes y := x + y$. In particular, if P is tropically convex and $x \in P$, then $\lambda \otimes x \in P$ for all λ. Thus, we can identify each tropically convex set $P \subseteq \mathbb{R}^d$ with its image in the *tropical projective space*, defined as the quotient vector space:

$$\mathbb{TP}^{d-1} := \mathbb{R}^d / \mathbb{R}^d(1, 1, \dots, 1).$$

Let $e_1, \dots, e_d$ be a basis of V. Let $[L]$ be a lattice class such L is in diagonal form with respect to the basis $e_1, \cdots, e_d$, i.e., $L = T^{m_1}Re_1 + \cdots + T^{m_d}Re_d$ for certain $m_1, \dots, m_d \in \mathbb{Z}$. We associate to $[L]$ the residue class of $(-m_1, \dots, -m_d)$ in $\mathbb{TP}^{d-1}$. A point in $\mathbb{TP}^{d-1}$ is a *lattice point* if it is represented by a vector in $\mathbb{Z}^d$. The association

$$[T^{m_1}Re_1 + \cdots + T^{m_d}Re_d] \longmapsto (-m_1, \dots, -m_d) + \mathbb{R}^d(1, 1, \dots, 1) \tag{5.3.1}$$

is a bijection between the set of lattices classes in diagonal form with respect to the basis $e_1, \dots, e_d$ and the set of lattice points of $\mathbb{TP}^{d-1}$.

We fix a configuration $\Gamma = \{u^{(1)}, \dots, u^{(n)}\}$ of lattice points in $\mathbb{TP}^{d-1}$. The point $u^{(i)} = (u_{i1}, \dots, u_{id})$ represents the diagonal lattice $L^{(i)} = T^{-u_{i1}}Re_1 + \cdots + T^{-u_{id}}Re_d$. Denote by $\mathrm{tconv}(\Gamma)$ the smallest tropically convex subset of $\mathbb{TP}^{d-1}$ containing Γ.

Lemma 5.3.5. *[4]*[Lemma 4.1] Let $\Gamma = \{u^{(1)}, \dots, u^{(n)}\}$ and $L^{(i)}$ be as above. The bijection in 5.3.1 induces a bijection between lattice points in $\mathrm{tconv}(\Gamma)$ and the lattices classes in the convex closure of the set of lattices classes $\{[L^{(1)}], \dots, [L^{(n)}]$.

5.3.1 Mustafin varieties and prelinked Grassmannian

Let G be a quiver, d an integer, and S a scheme. Suppose we are given the following data $\mathcal{E}$: a vector bundle $\mathcal{E}_v$ of rank d over S for each vertex v of G, and morphism $f_a : \mathcal{E}_v \to \mathcal{E}_u$ for each arrow a of G with tail v and head u.

Suppose that we also have the following condition satisfied: For each two paths γ and γ' in G connecting the same two vertices, there are sections $s, s' \in \mathcal{O}_S(S)$ such that

$$s \cdot f_\gamma = s' \cdot f_{\gamma'}.$$

Moreover, suppose that γ (resp. γ') is minimal, then s' (resp. s) is invertible.

Let $r < d$ be an integer. Then define the *prelinked Grassmanian* $LG(r, \mathcal{E}_\bullet)$ to be scheme representing the functor $\mathcal{LG}(r, \mathcal{E}_\bullet)$ which associates to a S-scheme T the set of rank-r vector subbundles $\mathcal{F}_v \subseteq \mathcal{E}_v$, one for each vertex v of G, such that $f_a(\mathcal{F}_v) \subseteq \mathcal{F}_u$ for each arrow a of G, where v is tail and u is the head of a. Representability of $\mathcal{LG}(r, \mathcal{E}_\bullet)$ was shown in [25][Prop. A.1.3].

For a convex configuration of lattices Γ, the Mustafin variety $\mathcal{M}(\Gamma)$ is a prelinked Grassmanian. We explain:

Let Γ be a convex configuration of lattices in V. We fix the following data:

1. The scheme $S := \mathrm{Spec}(R)$.

2. A reference lattice $L = Re_1 + \cdots + Re_d \subseteq V$.

3. The quiver $G = G(\Gamma)$, defined as follows: The set of vertices of G is the set of lattice classes in Γ. For each vertex v, we denote by $[L_v]$ the corresponding lattice class. As for the set of arrows, for each pair of vertices whose corresponding lattice classes are adjacent, we connect them by a pair of arrows, one in each direction.

4. For each vertex v of G, let L_v be a representative of the lattice class $[L_v]$, and let $g_v \in \mathrm{GL}(V)$ such that $g_v L_v = L$.

5. If $[L_v]$ and $[L_w]$ are adjacent, and $m_{v,w}$ is the minimum integer such that $T^{m_{v,w}} L_v \subseteq L_w$, we define $g_{v,w} = g_w T^{m_{v,w}} g_v^{-1}$.

6. For each pair of vertices v and w, let $f_{v,w} \in \mathrm{GL}(V)$ be a representative of the class of $g_{v,w}$ in $\mathrm{PGL}(V)$ such that its matrix representation $A_{v,w} = (a_{ij})_{i,j}$ with respect to the basis $e_1, \ldots, e_d$ of V satisfies $\mathrm{val}(a_{ij}) \geq 0$, and for at least one pair (i', j'), we have $\mathrm{val}(a_{i',j'}) = 0$.

7. For each vertex v of G, we put $\mathcal{E}_v := L \otimes \mathcal{O}_S$.

8. For each arrow $\mathfrak{a}$ of G from v to w, we declare the map $f_{\mathfrak{a}} : \mathcal{E}_v \to \mathcal{E}_w$ to be $f_{v,w} \otimes \mathcal{O}_S$.

We denote the prelinked Grassmannian of rank-r associated to the data $\mathcal{E}_\bullet$ above by $\mathrm{LG}(r, \Gamma)$. In the case $r = 1$, we write $\mathbb{LP}(\Gamma)$.

Corollary 5.3.6. *[14][*Cor. 4.6] Let Γ be a convex configuration of lattices, $\mathcal{M}(\Gamma) \simeq \mathbb{LP}(\Gamma)$

5.4 Mustafin varieties and linked projective spaces.

In this section, we will show that linked projective spaces are normal crossings schemes. In fact, they are locally special fibers of Mustafin varieties.

Let $\mathfrak{V}$ be an exact finitely generated linked net of vector spaces of dimension $r+1$ over a $\mathbb{Z}^n$-quiver Q.

Assume that $\mathfrak{V}$ is generated by a polygon Δ. We will show that $\mathbb{LP}(\mathfrak{V})$ is the special fiber of a Mustafin variety $\mathcal{M}(\Gamma)$ for a certain convex lattice configuration Γ.

Let $v \in \Delta$. There is a unique ordering $v_1, \ldots, v_m$ of the vertices of Δ such that $v_1 = v$ and $v_1, \ldots, v_m$ form an oriented m-gon. For convenience, put $v_{m+1} := v_1$. As $\mathfrak{V}$ is exact, it follows from [10][Prop. 10.1] that $\mathfrak{V}$ is the direct sum of finitely generated exact linked nets $\mathfrak{M}_0, \ldots, \mathfrak{M}_r$ of vector spaces of dimension 1. Since $\mathfrak{V}$ is 1-generated by Δ, so are the $\mathfrak{M}_j$. By [10][Thm. 7.8], the net $\mathfrak{M}_j$ is generated by v_{ℓ_j} for some $\ell_j \in \{1, \ldots, m\}$ for each $j = 0, \ldots, r$; let $s_j \in V_{v_{\ell_j}}$ be a generator. Reorganize so that $\ell_0 \leq \ell_1 \leq \cdots \leq \ell_r$. For each $\ell = 1, \ldots, m$, let $r_\ell := \#\{j \mid \ell_j = \ell\}$. We have $\sum r_\ell = r + 1$. The s_j induce a basis for $V_{v_i}^{\mathfrak{V}}$ for $i = 1, \ldots, m$, and thus a decomposition $V_{v_i}^{\mathfrak{V}} = V_{i,1} \oplus \cdots \oplus V_{i,m}$, where $V_{i,\ell}$ is the subspace generated by the $\varphi_{v_i}^{v_\ell}(s_j)$ for all j with $\ell_j = \ell$, for $\ell = 1, \ldots, m$; each $V_{i,\ell}$ has dimension r_ℓ.

For each $i = 1, \ldots, m$, the map $\varphi_{v_{i+1}}^{v_i}$ can then be represented by a diagonal matrix M_i. For $i = 1, \cdots, m-1$, all of its entries are 1 but those in positions $r_1 + \cdots + r_i + j$ for $j = 0, \ldots, r_{i+1} - 1$, which are 0. The matrix M_m has all of its entries 1 but those in positions $0, \ldots, r_1 - 1$.

Put $R := \mathbf{k}[[T]]$. For each $i = 1, \ldots, m$, define $M_i(T)$ as the diagonal matrix having the same entries as M_i except for the 0 on the diagonal, that are replaced by T. Define, for $i = 1, \ldots, m$,

$$L_i := \bigoplus_{j=0}^{r} T^{-\alpha_{ij}} R,$$

where

$$\alpha_i := \sum_{j=0}^{r_1+\cdots+r_i-1} e_j = e_0 + e_1 + e_{r_1+\cdots+r_i-1} \in \mathbb{Z}^{r+1}.$$

Put $\Gamma := \{[L_1], \ldots, [L_m]\}$. Let $L := \bigoplus_{j=0}^{r} R$.

Lemma 5.4.1. *The collection of lattices Γ is convex.*

Proof. All lattices in Γ are in the form of 5.3.1. Hence, it follows from [4][Lem. 4.1] that Γ is convex if and only if the collection of all residue classes of the α_j are the lattice points of their tropical convex hull. Let $C := \operatorname{tconv}(\alpha_1, \ldots, \alpha_m) \subseteq \mathbb{R}^{r+1}/\mathbb{R}$ be the tropical convex hull of the set of points α_i. Let $[\gamma] \in C$ be a lattice point in C. By definition, $[\gamma]$ is represented by a point $\gamma \in \mathbb{Z}^{r+1}$. Since $[\gamma] \in C$, there is a representative that can be written as a tropical linear combination of the $\alpha_1, \ldots, \alpha_m$. Then γ is a tropical linear combination of $\alpha_1, \ldots, \alpha_m$. Indeed, assume that there is $\lambda \neq 0$ such that

$$\lambda \otimes \gamma = \bigoplus_{i=1}^{m} \lambda_i \otimes \alpha_i.$$

Multiplying by the tropical inverse λ^{-1} of λ, we obtain

$$\gamma = \bigoplus_{i=1}^{m} \lambda^{-1} \otimes \lambda_i \otimes \alpha_i.$$

We may thus assume that

$$\gamma = \bigoplus_{i=1}^{m} \lambda_i \otimes \alpha_i.$$

Then, for each $i = 1, \ldots, m$ and $j = r_1 + \cdots + r_{i-1}, \ldots, r_1 + \cdots + r_i - 1$,

$$\gamma_j = \tau_i := \min\{\lambda_1, \ldots, \lambda_{i-1}, \lambda_i + 1, \lambda_m + 1\}.$$

Then

$$\tau_1 \geq \tau_2 \geq \tau_3 \geq \cdots \geq \tau_m \geq \tau_1 - 1.$$

Let $i \in \{1, \ldots, m\}$ be the largest integer such that $\tau_i = \tau_1$. Then $\tau_j = \tau_1$ for $j \leq i$ and $\tau_j = \tau_1 - 1$ for $j > i$. It follows that $\gamma = \tau_1 \otimes \alpha_i$. Thus $[\gamma] = [\alpha_i]$, as required. $\square$

Let G be the quiver whose vertex set is the set of lattice classes in Γ. The arrow set of G is

$$\{([L_i], [L_j]) \mid [L_i] \text{ is adjacent to } [L_j]\}.$$

Let $Q(\Delta)$ be the quiver with set of vertices Δ, such that every pair of distinct vertices is connected by a pair of arrows, one in each direction, as in Section 3.6.

Lemma 5.4.2. *Let G and $Q(\Delta)$ be as above. Then $G = Q(\Delta)$.*

Proof. We need only show that any two lattices in Γ are adjacent. Let $[L_i], [L_j] \in \Gamma$ be two lattice classes. We may assume $i \geq j$. Then $TL_j \subseteq L_i \subseteq L_j$, as required. $\square$

For each $i = 1, \ldots, m$, let g_i be the diagonal matrix whose entries in the diagonal are 1 but those in position j for $j = 0, \ldots, r_1 + \cdots + r_i - 1$, which are T. Note that $g_i L_i = L$.

For $i, j \in \{1, \ldots, m\}$, let $m_{i,j}$ be the minimum integer such that $T^{m_{ij}} L_i \subseteq L_j$. Then

$$m_{ij} = \begin{cases} 0 & i \leq j, \\ 1 & i > j. \end{cases}$$

Also, put $g_{i,j} := g_j T^{m_{ij}} g_i^{-1}$.

For each v_i, v_j, we fix a representative f_{ij} of the equivalence class of $g_{i,j}$ in $\mathrm{PGL}(K^{r+1})$ such that the minimum of the valuations of its entries is 0. We fix the vector bundle at every vertex $[L_i]$ of G to be $L \otimes \mathcal{O}_R$. For the arrow a from $[L_i]$ to $[L_j]$, we declare the map f_a to be $f_{i,j} \otimes \mathcal{O}_R$. We denote the associated prelinked Grassmannian of rank-1 subbunbdles by $\mathbb{LP}(\Gamma)$, as in Section 5.3.1.

Lemma 5.4.3. *Notation as above. Then* $\mathbb{LP}(\Gamma)_{\mathbf{k}} = \mathbb{LP}(\mathfrak{V})$.

Proof. By the definition of prelinked Grassmannian, we have

$$\mathbb{LP}(\Gamma)_{\mathbf{k}} = \left\{ (s_i) \in \prod_{i=1}^{m} \mathbb{P}^r \;\middle|\; \overline{f_{i,j}}(s_i) \wedge s_j = 0 \text{ for all } i, j \in \{1, \ldots, m\} \right\},$$

where the bar denotes the residue class in $\mathbf{k}$. Therefore, we need only show that $\overline{f_{i,j}} = M_i$ for $i = 1, \ldots, m$, where $f_{m,m+1} := f_{m,0}$. Indeed, for $i = 1, \ldots, m-1$, we have

$$\overline{f_{i,j}} = \overline{g_{i+1} g_i^{-1}} = M_i,$$

whereas for $i = m$ we have

$$\overline{f_{m,0}} = \overline{g_0 T g_m^{-1}} = M_m.$$

$\square$

Theorem 5.4.4. *Let $\mathfrak{V}$ be a pure nontrivial exact finitely generated linked net of vector spaces over a $\mathbb{Z}^n$-quiver. Then $\mathbb{LP}(\mathfrak{V})$ is a normal crossings scheme.*

Proof. Let $\mathfrak{M} \in \mathbb{LP}(\mathfrak{V})$. Then by [11][Thm. 3.6], there is a polygon Δ generating $\mathfrak{M}$. Since $P(\Delta) = \Delta$, we may consider the representation $\mathfrak{V}_\Delta := (\mathfrak{V}|_\Delta)^E$, as in Section 3.6, which is an exact nontrivial linked net of vector spaces over Q generated by Δ, and hence, by Lemma 5.4.3, $\mathbb{LP}(\mathfrak{V}_\Delta)$ is the special fiber of a Mustafin variety. Hence, by [13][Prop. 2.25] and Remark 5.2.6, $\mathbb{LP}(\mathfrak{V}_\Delta)$ is a strict normal crossings scheme. It is, then, sufficient to show that there is a Zariski open neighborhood of $\mathfrak{M}$ in $\mathbb{LP}(\mathfrak{V})$ which is isomorphic to an open subscheme of $\mathbb{LP}(\mathfrak{V}_\Delta)$.

Let H be the minimum set of vertices generating $\mathfrak{V}$. Then $P(H) = H$ and $\Delta \subseteq H$. Given a vertex v of Q denote by w_v its shadow in Δ. The $\mathfrak{X} \in \mathbb{LP}(\mathfrak{V})$ generated by Δ form an open subscheme U given by $\varphi_v^{w_v}(V_{w_v}^{\mathfrak{X}}) \neq 0$ for each $v \in H$. For each such $\mathfrak{X}$, there is a corresponding subrepresentation $\mathfrak{D}$ of $\mathfrak{V}_\Delta$ generated by all $V_u^{\mathfrak{X}}$ for $u \in \Delta$. Then $\mathfrak{D} \in \mathbb{LP}(\mathfrak{V}_\Delta)$. Indeed, given a vertex v of Q, we have that $V_v^{\mathfrak{D}} = \varphi_\mu^{\mathfrak{D}}(V_{w_v}^{\mathfrak{X}})$, where μ is an admissible path connecting w_v to v. Since $\varphi_\mu^{\mathfrak{D}}$ is the identity, $\mathfrak{D}$ is a pure subrepresentation of $\mathfrak{V}_\Delta$ of dimension 1, and thus $\mathfrak{D} \in \mathbb{LP}(\mathfrak{V}_\Delta)$.

Let

$$\Theta : U \to \mathbb{LP}(\mathfrak{V}_\Delta)$$

be the map taking $\mathfrak{X} \in U$ to $\mathfrak{D}$, as above. It is a scheme morphism. Also, the image of θ is the open subset U' of $\mathbb{LP}(\mathfrak{V}_\Delta)$ given by $\varphi_v^{w_v}(V_{w_v}^{\mathfrak{D}}) \neq 0$ for each $v \in H$. The induced map $\theta : U \to U'$ is an isomorphism. Indeed, given $\mathfrak{D} \in U'$, we let $\mathfrak{X}$ be the subnet of $\mathfrak{V}$ generated by $V_u^{\mathfrak{D}}$ for $u \in \Delta$. As before, for each vertex v of Q, we have $V_v^{\mathfrak{X}} = \varphi_v^{w_v}(V_{w_v}^{\mathfrak{D}})$, which is of dimension 1 because $\mathfrak{D}$ is. Hence, $\mathfrak{X} \in \mathbb{LP}(\mathfrak{V})$. The assignment $\mathfrak{D} \mapsto \mathfrak{X}$ is a scheme morphism and the inverse of Θ. $\square$

Bibliography

[1] Omid Amini and Eduardo Esteves. Tropicalization of linear series and tilings by polymatroids. *ArXiv:2405.04306*, 2024.

[2] Omid Amini, Eduardo Esteves, and Eduardo Garcez. Residue polytopes. *Arxiv:2410.13554*, 2024.

[3] Matthew Baker and Serguei Norine. Riemann–roch and abel–jacobi theory on a finite graph. *Advances in Mathematics*, 215(2):766–788, 2007.

[4] Dustin Cartwright, Mathias Häbich, Bernd Sturmfels, and Annette Werner. Mustafin varieties. *Selecta Mathematica*, 17(4):757–793, 2011.

[5] Dustin Cartwright and Bernd Sturmfels. The Hilbert scheme of the diagonal in a product of projective spaces. *International Mathematics Research Notices*, 2010(9):1741–1771, 12 2009.

[6] Juliana Coelho and Marco Pacini. Abel maps for curves of compact type. *Journal of Pure and Applied Algebra*, 214(8):1319–1333, 2010.

[7] A. J. de Jong. Smoothness, semi-stability and alterations. *Publications Mathématiques de l'Institut des Hautes Études Scientifiques*, 83(1):51–93, 1996.

[8] David Eisenbud and Joe Harris. Limit linear series: Basic theory. *Inventiones mathematicae*, 85(2):337–371, 1986.

[9] Eduardo Esteves, Piere Rodriguez, and Eduardo Vital. Degenerations of maps to projective spaces. *ArXiv:2507.00901*, 2025.

[10] Eduardo Esteves, Renan Santos, and Eduardo Vital. Quiver representations arising from degenerations of linear series, I. *Communications in Algebra*, 2025.

[11] Eduardo Esteves, Renan Santos, and Eduardo Vital. Quiver representations arising from degenerations of linear series, ii. *Bulletin of the Brazilian Mathematical Society, New Series*, 57(1):4, 2025.

[12] Gerd Faltings. *Toroidal Resolutions for Some Matrix Singularities*, pages 157–184. Birkhäuser Basel, Basel, 2001.

[13] Felix Gora. Local models, mustafin varieties and semi-stable resolutions. *Münster Journal of Mathematics*, 16:265–300, 2023.

[14] Marvin Anas Hahn and Binglin Li. Mustafin varieties, moduli spaces and tropical geometry. *Manuscripta Mathematica*, 166(1):167–197, 2021.

[15] Joe Harris and David Mumford. On the Kodaira dimension of the moduli space of curves. *Invent. math.*, 67:23–86, 1982.

[16] U. T. Hartl. Semi-stability and base change. *Archiv der Mathematik*, 77(3):215–221, 2001.

[17] Xiang He and Naizhen Zhang. Degenerations of grassmannians via lattice configurations. *International Mathematics Research Notices*, 2023(1):298–349, 10 2021.

[18] Michael Joswig, Bernd Sturmfels, and Josephine Yu. Affine buildings and tropical convexity. *Albanian Journal of Mathematics*, 2007.

[19] S. Keel and J. Tevelev. Geometry of chow quotients of grassmannians. *Duke Mathematical Journal*, 134:259–311, 2006.

[20] Georg Kempf, Finn Knudsen, David Mumford, and B. Saint-Donat. Toroidal embeddings. I. *Lecture Notes in Mathematics*, 339, 1973.

[21] Binglin Li. Images of rational maps of projective spaces. *International Mathematics Research Notices*, 2018(13):4190–4228, 02 2017.

[22] L. Mainò. Moduli space of enriched stable curves. *Ph.D. Thesis, Harvard*, 1998.

[23] David Mumford. An analytic construction of degenerating curves over complete local rings. *Compositio Mathematica*, 24(2):129–174, 1972.

[24] G. A. Mustafin. Nonarchimedean uniformization. *Math. USSR-Sb.*, 34(2):187–214, 1978. Zbl: 0411.14006, 0407.14006.

[25] Brian Osserman. A limit linear series moduli scheme (Un schéma de modules de séries linéaires limites). *Annales de l'institut Fourier*, 56(4):1165–1205, 2006.

[26] Brian Osserman. Limit linear series and the Amini–Baker construction. *Mathematische Zeitschrift*, 293(1):339–369, 2019.

[27] Brian Osserman. Limit linear series for curves not of compact type. *Journal für die reine und angewandte Mathematik (Crelles Journal)*, 2019(753):57–88, 2019.